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FELIX

Demands Refinement Framework

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ABSTRACT

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This work focuses on the process of refining new demands in software development projects, considering recurring challenges related to lack of standardization, misalignment between teams, and risks identified late in the project lifecycle. The main objective is to propose the FELIX Framework, a new refinement process that promotes greater governance, predictability, and alignment between business and technology areas from the initial stages of initiatives. The methodology adopted is based on the analysis of existing refinement practices, the observation of organizational processes, and the consolidation of a structured model that defines rituals, artifacts, and responsibilities, with a focus on reducing ambiguities and strengthening early decision-making. As a result, FELIX contributes to increasing the predictability of deliveries and strengthening the autonomy of the Product Owner by providing greater clarity and transparency in the refinement process.

Keywords: Product Owner, Solution Architecture, Tech Leader, Framework, Product Management, Agile Methodology.

SUMMARY

1. Introduction

During their internship at “BTG Pactual”, the authors of this paper identified a recurring challenge in the development of systems within the agile methodology: during the refinement of demands, the definition of the solution's architecture was concentrated on the Tech Lead, without the active participation of the Product Owner (PO), which made it difficult to understand the product and transition the tasks to the developers. Although POs are responsible for prioritizing the backlog and aligning deliveries with business needs, the lack of a clear direction in the organization of the architecture results in difficulties in the predictability of the process, in communication with stakeholders and in making strategic decisions before technical implementation.

This gap leads to problems such as misalignment between business expectations and technical constraints, rework due to unstructured architectural decisions and delays in delivery times. Often, Tech Leaders end up taking responsibility for the entire architecture, which overloads these professionals and limits the autonomy of the POs. This scenario compromises the efficiency of the teams, affecting deadlines, scope and transparency in communication.

With this in mind, this paper proposes the development of a framework that allows POs to map and organize the solution's architecture in a more strategic and autonomous way. With clear guidelines, this framework will make it easier to anticipate technical restrictions, better understand system integrations and reduce the need for constant reviews with Tech Leaders. First step, described in this document, involves interviews with POs and Tech Leaders, as well as a benchmarking study to analyze practices adopted by other companies.

1.1. Problem

At “BTG Pactual”, the absence of a structured model for Product Owners to actively participate in defining the solution's architecture creates a series of challenges. The lack of transparency in defining new projects and the revision of previously established agreements due to technical issues can impact deadlines, scope and alignments with stakeholders. In addition, the centralization of architectural decisions in Tech Leaders can result in bottlenecks and reduce the agility of the development process.

This problem highlights the need for a framework that enables POs to define the solution's architecture efficiently and strategically, ensuring greater predictability and risk mitigation.

1.2. Objectives

The aim of this work is to develop a framework that allows product owners to structure the solution's architecture autonomously, reducing exclusive dependence on technical leaders and ensuring greater predictability, transparency and alignment between the business vision and architecture decisions.

1.2.1. Specific Objectives

- Identify the challenges faced by POs in managing solution architecture through bibliographic research and benchmarking;
- Develop guidelines and tools to help POs define architectural solutions aligned with business needs;
- Create a structured model for documenting and communicating the solution architecture among stakeholders;
- Test and validate the framework in real scenarios, assessing its impact on project management and the predictability of value delivery;

- Refine and consolidate the framework based on the feedback obtained, ensuring its applicability in different organizational contexts.

2. Solution Development

2.1 Methodology

This study adopts a qualitative and exploratory approach, based on action research, to investigate and develop a framework that expands the role of the Product Owner (PO) in solution architecture. The choice of action research is justified by its iterative and participatory nature, allowing challenges to be analyzed in successive cycles of planning, action, observation and reflection (LEWIN, 1946).

The methodology is structured in three main stages:

1. Literature review - analysis of academic studies and market practices on the role of the PO in architectural decisions.
2. Collection of empirical data - interviews with POs and Tech Leaders to map challenges and opportunities.
3. Developing and validating the framework - applying the principles of action research to continually adjust the proposed framework.

Below we detail the action research adopted and its practical application in this study.

2.1.1 Action Research Methodology

Action research was first proposed by Kurt Lewin (1946) and is characterized by a continuous cycle of observation, planning, action and reflection. This method allows for active intervention in the problem studied, enabling the empirical validation of solutions and their adaptation as new discoveries emerge throughout the process.

According to McNiff (2017) and Coghlan & Shani (2018), action research drives organizational change by integrating researchers and participants in the investigative process, ensuring that the solutions developed are contextually relevant and applicable. In addition, studies such as those by Brydon-Miller et al. (2019) and Greenwood (2022) highlight the importance of the active participation of stakeholders, reinforcing the need for the solutions developed to be co-created to ensure their long-term sustainability.

The methodological cycle adopted in this study follows the four classic stages of action research:

1. Observation - identifying emerging patterns and assessing the impact of the proposed strategies.
2. Planning - theoretical survey and identification of the challenges faced by POs in solution architecture.
3. Action - interviews and analysis of practices adopted by organizations with autonomy models for POs.
4. Reflection and Adjustments - analysis of the data collected and continuous refinement of the framework.

Figure 1 illustrates this methodological cycle applied to the research:

Figure 1 – The Action Research Cycle and stages

Source: Researchgate, López Franco, 2020

This iterative approach ensures that the framework is developed empirically and validated in practice, promoting greater transparency in decision-making and alignment between business needs and architectural definitions.

2.1.2 Action Research Approach

With an iterative research approach, the study is moving into the practical phase, involving professionals from the sector to validate the challenges and needs identified. The research is being conducted in a progressive and adaptive manner, allowing for continuous adjustments as new data is collected.

The empirical research included interviews with market professionals, such as Product Owners and Tech Leaders, with the aim of understanding the challenges and needs related to the definition of daily processes and responsibilities in the architecture. Based on the data obtained, the research follows a structured path for the initial development of the theoretical framework. The aim is to ensure that the framework is aligned with the real demands of the market, promoting greater transparency in decision-making and reducing risks in systems development.

The main methodological aspects applied to the study include:

1. Identification of real problems faced by POs in their interaction with the solutions architecture.
2. Collection and analysis of empirical data through interviews and review of organizational processes.
3. Iterative refinement of the framework, ensuring alignment with market practices.

This process makes it possible to continually validate the proposed model, ensuring that it is theoretically grounded, empirically validated and applicable to the reality of the agile environment.

As highlighted by McNiff (2017) and Coghlan & Shani (2018), action research drives organizational change by integrating researchers and participants into the process, ensuring that the solutions developed are practically relevant and applicable to real-world scenarios. In this study, action research is used to test and refine the framework in collaboration with POs and agile technical leaders, making it more adaptable to industry needs.

One of the key challenges in this approach is the potential researcher influence on results. To mitigate bias, structured forms and standardized data collection techniques were employed, allowing interviewees to express their experiences autonomously. This ensures that the findings reflect the real needs of practitioners rather than researcher assumptions.

According to Brydon-Miller et al. (2019) and Greenwood (2022), action research emphasizes active stakeholder participation in co-creating solutions. This methodology enables the study to not only understand existing challenges faced by POs but also to collaboratively build and validate a framework that addresses the gaps between business strategy and technical architecture.

The following figure presents an illustration of the action research cycle, highlighting the continuous iteration across planning, action, observation, and reflection. By integrating literature review, empirical insights, and action research cycles, this methodology ensures that the developed framework is theoretically sound, empirically validated, and practically applicable to the agile context.

2.2. Project Context

This project is being developed based on the internal processes of “BTG Pactual”, seeking to optimize collaboration between the different functions within the agile environment. The clear definition of roles and responsibilities, in line with the agile methodologies adopted by the company, is fundamental to guaranteeing the efficiency of the process and the success of deliveries.

2.2.1 Specific Roles

The roles within an agile environment are fundamental to the success of a project, as they guarantee the effective application of the framework's principles and practices. Each role is responsible for specific tasks which, when performed well, contribute to optimizing the team's performance and delivering consistent value to the client. Clearly defining these roles is important to creating a collaborative, high-performance structure that is aligned with the project's strategic objectives. The roles described below are based on the processes of “BTG Pactual,” aligning with its operational needs and internal practices.

2.2.1.1 Role of the Scrum Master

The Scrum Master acts as a facilitator and coach for the team, ensuring the correct application of Scrum practices. According to Schwaber and Sutherland (2020), “the Scrum Master is responsible for promoting and supporting Scrum as defined in the Scrum Guide, helping everyone to understand the theory, practices, rules and values of the framework” (p. 6). Their role is essential for the formation of a cohesive and productive team, ensuring the adoption of agile principles.

In addition, Scrum Masters face significant challenges in organizational environments that are not yet fully adapted to the agile mindset. Hoda and Noble (2017) point out that “Scrum Masters often face difficulties in balancing their facilitator role with the leadership expectations imposed by more traditional organizations” (p. 45). This balance is important for the Scrum Master to perform their role effectively, without compromising the autonomy and collaboration of the team.

Studies indicate that the Scrum Master's experience can directly influence team performance. Moe et al. (2019) point out that “the presence of an experienced Scrum Master can significantly increase team cohesion, resulting in more consistent deliveries aligned with customer objectives" (p. 12). This positive impact reinforces the importance of the Scrum Master in facilitating communication and removing impediments that could compromise the team's productivity.

At BTG Pactual, there is no specific person dedicated to the role of Scrum Master. Instead, this role is mainly performed by the Tech Lead, who takes responsibility for facilitating the application of Scrum practices, guiding the team and ensuring that the agile principles are followed. The Tech Lead, as well as leading the technical part and getting directly involved with the stakeholders, also plays the role of facilitator, removing impediments and promoting a collaborative environment, which ensures that the team runs smoothly and that deliveries are aligned with the project's objectives.

2.2.1.2 Role of the Product Owner

The Product Owner, in turn, is responsible for maximizing the value of the product developed by the team. According to Cohn (2019), "the Product Owner must ensure that the backlog is always up to date, prioritized and aligned with the organization's strategic objectives, serving as the main bridge between stakeholders and the development team" (p. 23). This position requires advanced communication, negotiation and prioritization skills to balance different interests and ensure high-value deliveries.

However, playing this role is not without its challenges. Pichler (2018) states that "one of the biggest challenges facing Product Owners is balancing the conflicting demands of stakeholders while maintaining a focus on the value of the product for the end customer" (p. 34). As such, the ability to make strategic decisions without compromising the product vision is essential to the success of the project.

Effective communication between the Product Owner and the development team is a determining factor in the successful execution of Scrum. Gregory et al. (2021) note that "clear communication between the Product Owner and the development team is essential to avoid misunderstandings and ensure that deliveries meet stakeholder expectations” (p. 18). This continuous alignment strengthens collaboration and contributes to an efficient workflow.

At BTG Pactual, the Product Owner function is structured independently, with a dedicated team that is not directly integrated with the developer team. Each Product Owner is assigned to manage one or more teams, and is responsible for coordinating and ensuring that demands are carried out effectively. The role of this function is fundamental as a bridge between the business team and the developers, facilitating communication and the flow of information. The Product Owner ensures that priorities are clearly defined, activities are constantly updated and business objectives are aligned with the product's progress and deliveries. This allows the company to meet market needs in an agile and strategic way, without losing focus on the value delivered to the end customer.

2.2.1.3 Role of the Tech Lead

According to Unisinos (2024), the Tech Lead is the professional responsible for leading the technical part of a project, defining the technologies and architectures used. They also act as a link between the development team and other departments in the company, promoting alignment and ensuring that compatible methodologies are adopted. Their role goes beyond carrying out technical tasks, taking a guiding role in identifying and solving problems.

Alura (2024) stresses that the Tech Lead must have a balance between technical and interpersonal skills. Their main duties include defining the technical roadmap, supervising the team to ensure consistent deliveries and creating a collaborative environment. In addition, they must align technical goals with the company's strategic objectives, acting as a bridge between technical teams and stakeholders.

To perform this role effectively, according to Unisinos (2024), it is essential that the Tech Lead has advanced knowledge of programming languages such as Python and Java, experience with agile methodologies (Scrum or Kanban) and interpersonal skills such as clear communication and effective leadership. These factors are important to the success of projects and strategic alignment within organizations.

At BTG Pactual, the Tech Lead's role is even more comprehensive and strategic. They are responsible for multiple areas and play a collaborative role, going beyond technical leadership to actively engage with stakeholders. In partnership with the Product Owner (PO), the Tech Lead is responsible for building a roadmap that is coherent and viable for the development team, aligning the needs of the business with the capabilities of the team. In addition, the Tech Lead manages various project fronts, ensuring that initiatives are conducted in an integrated and efficient manner, promoting continuous product progress with a clear and aligned vision between the technical and business teams.

2.2.1.4 Role of the Stakeholder

Stakeholders are individuals, groups or organizations with an interest in the project. They can be internal (such as staff and managers) or external (such as clients and suppliers). According to Strong (2022), understanding and managing their expectations is important to aligning objectives, minimizing risks and maximizing benefits.

They can be classified as internal, including employees and managers, or external, encompassing customers, investors and communities. Primary stakeholders are directly impacted, such as end users, while secondary stakeholders have an indirect interest, such as regulators. Euax (2022) points out that identifying key stakeholders is essential for prioritizing efforts, as they have the greatest influence on the project.

Effective stakeholder management involves identifying, analyzing and engaging them. According to Docusign, tools such as stakeholder mapping make it easier to define roles, responsibilities and levels of influence, helping to avoid conflicts and strengthen relationships.

At BTG Pactual, the stakeholders are mainly responsible for understanding product evolution needs and ensuring that changes are aligned with the strategic objectives of the business. This role is played by business representatives, who monitor the market and customer demands. They are responsible for passing on the necessary developments to the Tech Lead and Product Owner (PO), who in turn translate this information into technical development and the definition of product priorities. In this way, the stakeholders act as key agents in defining the product's direction and aligning the market's needs with the team's capabilities.

2.2.1.5 Role of the Solutions Architect

The solution architect plays an essential role in agile projects, being responsible for translating business needs into effective technical solutions. They act as a link between stakeholders and development teams, ensuring that the system's architecture meets both functional and non-functional requirements, such as performance, security and scalability. According to Gertel (2021), this professional must have a strategic vision to define architectural guidelines, as well as promoting an incremental and evolutionary design that allows adaptations according to the project's needs.

In addition, the solutions architect plays an important role in stakeholder engagement, ensuring that all interested parties share a clear vision of the project's objectives. According to Menon, Sinha and MacDonell (2021), this continuous interaction between the architect and stakeholders prevents misalignments and allows for more assertive technical decision-making. Another relevant aspect is their technical leadership, since they act as mentors to the development team, sharing knowledge, promoting good practices and helping to solve complex problems. As Altmann (2021) points out, this mentoring contributes to the team's professional growth and improves the quality of deliveries.

Collaborative decision-making is also part of their responsibilities, as the solution architect must encourage the active participation of the development team, ensuring that technical choices are feasible and aligned with the team's capacity. This also involves the selection of suitable technologies, often involving the creation of Proofs of Concept (PoCs) to validate the viability of a solution before its full implementation (Arquitetocloud, 2021). In addition, quality assurance and risk management are fundamental functions of this professional, who needs to anticipate possible flaws in the architecture and implement mitigation strategies to ensure the stability and security of the solution.

At BTG Pactual, there is no specific position dedicated to the solutions architect. Instead, this role is shared between the Tech Lead and the business team. The Tech Lead, with the strategic support of the business team, plays a crucial role in defining the system architecture, and is responsible for ensuring that the technical solutions meet the demands of the project, while respecting the technical limitations and capabilities of the team.

2.2.2 POs' Main Responsibilities

The theoretical foundation of this study is based on a literature review on the responsibilities and challenges of the Product Owner (PO) in agile environments, with an emphasis on Scrum. While there are various perspectives on the role of the PO, this study specifically focuses on the challenges highlighted by Schwaber (2004), Kristinsdóttir (2014), and Sverrisdottir, Ingason, and Jonasson (2014). These references support the analysis of how the role of the PO has evolved in different organizational contexts and how it can face the challenges inherent in its function. It is clear that throughout the existence of this study, new references may emerge and contribute to further discussions and confront different approaches over time.

There are multiple perspectives on the role of the Product Owner in agile methodologies, and different frameworks and organizations interpret this role in various ways. However, this study focuses on the challenges faced by POs in real-world applications, as reported in the selected references. By narrowing the scope to these specific challenges, this research aims to provide a deeper understanding of the practical difficulties POs encounter, rather than attempting to cover all possible variations of the role.

The role of the Product Owner (PO) has emerged as one of the most important within agile methodologies, being responsible for ensuring that product development is aligned with business objectives and user needs. According to Schwaber (2004), the PO represents the interests of the stakeholders and defines the prioritization of the product backlog, guaranteeing the delivery of continuous value. This role is multifaceted and involves both interaction with stakeholders and direct collaboration with development teams.

Sverrisdottir, Ingason, and Jonasson (2014, p. 258) point out that, in practice, the interpretation of the PO's role varies considerably between organizations. Some companies adopt a model in which there are two Product Owners for a single product, with one responsible for the business aspects and the other for the technical aspects of development. This approach can facilitate communication between stakeholders and technical teams but also deviates from the original Scrum model, which defines the PO as a single person responsible for all these areas. The Product Owner is, among many things, responsible for establishing and communicating a clear vision of the product to the entire team. As described by Kristinsdóttir (2014), this responsibility includes both communicating a strategic vision and constantly adapting to changes in the market and the organization.

2.2.2.1. Product Backlog Management

The PO is responsible for prioritizing the product backlog, ensuring that the most valuable features are delivered first. Schwaber (2004) points out that this prioritization is based on both customer needs and return on investment (ROI). At Spotify, the POs reported difficulties in directly measuring the value of the work done, since many deliveries were to internal customers. Some Product Owners used qualitative feedback and indirect metrics, such as stakeholder satisfaction and user base growth, to measure the effectiveness of the features delivered (Kristinsdóttir, 2014, p. 9).

Sverrisdottir, Ingason and Jonasson (2014, p. 263) observed that the frequency of backlog review and maintenance varies significantly between the POs interviewed. In some cases, the backlog is reviewed daily, while in other organizations this activity occurs only once a month:

> "Review of the PB is very different between the participants. Most people seem to look at the backlog weekly, daily or even many times a day, as there may be significant changes on the backlog. For one participant this is done only once per month, when a product is put into operation" (Sverrisdottir, Ingason and Jonasson, 2014, p. 263).

2.2.2.2 Engagement with Stakeholders

The Product Owner acts as a bridge between the stakeholders and the development team. According to Kristinsdóttir (2014, p. 7), "the Product Owners did agree that the teams should know who their customer is, but they did not all find it to be their responsibility to identify those customers and their needs or to communicate those needs to their team." This shows that there is a variation in the perception of the PO's responsibilities within different organizations.

Sverrisdottir, Ingason and Jonasson (2014, p. 264) reinforce that managing expectations is one of the main duties of the PO, and that their communication skills are fundamental to maintaining alignment between the technical teams and the demands of the business:

> "Most participants indicated that expectations management was one of the most important duties of the PO. Examples of vital characteristics of the PO to be able to fulfill their role are planning and communication skills. This was considered necessary to create a common understanding between the business manager and the IT department." Sverrisdottir, Ingason and Jonasson (2014, p. 263)

2.2.2.3. Facilitating Collaboration with the Development Team

Agility depends on effective communication between the PO and the development team. Kristinsdóttir (2014, p. 10) points out that one of the challenges faced by Product Owners at Spotify was the limited time for interaction with their teams. Some POs were able to devote only 5% of their time to direct collaboration, while others spent up to 70% of their time working side by side with developers. This variation had a direct impact on the clarity of demands and the speed of deliveries.

Sverrisdottir, Ingason and Jonasson (2014) complement this analysis by pointing out that the time dedicated by POs to interactions with the team varies considerably. While some POs spend most of their time aligning priorities with stakeholders, others work more closely with developers, ensuring that the technical specifications are clear and aligned with the product vision (Sverrisdottir, Ingason and Jonasson, 2014, p. 262).

2.2.2.4 Decision Making and Prioritization

The Product Owner (PO) must make quick, data-driven decisions to ensure the effectiveness of agile development. However, as noted by Kristinsdóttir (2014, p. 9), there are significant challenges in this practice, as many POs reported difficulties in accessing objective metrics and quantitative data, leading them to rely on intuition and qualitative feedback. "Most of the Product Owners found it difficult to measure the value of the work their team was doing because often there was no transaction of money taking place" (KRISTINSDÓTTIR, 2014, p. 9).

Spotify tracks usage and user satisfaction as impact metrics for changes to the software. However, as many of these changes are incremental, users don't always notice the changes made, which makes it difficult for POs to assess the real impact of the teams' work (Kristinsdóttir, 2014, p. 9-10).

Sverrisdottir, Ingason and Jonasson (2014, p. 263) point out that the autonomy of the PO in making decisions can be a determining factor in the success of the project:

> "All participants said that it is either fairly important or very important that the PO has unrestricted authority to make decisions in the project. The role of product owner is to make decisions and this is often essential to be able to finish the project." Sverrisdottir, Ingason and Jonasson (2014, p. 263).

2.2.2.5. Managing expectations and communication

An important aspect of the PO's role is managing expectations and aligning visions between the different stakeholders. As highlighted by Kristinsdóttir (2014, p. 13), "Product Owners need to have one foot in the daily work and one foot in the future, and to lead people to work on the right things at any given moment." This reinforces that the role requires a constant balance between operational execution and the strategic vision of the product. Sverrisdottir, Ingason and Jonasson (2014, p. 265) add that the PO needs to be a communicative and influential leader, as their ability to negotiate priorities and align expectations between stakeholders and technical teams is a critical factor in the success of the project.

2.2.3 Main Challenges

The Product Owner is responsible for establishing and communicating a clear vision of the product to the entire team, ensuring that the backlog is properly prioritized and that deliveries are aligned with the organization's strategic objectives. However, the literature shows that the practical application of this role varies significantly between organizations, which creates challenges for its effective execution.

In the BTG context, there is evidence of challenges related to communication between POs and developers, as well as the participation of POs in the solution architecture, which does not seem to be a consolidated organizational practice. This issue still needs to be investigated further to understand its real extent and impact. The literature shows that similar challenges occur in different organizations. Kristinsdóttir (2014, p. 10) observes that the limitation of the time dedicated by POs to interacting with development teams can directly impact the clarity of demands and the speed of deliveries: “Every Product Owner found it extremely important to spend time with the team every day, as much as they physically could, but some struggled with that as they worked with more than one team and had to attend various meetings."

Another recurring challenge pointed out in the literature is the lack of a standardized understanding of the role of the PO within the organization. Sverrisdottir, Ingason and Jonasson (2014, p. 257) point out that this variability in the interpretation of the PO's role is common in different companies, leading to difficulties in the efficient implementation of Scrum and in backlog decision making.

The difficulty in measuring the value generated by deliveries is also an obstacle present in many organizations. Kristinsdóttir (2014, p. 8) points out that, in the absence of direct monetary transactions, many POs find it difficult to objectively measure the impact of deliveries. This problem can manifest itself when there are no uniformly established success metrics between the different teams, making it difficult to effectively prioritize the backlog.

The management and review of the backlog also represent relevant challenges. Sverrisdottir, Ingason and Jonasson (2014, p. 263) observed that "review of the PB is very different between the participants. Most people seem to look at the backlog weekly, daily or even many times a day, as there may be significant changes on the backlog. For one participant this is done only once per month, when a product is put into operation."

Another critical factor addressed in the literature is the autonomy of the PO in decision-making. Sverrisdottir, Ingason and Jonasson (2014, p. 264) point out that "all participants said that it is either fairly important or very important that the PO has unrestricted authority to make decisions in the project. The role of product owner is to make decisions and this is often essential to be able to finish the project." The lack of this autonomy can lead to delays and compromise the ability to adapt agilely to the market.

Finally, managing expectations and efficient communication between stakeholders and the development team are key challenges. Sverrisdottir, Ingason and Jonasson (2014, p. 264) point out that "most participants indicated that expectations management was one of the most important duties of the PO. Examples of vital characteristics of the PO to be able to fulfill their role are planning and communication skills." Kristinsdóttir (2014, p. 13) reinforces the importance of continuous communication by stating that "Product Owners need to have one foot in the daily work and one foot in the future, and to lead people to work on the right things at any given moment".

Thus, it can be seen that the challenges faced by POs in different organizational contexts involve aspects such as communication, backlog prioritization, autonomy and participation in the solution architecture. In the case of BTG, investigating these challenges will allow us to better understand the specificities of this environment and how these factors can impact the efficiency of the PO's role in product development.

2.3. Literature survey

The academic study of the roles and responsibilities of those involved in the architecture of agile solutions and methodologies relies on various academic and professional contributions. This section presents the main authors and their approaches within the academic context, providing a relevant theoretical overview for the comparative analysis of different roles and their respective responsibilities. This theoretical survey is essential to support the proposal of a framework for Product Owners, allowing the integration of agile practices and solution architecture from an academic perspective.

To ensure a comprehensive and systematic approach to the literature review, we adopted a structured strategy to identify and analyze relevant academic and professional sources. The aim was to map the current landscape of roles and responsibilities in agile methodologies, with a particular focus on Product Owners (POs), Tech Leads and Solution Architects. The search was guided by clear inclusion criteria: renowned academic articles and reports published in the last 10 years, although we did find articles related to these authors and in these readings that helped us to better understand our research. Platforms such as Google Scholar and Scopus were prioritized for their reliability and breadth of coverage. Keywords such as "responsibilities", "roles", "agile methodologies", "Product Owner", "Tech Lead" and

"Solution Architecture" were used to create search strings, ensuring a focused but comprehensive retrieval of relevant material.

The analysis was organized into thematic categories, including specific roles (e.g. Scrum Master, PO), core responsibilities, common challenges and emerging best practices. This categorization allowed for a clear comparison of different approaches and facilitated the identification of gaps and trends in the literature. This approach - combining academic rigor with real-world knowledge - ensured a balanced perspective on the topic.

Finally, the preliminary conclusions were consolidated to inform the development of the frameworks scenario. By comparing the practices identified with the specific challenges faced at the institution, we were able to visualize the real needs and were able to see in some ways that they are expressed in the frameworks, thus aligning the reality of the institution with established academic norms.This structured methodology not only increases the credibility of our research, but also allows us to explore a clear roadmap for integrating theoretical knowledge into practical solutions.

2.3.1.Roles in Solution Architecture

Solution architecture involves multiple players, from Product Owners to software architects and technology managers. In the academic context, the following studies stand out:

Parducci and Oliveira (2013), in the book TOGAF: IT Solutions Architecture for Enterprises, address how structuring enterprise architecture can improve organizational efficiency. The TOGAF framework defines clear roles for solution architects and enterprise architects but lacks specific guidelines for Product Owners, a point widely explored in academic research.

The Open Group - Responsible for developing TOGAF, one of the most widely adopted frameworks for enterprise architecture and solutions, defines roles such as

Architecture Board and Solution Architects, but without a specific academic approach to the autonomy of POs in defining architecture.

2.3.2.Scrum and the Distribution of Responsibilities

Scrum is one of the most studied agile methodologies in academia, with several scientific articles exploring the clear definition of the roles involved:

Schwaber (2020), one of the creators of Scrum, points out that the Product Owner is responsible for maximizing the value of the product, but his influence on the architecture of the solution depends on collaboration with the technical team. Academic research indicates that this influence can be more active than originally proposed by the framework.

Kawamoto (2017), in her thesis *Scrum-DR: An Extension of the Scrum Framework Adherent to the Capability Maturity Model Using Design Rationale Techniques*, adapts Scrum to integrate predictability and stability into CMMI-DEV, indicating that the PO can be more involved in defining architectural requirements, a frequent theme in academic studies on hybrid methodologies.

Garcia (2015) adapted Scrum for different projects, pointing out that responsibility for the architecture tends to be concentrated on the developers, with little direct involvement from the Product Owners, a challenge identified by academic research into agile governance.

2.3.3 Kanban and Architecture Management

The Kanban method has been widely studied in the academic context for its applicability to agile project management:

Anderson (2020) defines Kanban as an evolutionary system that allows incremental changes to the architecture without a rigid definition of roles, which can make it difficult for POs to actively participate in the solution architecture, an aspect analyzed by academic studies on decentralized governance.

Ōno (1997), creator of Kanban at Toyota, focuses on continuous improvement, but does not specify how the roles relate to the solution architecture, a gap often discussed in academic research on the application of Kanban in software.

Bozheva and Jones (2021), co-authors of the *Kanban Maturity Model*, highlight the need for cultural evolution in organizations so that different profiles, including Product Owners, can contribute to architectural decisions, a recurring theme in academic literature on change management in IT.

2.3.4 Lean Software Development and Shared Responsibility

The Lean approach has been widely explored in academic research for its contribution to optimizing software development:

Poppendieck and Poppendieck (2019) introduce Lean principles applied to software development, arguing that architectural decisions should be made in a decentralized manner, involving Product Owners, developers, and architects. This approach has been validated by academic studies on large-scale agile development.

Beck (2004), creator of Extreme Programming (XP), argues that architecture should emerge from interactions between teams, suggesting that Product Owners can participate in decisions as long as they have sufficient technical knowledge. Academic research reinforces this view by analyzing collaboration models between POs and architects.

2.3.5 Agile Frameworks and Roles in Architecture

The evolution of agile methodologies has resulted in hybrid frameworks that combine different approaches to defining roles and responsibilities and are widely discussed in academic circles:

Ambler and Lines (2020), creators of the Disciplined Agile Framework (DAF), propose a model that incorporates elements of Scrum, Kanban, and Lean, allowing greater flexibility in defining roles. However, the DAF still keeps the architecture of solutions under the responsibility of software architects, a point often questioned in academic studies on PO autonomy.

Larman and Vodde (2016), creators of Large Scale Scrum (LeSS), emphasize the need to align architecture with business strategy but suggest that Product Owners should focus on prioritizing value, leaving architectural decisions to technical experts—an approach that has been critically analyzed in several academic publications.

A review of the academic literature shows that different agile and solution architecture approaches have significant variations in the definition of the roles and responsibilities of those involved. While frameworks such as TOGAF and DAF keep the architecture under the responsibility of technical specialists, approaches such as Lean and XP propose greater decentralization, allowing the active participation of Product Owners in the architecture of solutions, a recurring theme in academic research on agile governance.

Based on this academic analysis, the development of a framework for Product Owners should consider a balance between autonomy and technical support, allowing these professionals to make architectural decisions without compromising the technical coherence of the solutions. The next section of the paper will explore how to structure this framework to guarantee scalability, continuity and governance in the management of critical systems, always from an academic perspective based on scientific research.

2.4. Management approaches and solution architecture

In this chapter, we explore the main management frameworks used to organize and conduct solution development, with particular emphasis on agile approaches and associated organizational role definitions. Next, the different types of solution architectures commonly adopted are presented, highlighting their characteristics, advantages and limitations. Finally, comparisons are made between architectural and methodological models, with the aim of providing a critical view of how the choice of these approaches directly impacts efficiency, governance and the ability of organizations to adapt to business changes and challenges.

2.4.1 Types of Management Frameworks

In this section, the aim is to present the management frameworks used by BTG Pactual in the daily processes of refining, organizing and structuring the activities carried out by the developers. The different approaches adopted will be explored and how each framework contributes to improving productivity, delivering value and adapting teams to project demands. Among the frameworks used are: Scrum, Kanban, Lean Software Development, Disciplined Agile, Scaled Agile Framework (SAFe), Large Scale Scrum (LeSS), Dynamic Systems Development Method (DSDM) and Lean Inception.

Scrum is one of the most widely used frameworks, organizing work in sprints with defined roles, such as Product Owner, Scrum Master and Development Team. It encourages collaboration and the continuous delivery of value. As Scrum.org points out, “Scrum is a simple framework for managing and doing work in teams” (Scrum.org, 2021). However, Scrum can be inflexible in environments with highly dynamic requirements and depends on the active presence of defined roles. To scale Scrum, frameworks such as Nexus are employed, extending Scrum to allow multiple teams to work on a single Product Backlog (Scrum.org, 2021).

Kanban, on the other hand, focuses on visualizing the workflow, limiting work in progress and maximizing the continuous delivery of value. It doesn't use sprints or fixed roles, offering flexibility for constant change. A 2020 study on Kanban highlights its effectiveness in improving workflow and reducing cycle time (Anderson, 2020). Kanban is ideal for teams that need flexibility and a continuous view of the workflow, but can be challenging for teams that need a more rigid structure.

Lean Software Development follows the principles of lean production, focusing on creating value and eliminating waste. According to a 2019 article, “Lean is about creating value for the customer while minimizing waste” (Poppendieck, 2019). Its implementation can be challenging in complex environments, but it is effective for improving processes and reducing costs.

Disciplined Agile (DA) integrates various agile practices, such as Scrum, Kanban and Lean, to cover the entire software development lifecycle. According to Scott W. Ambler and Mark Lines in their book Choose Your WoW!: A Disciplined Agile Delivery Handbook for Optimizing Your Way of Working (2020), “DA allows teams to choose the right practices for their context” (Ambler & Lines, 2020). DA is ideal for organizations looking for a flexible and holistic approach, although its complexity can be challenging due to the integration of multiple practices.

Scaled Agile Framework (SAFe) was developed to help large organizations implement agile practices on a large scale, integrating Lean, Agile and DevOps concepts. Dean Leffingwell, in the book SAFe 5.0: The World's Leading Framework for Enterprise Agility (2020), states that “SAFe provides a framework for organizations to implement Agile at scale” (Leffingwell, 2020). Although effective, its implementation can be complex and require a significant investment in training and certification.

Large Scale Scrum (LeSS) is a minimalist approach to scaling Scrum while maintaining its basic principles. Craig Larman and Bas Vodde, in the book Large-Scale Scrum: More with LeSS (2016), point out that “LeSS is simple and flexible, allowing organizations to maintain the simplicity of Scrum at scale” (Larman & Vodde, 2016). It is ideal for organizations that want to scale without adding excessive complexity, but may not be suitable for very complex environments.

Dynamic Systems Development Method (DSDM) is an agile framework that prioritizes the continuous delivery of value, with an emphasis on collaboration and risk management. Although there are few specific recent references to DSDM, it is recognized as an approach that focuses on customer satisfaction and proactive risk management, and is mentioned in several agile frameworks.

Lean Inception combines Lean principles with the inception phase of projects, focusing on clearly defining requirements and eliminating waste right from the start. It is mentioned in agile project initiation contexts as an approach that helps to clearly define requirements from the start of the project.

In summary, each framework discussed has its own particularities and contributes differently to BTG Pactual development process. While Scrum offers structure and a focus on collaboration, Kanban provides flexibility and continuous visualization of the workflow. Lean Software Development prioritizes the elimination of waste, and Disciplined Agile allows for a personalized and holistic approach. Frameworks such as SAFe and LeSS help scale agile practices in large organizations, while DSDM and Lean Inception offer approaches focused on the continuous delivery of value and clear definition of requirements. Combining these frameworks allows the company to adapt to the needs of the project, maximizing productivity and value delivery.

2.4.1.1 Agile Frameworks and Organizational Roles

The agile frameworks that were outlined define different approaches to the organization of work, including specific roles that ensure the continuous and efficient delivery of value. In this section, we explore how the roles of Product Owner (PO), Solution Architect, and Tech Lead are distributed in the main frameworks used by BTG Pactual.

Scrum

- **Product Owner (PO):** Responsible for the Product Backlog, defining priorities and ensuring that the development team works on the items of greatest value to the business. Acts as the main point of contact between stakeholders and the development team.

* **Solution Architect:** Scrum does not formally define this role, but in companies that use it, the solution architect can act as a technical expert external to the Scrum team, ensuring that architectural decisions are aligned with product requirements.
* **Tech Lead:** In Scrum, there is no official Tech Lead role, but this professional usually assumes a technical leadership position within the Development Team, ensuring good code practices, quality and technical alignment with the product vision.

Kanban

* **Product Owner (PO):** There is no formal PO role in Kanban, but the work of prioritizing the backlog still takes place, often being distributed among the team or maintained by a product manager.
* **Solution Architect:** As Kanban is highly flexible, the architect's role varies according to the needs of the project. They can act as a technical consultant to ensure that ongoing changes do not negatively impact the system's architecture.
* **Tech Lead:** Works to continuously improve the workflow, ensuring that deliveries are efficient and that there are no technical bottlenecks in the development process.

Lean Software Development

* **Product Owner (PO):** Emphasizes the continuous delivery of value and the elimination of waste. The PO here needs to be aligned with Lean principles, ensuring that only essential requirements are prioritized.
* **Solution Architect:** Works to eliminate waste in the architecture, promoting scalable and sustainable solutions, with a focus on simplicity and efficiency.
* **Tech Lead:** Leads the team in the use of Lean practices, ensuring that technical solutions are developed efficiently and without wasted effort.

Disciplined Agile (DA)

- **Product Owner (PO):** In DA, this role can be more flexible, being called Product Manager at strategic levels. In addition to prioritizing the backlog, they can also manage the product portfolio.
- **Solution Architect:** Has an essential role, as DA emphasizes the need for a well-defined architecture to support different agile approaches. Can act both centrally and decentrally, depending on the context.
- **Tech Lead:** Responsible for ensuring the application of appropriate technical practices within the approach chosen by the team (Scrum, Kanban, Lean, etc.), facilitating collaboration between developers.

Scaled Agile Framework (SAFe)

- **Product Owner (PO):** In SAFe, there are both Product Owners and Product Managers. The PO works within the agile team, refining the team's backlog and ensuring alignment with the Product Manager, who has a broader view of the product strategy.
- **Solution Architect:** In SAFe, the role of Solution Architect/Engineer is formalized, ensuring that the architecture is well planned to support large-scale business requirements. They work closely with System Architects and Enterprise Architects.
- **Tech Lead:** Generally referred to as System Team Lead, they work on the technical side within agile teams, ensuring the correct implementation of architectural solutions and facilitating integration between teams.

Large Scale Scrum (LeSS)

- **Product Owner (PO):** In LeSS, there is only one PO for all Scrum teams, ensuring alignment and centralized prioritization for the entire organization.
- **Solution Architect:** This role is not formally defined in LeSS, as the framework seeks to keep Scrum as pure as possible. However, architects can act as facilitators to help teams make decentralized technical decisions.

* **Tech Lead:** As in Scrum, there is no formal Tech Lead role, but usually a more experienced developer takes on this role to ensure good practices and code quality.

Dynamic Systems Development Method (DSDM)

* **Product Owner (PO):** In DSDM, this role is called Business Visionary or Business Ambassador, ensuring that the business requirements are understood and implemented correctly.
* **Solution Architect:** Acts as Technical Coordinator, ensuring that technical and architectural decisions are aligned with the project's objectives.
* **Tech Lead:** Works to ensure the application of good technical practices and to optimize development efficiency within the agile team.

Lean Inception

* **Product Owner (PO):** Plays a fundamental role in the initial phase of the project, ensuring that the requirements and objectives are well defined before development begins.
* **Solution Architect:** Works with the team to ensure that the initial product definition takes important technical aspects into account.
* **Tech Lead:** Contributes to strategic technical decisions to avoid rework in the future, helping to establish a solid foundation for development.

2.4.2. Types of Solution Architectures

Solution Architecture is the discipline responsible for designing and implementing technological strategies to meet an organization's business needs. According to The Open Group (2018), “solution architecture defines the structure and behaviour of applications, data and technological infrastructure, ensuring that they are aligned with organizational goals”.

The architectures considered were designed based on the context in which “BTG Pactual” operates and the way it carries out its processes. This approach ensures that the proposed solutions are aligned with the company's operational reality, addressing specific challenges and maximizing the effectiveness of technological implementations.

Solution Architecture must have certain characteristics, such as a focus on the business, which, according to Ross, Weill and Robertson (2006), means ensuring that the technologies adopted contribute directly to the company's strategic objectives, avoiding isolated solutions that are not aligned with the business. Strategic planning is also important because, according to Bass, Clements and Kazman (2012), a good architecture allows organizations to identify weaknesses and opportunities in the current infrastructure, ensuring that new solutions are scalable and sustainable. In addition, Solution Architecture must guarantee technological integration, as Lankhorst (2017) points out, ensuring that different systems can communicate efficiently, promoting interoperability and reducing organizational silos.

The following section will present the solution architectures that support the proposed technological strategies.

SOA - Service-Oriented Architecture

Service Oriented Architecture (SOA) is a software design paradigm that aims to facilitate the construction of scalable and flexible systems, allowing business functionalities to be made available as reusable and interoperable services. SOA is based on three fundamental principles:

- **Modularity and Component Reuse:** Systems are divided into independent services, which can be developed, deployed and maintained efficiently, avoiding duplication of effort and facilitating system evolution.
- **Interoperability and Communication between Services:** Services communicate through standardized interfaces, such as APIs or web services, allowing different applications to consume them regardless of the technology used.
- **Abstraction and Encapsulation of Functionalities:** Each service encapsulates a specific functionality, hiding implementation details and exposing only the

interface necessary for its use, making it easier to maintain and evolve the system.

Architecture Based on Microservices

Microservice-based architecture is a style of software design that divides a system into a collection of small, independent services, each responsible for a specific functionality. These services communicate via well-defined interfaces, usually APIs, and are designed to be scalable, flexible and easy to maintain.

- **Independence and Loose Coupling:** Each microservice is developed, deployed and maintained independently, with communication via APIs, which allows for loose coupling between services.
- **Scalability and Flexibility:** Microservices can be scaled individually, which facilitates resource management and improves the efficiency of the system as a whole.
- **Fault tolerance:** If one service fails, it doesn't affect the others, allowing the system to continue working even in the event of partial failures.
- **Rapid Development and Deployment:** Teams can work in parallel on different services, speeding up the development and deployment cycle.

Monolithic Architecture

Monolithic architecture is a traditional style of software design where all the functionalities of an application are integrated into a single unit. This means that the user interface, business logic and data access are combined into a single executable program. Monolithic architecture is often used in smaller projects or startups, due to its simplicity and ease of development.

- **Single Deployment Unit:** The entire system is deployed as a single unit, which makes initial deployment easier, but can be complicated for upgrades, as it requires re-deploying the entire application.

- **Tight Coupling:** The different components of the system are tightly coupled, which can make it difficult to modify and maintain the system over time.
- **Vertical Scalability:** Scalability is achieved by increasing the capacity of the servers where the application is hosted, which can be limited and expensive for large-scale systems.

Event-Driven Architecture

Event-driven architecture (EDA) is a software design model based on publishing, processing and reacting to events in real time. This approach allows systems to operate independently, reacting to events as they occur, without the need for direct and immediate interactions between services.

- **Events as the Central Point:** Events are the main focus of EDA. They represent significant facts that occur in the system, such as a purchase made or a change of state.
- **Asynchronous Communication:** Services communicate through events, which allows for loose coupling between them. This means that event producers don't know who the consumers are, and vice versa.
- **Reaction to Events:** Each service decides how to react to an event, or whether to react at all. This provides flexibility and autonomy for the services, allowing them to operate independently.

Architecture in N Layers

The N-layer architecture is an extension of the layered architecture, where the system is organized into a variable number of layers, each with a specific responsibility. This approach allows for a more flexible and scalable structure than the traditional three-tier architecture, which only includes presentation, application and data layers.

- **Hierarchical Organization:** The layers are organized hierarchically, with each layer interacting only with adjacent layers, which facilitates maintenance and system modularity.

- **Separation of Responsibilities:** Each layer has a clearly defined function, such as presentation, business logic, data persistence, etc., which promotes separation of concerns and code reuse.
- **Flexibility and Scalability:** The N-layer architecture allows new layers to be added or removed as necessary, without affecting the rest of the system, which makes it easier to adapt to changes in business requirements.

2.4.2.1 Architectural Comparison Summary

Below is a summary comparison of different software architectures, highlighting their advantages, disadvantages and common use cases. The table 1 condenses the key points, making it easy to quickly compare each approach.

Table 1 : Key points of the different solution architectures

Architecture	Advantages	Disadvantages	Use Cases
Service-Oriented Architecture (SOA)	- Modularity and service reuse - Easy integration with legacy systems - Scalability and flexibility	- More complex to implement - Requires governance to maintain standards and compliance	- Large corporations with multiple heterogeneous systems - Systems requiring high interoperability
Microservices Architecture	- Independent development and deployment - Better scalability and fault tolerance - Easy adoption of different technologies	- More complex management and monitoring - Requires a robust communication system (e.g., API Gateway, Service Mesh)	- Systems requiring high scalability - Companies aiming for faster development and delivery cycles

Monolithic Architecture	- Simpler to develop and deploy - Lower internal latency since all components are in the same environment	- Difficult to scale horizontally - Complex to maintain in large projects	- Applications with few modules - Small teams and short deadlines
Event-Driven Architecture (EDA)	- High scalability and flexibility - Better decoupling between components	- More complex event handling - Difficult to trace and debug	- Systems requiring high availability and resilience - Real-time data processing
N-Tier Architecture	- Better code organization - Easier to maintain and evolve	- Can introduce unnecessary latencies - Harder to scale compared to microservices	- Traditional corporate applications - Systems requiring a clear separation of responsibilities

2.4.2.2 Methodology Comparison

As well as analyzing the advantages and disadvantages of each architecture, it is also essential to compare them on the basis of important technical criteria such as scalability, complexity and coupling. These factors directly influence the performance, maintainability and suitability of each approach for different business scenarios. The following table 2 provides a structured comparison, helping to identify the best methodology for specific system requirements.

Table 2 : Differences in methodologies

Methodology	Scalability	Complexity	Coupling	Best Use Case

SOA	High	High	Low	Large corporations, system integration
Microservices	High	High	Very Low	Distributed systems, rapid scalability
Monolithic	Low	Low	High	Small applications
Event-Driven	Very High	High	Very Low	Asynchronous systems, real-time
N-Tier	Medium	Medium	Medium	Enterprise systems

Choosing the right methodology for architecting solutions depends on factors such as scalability, system complexity, integration needs and business model. Modern architectures, such as microservices and event-driven, are more suitable for distributed and highly scalable systems, while simpler solutions can benefit from a monolithic or N-layer approach. The correct definition has a direct impact on the efficiency and success of the solution developed.

2.5. Interview with Expert

The aim of this section is to present the research carried out. The main focus is to understand the daily work of the project's actors, how demands are organized and refined, as well as to analyze their responsibilities and the degree of autonomy they have.

A preliminary survey was carried out during sprint 2 (17/02 to 28/02) with the developers and POs. During sprint 3 (03/03 to 14/03), the questions applied to these actors were reviewed in order to identify possible gaps or points for improvement. In addition, the final version of the questionnaire for Tech Leads and Solution Architects was drawn up. In sprint 4 (17/03 to 28/03), the aim is to hold an official interview with the three groups of actors and apply for the final questionnaire, covering POs, developers, and Tech Leads / Solution Architects.

2.5.1 Interview with developers

To ensure that the development process is efficient and that demands are met clearly and precisely, it is important to involve developers in researching and mapping their activities. They are often the ones responsible for carrying out the tasks, so understanding how demands reach them and how they are interpreted is essential. A good understanding of the needs and challenges faced by developers can optimize workflow, avoid misunderstandings and promote more efficient communication between teams. By identifying possible flaws in the way demands are passed othe n, it is possible to adopt improvements that have a positive impact on productivity and the quality of product development.

The aim of this research was to understand how work demands are assigned in the day-to-day lives of professionals, focusing in particular on the flow of tasks for developers. It sought to understand the channels through which demands reach them, such as through platforms like Monday, Jira or Trello, or through direct communication with leaders, Tech Leads, Product Owners (PO), or other means. In addition, the study sought to assess whether the current process is effective or could be improved, and how any changes could impact product development.

The questions submitted to the participants were:

1. How do work demands come to you on a daily basis? Describe the process, including where you receive it and how this assignment takes place.

2. Do you think the way demands come to you currently works well or could it be improved?

   a. It works well, I don't see any need for change.
   b. It works, but it could be better.
   c. It doesn't work well, it needs improvement.

3. If you think it could be improved, how could this process be made more efficient?

4. How often do you receive work demands?
   a. Daily
   b. Weekly
   c. Monthly
   d. Other…

5. Do you feel you have the autonomy to organize your own demands or are you totally dependent on others?
   a. I have complete autonomy
   b. I have some autonomy, but I follow a fixed direction
   c. I am totally dependent on others

The answers to both surveys are available in Appendix C. The following is a comparative analysis of these responses, with the aim of identifying patterns, divergences and relevant perceptions about the process of receiving demands from developers.

Each section of the analysis corresponds to a specific question in the survey - for example, Analysis 1 “Origin and flow of demands” refers to Question 1, and so on - ensuring a structured, question-oriented comparison.

Analysis question 1 - Origin and flow of demands

Most of the demands come through direct communication channels, such as face-to-face or virtual meetings, emails and management tools such as Jira, Azure

DevOps and Confluence. The responses indicate a strong dependence on communication tools such as Teams and the practice of daily meetings to align priorities. However, there is a lack of a centralized or unified system for monitoring all demands, which can lead to confusion and a lack of transparency between teams. Tasks are often assigned in a fragmented way, either by teams of Tech Leads or Product Owners (POs), or even on the team members' own initiative.

Although some teams use task management tools, such as Jira and Azure DevOps, many responses indicate that there is still a gap in the structured and transparent process of assigning demands. Some teams receive demands directly from leaders, without a clear formalization of priorities or deadlines, which is aggravated by communication often taking place via informal meetings and the use of dispersed tools. This suggests the absence of a formal model for managing delivery expectations and ensuring that everyone involved has visibility of the progress of tasks.

On the other hand, some answers indicate that employees show autonomy and proactivity, taking the initiative to look for new tasks, either by asking leaders directly or by analyzing the task boards themselves. This shows that, although there is a hierarchical structure, many teams have a culture of proactivity. However, a lack of clarity regarding the prioritization of these tasks can result in activities that are not aligned with the most urgent needs of the business.

Another relevant point is the lack of visibility and transparency in the workflow between teams. The lack of a centralized system and continuous communication makes visibility difficult, especially in larger projects involving multiple teams or stakeholders. Although some project management tools are used, many teams don't seem to have a consolidated view of the progress of tasks throughout the organization, which makes it difficult to monitor the progress of demands.Table 3 below summarizes the results of question 1.

Table 3 : summary of the question “origin and flow of demands”

Standard	Explanation	Impact
Communication through direct channels	Demands usually come in through meetings (face-to-face or virtual), emails or chat tools.	It can lead to a disorganized task assignment process without a centralized vision.
Use of management tools	Some teams use tools like Jira, Azure DevOps and Confluence to organize their work.	It helps to structure tasks, but can be insufficient when there is no constant follow-up.
Lack of formalization in the process	Tasks are often assigned informally, with little documentation or clear prioritization.	It can lead to a lack of transparency and visibility in the progress of the work, undermining management.
Dependence on Tech Leads and POs	Most tasks are assigned by Tech Leads or POs, limiting the autonomy of the teams.	It reduces agility, since developers depend on an extra layer of approval.
Proactivity in assignments	Some developers take the initiative to seek out tasks or offer solutions directly.	It can improve individual productivity, but it can cause misalignment with the overall strategy.

Analysis question 2 - Efficiency of the reception process

The responses reveal a clear trend regarding the perception of the current effectiveness of the process for receiving work demands. The majority of participants (13 out of 15) believe that the process works, but could be better. However, some responses indicate that the current process does not work well and needs improvement. In addition, there is a small group (3 people) who believe that the current way of receiving demands works well and there is no need for change.

The majority of respondents believe that the process is viable, but that there is room for improvement. This suggests that, despite some flaws or areas that could be optimized, the current model is not completely ineffective. However, the need for improvement may be linked to issues such as lack of visibility, formalization of attributions or an overload of unstructured communication. There are also responses indicating that the process is not fully meeting expectations, possibly due to the lack of a centralized system or communication failures between the parties involved (stakeholders, technical leaders and teams). This may be leading to a more disorganized workflow, compromising efficiency. On the other hand, the low number of responses that consider the current process to be effective suggests that a more structured model could be more effective. However, some people believe that the process is already sufficient for the size and complexity of the teams or projects. These answers probably come from teams with a smaller scope or less complex demands. Table 4 below summarizes the results of question 2.

Table 4: summary of the question Efficiency of the reception process

Standard	Explication	Impact
"It works well, I don't see any need for change."	The majority of responses point out that although the process works, there are points for improvement.	It signals the need for adjustments to increase efficiency and improve communication, without radical changes.
"It works, but it could be better."	A smaller number of participants consider that the current process meets the team's needs well.	It indicates a well-established process that is working for the size and complexity of the team.
"It doesn't work well, it needs improvement."	Some responses indicate that the task assignment system is not working efficiently and needs significant adjustments.	It reflects the need for more in-depth reviews of the demand management process, possibly involving better communication tools or practices.

Concern about micromanagement	One participant's response indicates that excessive control can be detrimental, leading to a decrease in team morale.	It warns of the risk of over-management, which can have a negative impact on team performance and motivation.

Analysis question 3 - Process improvements

Most respondents recognize that the process of receiving and assigning demands could be improved. Below are some recurring suggestions that could be implemented to make the process more efficient:

The use of systems that automatically organize demands after meetings was a frequent suggestion. Assigning tasks using more effective tools, such as integration with systems like Jira, Azure DevOps or others, was also mentioned. In addition, automating the sending and follow-up of tasks is a suggestion that came up to improve the process. These suggestions indicate the need to centralize the process in a more efficient task management platform, which allows for better integration between all the stages, from raising the demand to execution. Tools that can automatically organize and prioritize tasks would bring more agility and visibility to the process, reducing the dependence on manual communication.

Another recurring suggestion was to improve communication and collaboration. More direct interaction with stakeholders would help to better understand demands and clarify doubts. In addition, the active participation of developers in the discovery process could increase visibility of business needs and the impact of changes. More detailed specifications on the flows and impacts of changes were also pointed out as an important improvement. Improving communication between developers and stakeholders would help ensure that demands are aligned with real needs and reduce the risk of rework. In addition, allowing developers to participate more actively in the initial phases would help increase their understanding of the requirements and scope, which would generate a more efficient process.

With regard to the organization and definition of tasks, it was suggested that the activities be organized in such a way that they reflect the skills and experience of the developers, as well as creating more detailed and specific tasks, with a clear step-by-step approach, to make it easier for those involved. It was also mentioned that creating cards with all the information needed to carry out the tasks, avoiding the need to search for additional data, would be beneficial. Organizing tasks according to the experience and skills of the developers allows for more efficient allocation and improves workflow. More detailed and well-described tasks reduce the time spent searching for information and allow developers to start working more efficiently, reducing delivery time.

Improving visibility and transparency was also highlighted. It was suggested to increase the visibility of tasks and deadlines by creating clear systems for notifying and following up on activities. Centralizing tasks on a platform where developers can signal when they have started or completed an activity was also mentioned. In addition, monitoring the progress of tasks in real time was highlighted as an important improvement. Improving visibility and transparency in tasks helps avoid confusion and makes it easier to track progress, ensuring that everyone involved can see the progress of activities, quickly identify blockages and optimize communication between teams.

Finally, adjustments to the planning process were suggested. Improving the definition of the scope of changes and assessing the impact of each change was pointed out, as was the creation of a committee to assess risks and plan broader changes that impact several projects or teams. Having a clearer and more detailed planning process, with risk assessment, would help to improve accuracy in defining deadlines and scope. This type of process would also reduce the risk of work overload and help to adjust expectations. Table 5 below summarizes the results of question 3.

Table 5: summary of the question Process improvements

Standard	Explication	Impact
Centralization and Automation	Integrated task management tools that automatically organize and prioritize demands.	Increased visibility, organization and efficiency in the process of assigning and monitoring tasks.
Direct communication with stakeholders	Developers involved in the discovery process and in direct contact with stakeholders to better understand demands.	Reduced risk of rework, increased understanding of requirements, and alignment of expectations.
Organizing and Detailing Tasks	More detailed tasks, with clear steps aligned with the developers' skills.	Reduced time spent searching for information, increased efficiency and clarity in the work carried out.
Visibility and Transparency in Tasks	Creating centralized platforms with clear notifications about the status of tasks and deadlines.	Better monitoring of progress, less risk of tasks not being completed on time.
Clear prioritization of tasks	To-do list with clearly prioritized tasks to ensure that the most important ones are done first.	Increased productivity and focus on tasks with greater impact and urgency.
Planning and Risk Assessment	Better definition of the scope of changes, assessment of impacts and creation of a committee for risk analysis.	More efficient planning, with less risk of surprises or work overload for the teams.

Analysis question 4 - Frequency of demands

Analysis of the responses reveals that the majority of respondents receive daily demands, which is a reflection of the dynamic nature of the work environment, where new challenges arise frequently, especially in areas of development and project management. Some people mentioned other frequencies, and this gives us an idea of how different teams or functions can have different rhythms.

8 out of 15 people receive tasks on a daily basis, which is to be expected in agile development environments with a high volume of work. This may indicate that the teams have a constant workload and need to be continually updated on priorities and tasks. For these teams, the fluidity of the task assignment process is crucial to maintaining the pace of production without interruption.

Some respondents mentioned that they receive demands on a weekly basis. This suggests that the process of planning or reviewing tasks occurs with this regularity, which may be related to weekly sprints or alignment meetings. This frequency indicates that the workload is more predictable, with a defined schedule of deliveries and tasks to be carried out throughout the week.

Only one person mentioned that they receive demands on a monthly basis. This may be indicative of less dynamic work or of an area with longer production cycles, such as large-scale projects or those involving more strategic and long-term planning.

Analysis question 5 - Autonomy in organizing demands

The analysis of the answers reveals a diversity of levels of autonomy in the process of organizing work demands, which suggests different degrees of control that team members have over how tasks are assigned and carried out. The distribution of responses can be divided into the following categories:

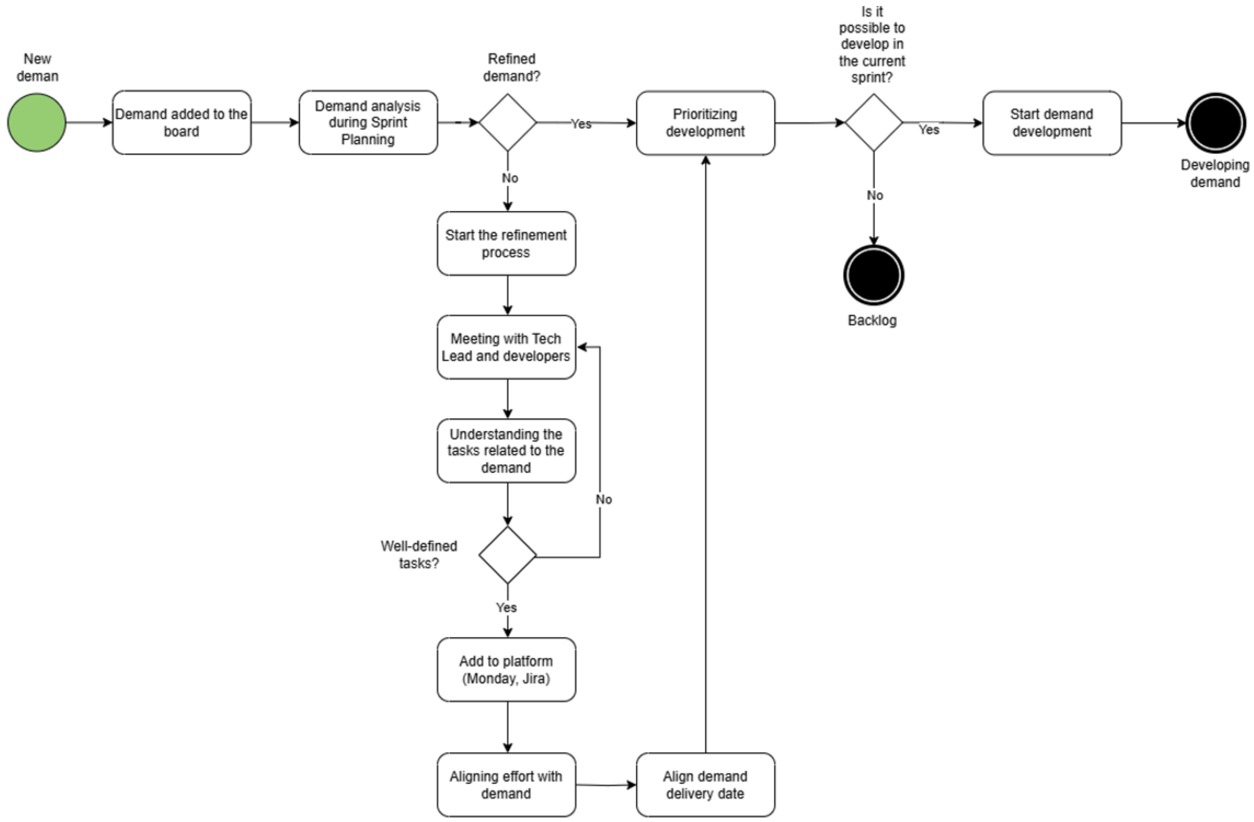
Seven people indicate that they have complete autonomy to organize their own demands. These people probably have a high level of responsibility, with the freedom to manage time and priorities on their own. This level of autonomy is positive, as it allows for greater flexibility and can increase motivation and engagement, allowing professionals to choose the tasks that best align with their skills and interests.

Seven people indicate that they have some autonomy, but still follow a fixed direction. These people probably have more control over their daily tasks, but still depend on some level of supervision or guidance, either from a manager or a more rigid structure for assigning tasks. This may indicate a balance between independence and the need for alignment with organizational goals and team coordination.

Only one person reported that they are totally dependent on the assignment of others to organize their demands. This suggests that this person is in a role where decisions about what to do, when to do it and how to do it are determined exclusively by others, possibly in a more hierarchical structure or with more centralized work processes.

Macro flowchart based on research

Figure 2 below describes the typical steps involved in receiving demands, based on a survey carried out with developers.



Source: Authors

The flowchart constructed from the interviews represents, in macro form, the main aspects identified during the process of receiving and developing demands. Among the highlights observed is the way in which demands are initially added to the board (such as Jira or Monday) and analyzed in the sprint planning meetings. This initial stage reflects the reality of many teams that deal with demands coming from different channels, often in a decentralized way. Next, a check is made to see if the demand has already been refined, and if not, a refinement process begins.

This refinement usually involves a meeting with the Tech Lead and the developers, who take a more active stance, typical of the roles traditionally assigned to the Product Owner. The developers help to understand the details of the demand and the tasks involved. After this alignment, another check is made to ensure that the tasks are well defined. If they are, the demand is entered into the management platforms and the necessary effort is aligned, taking into account the available resources and complexity. This point in the process highlights the direct involvement of developers in both the planning and organization of deliveries, something that ideally should be more structured and centred on the PO.

Finally, after the development has been prioritized, it is checked whether it is possible to carry out the demand in the current sprint. If so, development begins; if not, the demand is sent to the backlog. This part of the process demonstrates the teams' attempt to maintain agility and a continuous pace of delivery, but it also reveals challenges related to the prioritization and visibility of activities.

Assumption of PO Responsibilities by Developers

Analysis of the answers to the form revealed that developers often take on responsibilities that would ideally be assigned to Product Owners (POs). This overlap is largely due to the absence of well-defined processes, fragmented communication and a lack of detail in demands.

Among the main responsibilities of the POs that end up being absorbed by the developers is the effort to understand the demands. Many professionals report the need to seek out the context of the tasks on their own, analyzing boards or contacting stakeholders to clarify doubts, due to the lack of complete information at the time of transfer.

In addition, in some cases developers end up defining which tasks to carry out based on their own perception of priority, without formal guidance. It is also common for them to identify technical dependencies and align with other teams, functions that would normally fall to the PO during the planning phase.

Finally, developers often update the status of demands in the tools and monitor the progress of deliveries, activities that should be shared with the PO to ensure visibility and alignment with business objectives.

2.5.2 Interview with POs

The purpose of this section is to present the research carried out with the POs and their respective analyses. The study focuses on investigating the process of refining demands in the daily lives of POs, an essential stage in product management. This refinement consists of the detailed definition of demands, ensuring that the development teams have clear and precise information to implement the desired solutions. The research seeks to understand the origin of the demands, the agents involved in the process, the degree of autonomy of the POs in the refinement, as well as the methodologies and tools used for this purpose.

Two interviews were carried out with different POs, guaranteeing the anonymity of the participants. The first questionnaire was administered between 02/17 and 02/28, obtaining 5 responses, while the second took place between 03/17 and 03/28, with a total of 6 responses. The questions asked of the participants were as follows:

1. How do requests reach you? Describe where requests come from, who participates in this process and how demands are assigned.

2. When a demand arrives, is it already fully defined or do you need to refine it? Explain whether demands arrive ready-made or whether there is room for improvement.

3. How does the refinement of demands occur in your daily routine? Describe your refinement process, what steps you follow, who participates and what tools or methods you use.

4. Do you feel you have enough autonomy to refine demands in the best way? Why? Explain whether you have the freedom to adjust the demands as necessary or whether there are restrictions that hinder this process.

5. What are the main challenges you face when refining demands? List the main obstacles that hinder more efficient refinement, such as lack of information, time constraints, lack of alignment with stakeholders, etc.

The answers to the two surveys are available in Appendix A. The following is a comparative analysis of these responses, with the aim of identifying patterns, divergences and relevant insights into the process of refining demands in the day-to-day running.

Each section of the analysis corresponds to a specific survey question—for instance, Analysis 1 "Origin and flow of demands" refers to Question 1, and so on—ensuring a structured and question-oriented comparison.

Analysis question 1 - Origin and flow of demands

Demands can come from a variety of sources, including internal stakeholders (leadership, business areas), external stakeholders (customers) and internal teams (development, UX, operations). Some arrive in a structured way, through management tools and strategic planning, while others are passed on informally, via emails, WhatsApp or direct conversations.

The Product Owners' reports vary. Some describe a structured flow, with annual planning and periodic reviews, while others point to a disorganized scenario, with demands coming in chaotically. In some cases, prioritization takes place formally before Program Increments (PI), while in others, demands are received without clear prioritization criteria.

Market changes, competitive movements and emergency requests from customers often require replanning. In many responses, Tech Leads are primarily responsible for assigning demands and defining technical priorities. This scenario highlights one of the central challenges of the survey: the strong dependence on Tech Leads which can create bottlenecks and compromise the autonomy of POs, making predictability and transparency difficult in the decision-making process. Table 6 below summarizes the results of question 1.

Table 6 : summary of the question “origin and flow of demands”:

Standard	Explanation	Impact
Demands come from multiple channels	Emails, messages, meetings, agile tools, top-down from leadership	Makes traceability and standardization of the workflow difficult
Lack of prior alignment on priorities	Some POs mention quarterly/annual planning, but others receive demands without clear criteria	Causes abrupt scope changes and constant replanning
Stakeholders strongly influence prioritization	Demands come from commercial areas, clients, UX, operations, and leadership	Can compromise the strategic vision if well-defined criteria are not in place
Tech Leads act as gatekeepers	Tech Leads often assign demands and define effort	Reduces POs’ autonomy and may cause delivery bottlenecks

Analysis question 2 - Level of demand refinement

Most respondents point out that demands rarely arrive fully refined and ready for development. In most cases, only the central theme is known, requiring detailed work to define business rules, technical impacts and actual scope.

Simple demands usually come better defined, while more strategic and complex demands require multiple meetings, alignments and iterations before being implemented. The time dedicated to refinement grows in proportion to the complexity and technical dependencies involved.

In addition, many demands require continuous interaction with clients, business areas, UX, development and other teams. The lack of a standardized format for delivering requirements contributes to rework in the refinement phase, making the process less efficient.Table 7 below summarizes the results of question 2.

Table 7 : summary of the question “level of demand refinement ”

Standard	Explanation	Impact
Demands come in without sufficient detail	Most demands come in with only a general theme, requiring more in-depth analysis	Generates rework and the need for additional meetings
Complexity impacts the level of refinement required	The more complex the demand, the more time and effort is spent on refinement	Delays in defining scope and technical impacts
Stakeholders play an essential role in refinement	Refinement requires frequent contact with technical, business and customer teams	It requires a structured alignment process to avoid misalignment
Lack of a standard for receiving requests	Some companies have frameworks such as AGNS, but most work with different formats	Hinders predictability and efficiency of refinement

Analysis question 3 - Refinement process

Most respondents point out that refinement involves multiple stakeholders and takes place iteratively over time. Simple demands can be refined directly by the PO, with little or no external interaction, while more complex demands require multiple meetings, detailed documentation and the participation of several teams.

To support refinement, tools such as Jira, Confluence, Monday, Figma and BPMN are used to document and manage the process. In addition, some answers mentioned the use of chat groups and recorded meetings to facilitate the exchange of information. Refinement usually starts with a business analysis and then goes through a technical evaluation. The participation of tech leads and specialists is often cited as essential for mapping impacts on systems and business rules.

At the end of the process, many teams formalize the decisions and definitions in structured documentation, such as Wikis, Confluence, flowcharts or cards in Jira. This record ensures alignment between those involved and reduces the risk of misinterpretations during development. Table 8 below summarizes the results of question 3.

Table 8 : summary of the question “Refinement process ”

Standard	**Explanation**	**Impact**
Refinement takes place in stages and involves multiple stakeholders	First business understanding → technical analysis → validation and formalization	Ensures clarity and reduces the risk of rework
Simple vs. complex demands require different approaches	Simple: quick refinement by the PO. Complex: multiple meetings and detailed documentation	Helps balance time invested in refinement
Tools structure and document refinement	Jira, Confluence, Monday, BPMN, Figma, flowcharts, emails and chats are all common.	Facilitates traceability and communication between areas
Refinement is a continuous process, not a single step	Occurs throughout the sprint/planning and can be adjusted as new insights emerge	Maintains alignment and avoids surprises in delivery

Analysis question 4 - Level of autonomy

Many POs report that their autonomy is limited, especially in demands involving the backend, integrations or architectural decisions. In these cases, dependence on Tech Leads is significant, as technical feasibility needs to be assessed before moving forward. In the case of more superficial demands, such as interface improvements or adjustments to internal processes, autonomy tends to be greater.

When the client is already clear about what they want or there are regulatory requirements, there is little room for adjustment. On the other hand, when the demand comes in unstructured, the PO has more freedom to refine it, as long as they can back up their decisions with solid arguments.

Some POs mention that the lack of adequate time for refinement negatively impacts the process, compromising the quality of strategic planning and the ability to better question stakeholder needs. Furthermore, even when they have the autonomy to refine demands, many feel the need to validate decisions with managers or Tech Leads before moving forward.

POs who feel more autonomous generally stress the importance of maintaining an open communication channel with clients and other areas. This continuous contact allows for adjustments throughout the process and reduces the risk of wrong decisions that need to be corrected later. Table 9 below summarizes the results of question 4.

Table 9 : summary of the question “Level of autonomy”

Standard	**Explanation**	**Impact**
Autonomy depends on the technical complexity of the demand	The more technical the demand, the more dependence on tech leads and less freedom for adjustments	PO may feel lack of predictability and security in refinement
Regulatory or very specific demands have less flexibility	If a requirement comes ready-made from the client or is imposed by regulation, the PO has little room for refinement	PO acts more as a facilitator of the process than as a decision-maker
POs who document and argue well gain more autonomy	Explaining the reason for changes and documenting helps gain credibility with stakeholders	Facilitates changes in scope without major objections

POs who feel technically insecure tend to seek validation before moving forward	If a PO is unclear about the technical impact of a change, they need to consult tech leads or managers	This can lead to delays in refinement and increase the burden on tech leads

Analysis question 5 - Main challenges

POs face difficulties in obtaining technical details of requirements, either due to a lack of documentation or the constant need for interactions with IT teams. The absence of code documentation aligned with business documentation makes it difficult to verify requirements in an agile way and make informed decisions.

Business pressure for quick deliveries leads to incomplete refinements, which often continue during development. In addition, the limited availability of Tech Leads to support refinement results in less informed decisions and late adjustments.

Another recurring challenge is the arrival of demands without a well-defined purpose, prioritizing “what to do” without a clear understanding of “why”. This reduces the autonomy of the PO, who could contribute to defining the best solution if the objectives were more transparent. In addition, frequent changes of direction generate rework and loss of context for the team, since stakeholders are not always clear about what they really need.

A deeper understanding of the business, its internal politics, and strategic direction can help define better technical solutions—either directly, by aligning decisions with clear objectives, or indirectly, by anticipating constraints and stakeholder expectations.

POs also face difficulties in assessing the complexity of a demand before development begins. This lack of technical visibility compromises scope negotiation, making it difficult to analyze impacts and define viable alternatives. In addition, some changes can generate financial or contractual impacts that are not considered in the initial refinement, creating additional risks to the project.

Convergences and divergences between the theory and practice of the PO's role

The aim of this section is to revisit the points made about the responsibilities of Product Owners (POs) described in topic 3.2, based on the literature, and compare them with the data collected in the survey with POs. The intention is to identify convergences and divergences between theory and practice, understanding how the responsibilities attributed to OPs are manifested in the organizations analyzed.

- **Backlog management and prioritization:** The literature points out that the PO should have autonomy in prioritization based on business value and return on investment (Schwaber, 2004). However, in practice, many interviewees reported strong influence from stakeholders (leadership, commercial areas, operations), with prioritization often being defined outside of the PO's control. In some cases, the Tech Lead plays a determining role in the distribution of demands, which reduces the PO's autonomy - diverging from the model described by Schwaber and reinforcing the hybrid scenario described by Sverrisdottir et al. (2014), in which responsibilities are shared informally.

- **Involvement with stakeholders:** The theory emphasizes that the PO should be the link between the team and the stakeholders, communicating the needs of the users and the vision of the product (Kristinsdóttir, 2014). The research data shows that this function is present, but not always in a structured way. Some POs don't feel responsible for identifying stakeholders or translating their demands to the technical team, showing a gap between the ideal model and operational reality.

- **Collaboration with the development team:** Although Scrum proposes close working between the PO and the technical team, the study revealed a wide variation in the time dedicated by POs to direct collaboration with developers - reflecting the findings of Kristinsdóttir (2014). In environments where the PO actively participates in technical discussions, there is greater clarity in deliveries.

However, in contexts where the PO has low availability or relies heavily on the Tech Lead, problems of alignment and incomplete refinement arise.

- **Autonomy in decision-making:** The literature reinforces the importance of the PO's autonomy for the effectiveness of the agile process. However, the reports show that this autonomy is often limited, especially when it comes to technical demands. The need for constant validation with Tech Leads and managers reflects an organizational structure that does not yet give the PO full decision-making power, which corroborates the challenges pointed out by Sverrisdottir et al. (2014) regarding the importance of decision-making authority for project success.

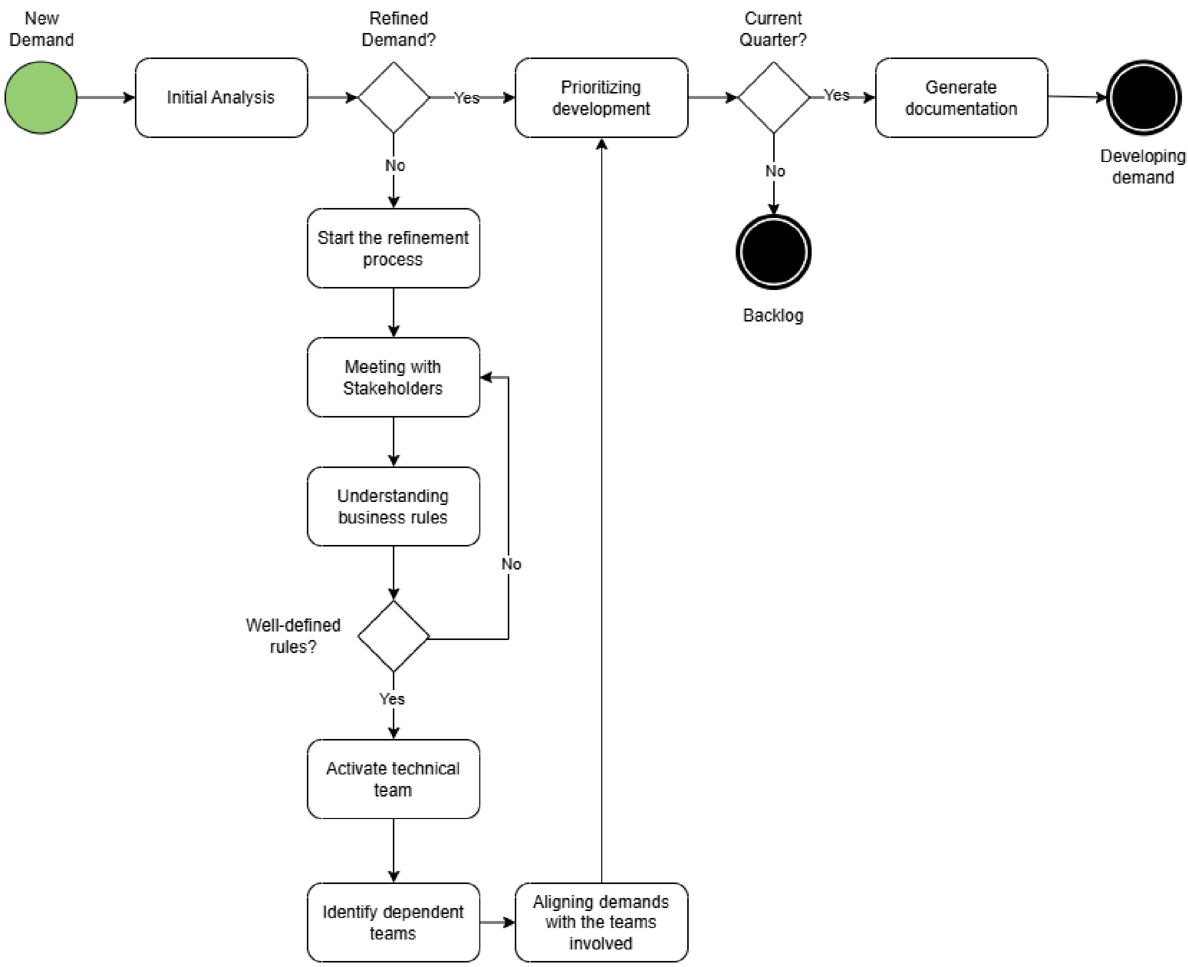
- **Communication challenges and expectations:** Managing expectations and the ability to negotiate are cited as essential skills in the literature. In practice, it was observed that POs face difficulties in this respect, mainly due to the pressure to deliver quickly and the lack of clarity in requests. Fragmented communication and the lack of clear criteria for prioritization reduce the effectiveness of the OP's role as a strategic articulator - a challenge already foreseen by Kristinsdóttir (2014).

- **Refinement process:** The iterative and collaborative refinement process described in the literature is partially confirmed in practice. Although the interviewees report the use of tools and methodologies to organize refinement, there is a lack of standardization and an excess of informality in the arrival of demands. This results in continuous rework and refinement during development, contrary to the idea of solid preparation prior to the start of implementation.

Macro flowchart based on research

Figure 3 below describes the typical steps involved in refining demands, based on research carried out with Product Owners (POs).

Figure 3 – Macro flowchart based on research



Source: Authors

The flowchart constructed from the interviews represents, in macro form, the main aspects identified in the work of the Product Owners throughout the process of arriving at and developing demands. Among the highlights observed:

- **Detailed Refinement:** The POs dedicate part of their time to refining the demands, which involves exploring different scenarios, identifying dependencies and understanding the business rules.
- **Extensive Collaboration:** The diagram also shows the strong collaborative action of the POs with different stakeholders - such as business areas, technical

leaders and other teams - reinforcing the role of continuous articulation throughout the demand cycle.

- **Expectations management:** The active management of expectations in relation to scope and deadlines, mentioned frequently in the interviews, is reflected in the interactions provided in the flowchart, showing the points of communication and alignment with stakeholders throughout the process.

2.5.3 Interview with Tech Leads

This section presents the analysis of the third round of interviews conducted between March 17 and March 28, focusing on the role of Solution Architects and Tech Leads in receiving, refining, and executing technical demands. The research seeks to understand how architectural decisions are made, how collaboration with Product Owners and stakeholders is structured, and what challenges arise in aligning business needs with scalable and maintainable technical solutions.

The six participants interviewed in this phase of the research shared their practices, constraints, and approaches to balancing fast-paced delivery with architectural integrity. Each subsection below corresponds to one of the survey questions, with observed patterns organized in the format: Standard, Explanation, and Impact.

Below are the questions that were aimed at this audience in question, the Teche Leaders

1. How do technical demands reach you? Describe the process for receiving technical demands and how these demands are communicated to you (meetings, management platforms, direct communication with stakeholders, POs, etc.).

2. How do you define and prioritize the technical approach for each demand you receive? What criteria do you use to decide on the best technical solution to meet the demand? How do you decide between different approaches or technologies?

3. Do you have the autonomy to define the architecture and technical solutions to the demands or are there limitations? Are there external factors (e.g. budget constraints, deadlines, stakeholder requirements) that affect your ability to define the ideal solution?
4. What tools or methodologies do you use to coordinate the execution of technical demands with your team? Tell us about the tools such as Jira, Azure DevOps, Trello, among others, that you use to manage the technical execution of demands.
5. Have you ever faced challenges related to changing requirements or priorities during the development cycle? How do you deal with unexpected changes and how do they affect the proposed technical solution? How do you adjust the team's planning to adapt?
6. What is your collaboration process with POs and stakeholders throughout the development cycle? When do you interact with POs or stakeholders during the process? How do these interactions impact the definition and execution of technical solutions?

Analysis question 1 – Origin and flow of technical demands

Technical demands come from a variety of sources: product owners, internal stakeholders (leadership, business teams) and even from the architects or developers themselves when they identify technical or improvement opportunities. These demands come through structured tools such as Confluence or Azure DevOps, but also through informal means such as meetings and chats.Table 10 below summarizes the results of question 1.

Table 10 : summary of the question “Origin and flow of technical demands”

Standard	**Explanation**	**Impact**
Technical demands come from multiple channels	Includes PO requests, business input, and direct identification by architects or devs	Requires constant triage and prioritization to avoid overload

Some demand sources bypass standard intake processes	Direct stakeholder requests or self-initiated improvements are common	Leads to context-switching and potential conflicts in prioritization
Lack of standardization in how demands arrive	Use of diverse tools and informal communication	Reduces traceability and hinders alignment across teams

Analysis question 2 – Criteria for defining the technical approach

Architects and Tech Leads describe a decision-making process that weighs reusability, complexity, scalability, and operational costs. Many rely on predefined templates or cloud-native patterns to ensure fast and reliable delivery—especially in tactical contexts. Evaluation criteria include business impact, time-to-market, and future maintainability. Table 11 below summarizes the results of question 2.

Table 11 : summary of the question “Criteria for defining the technical approach”

Standard	**Explanation**	**Impact**
Preference for reusable and standardized architectures	Use of known patterns (e.g., Lambdas, Cronjobs, managed services)	Speeds up delivery and reduces risk
Decisions driven by complexity vs. value vs. time	Balancing technical effort with business return	Helps justify trade-offs and define MVPs
Focus on long-term scalability and maintenance	Includes operational cost, system growth, and monitoring	Ensures sustainability of the solution

Analysis question 3 – Autonomy in defining the solution

While Tech Leads generally have autonomy, organizational constraints (non-approved technologies, leadership bias, cost ceilings) influence decision-making. Even with architectural freedom, decisions are often guided by internal standards and delivery targets.Table 12 below summarizes the results of question 3.

Table 12 : summary of the question “Autonomy in defining the solution”

Standard	**Explanation**	**Impact**

Autonomy varies depending on delivery context	Tactical projects allow more freedom than strategic, highly visible ones	Requires careful navigation of priorities and expectations
Internal politics and tech preferences constrain options	Certain tools are avoided due to leadership taboos or lack of approval	Limits experimentation and architectural evolution
Alignment with internal standards fosters consistency	Solutions must conform to team norms for easier collaboration	Promotes knowledge sharing and supportability

Analysis question 4 – Tools and coordination methods

The use of platforms such as Azure DevOps, Confluence, Monday, and OneNote is widespread for coordinating execution. However, the lack of integration between business specs and technical documentation is a recurring issue, creating extra effort during refinement and onboarding.Table 13 below summarizes the results of question 4.

Table 13 : summary of the question “Tools and coordination methods”

Standard	Explanation	Impact
Execution managed through multiple tools	Azure DevOps for tracking, Confluence for specs, OneNote for notes	Fragmented visibility of the full process
Separation between business and code documentation	Limited linkage between what’s built and why it was built	Reduces context for new developers or auditors
Methodologies adapted to team needs	Varying levels of adherence to Scrum or Kanban	Influences team discipline and predictability

Analysis question 5 – Communication of technical priorities

Translating technical implications into business language is a core challenge. Misunderstandings often arise around integration complexities and the effort involved in seemingly "simple" tasks. Tech Leads often take on the role of educators, guiding POs and stakeholders toward feasible timelines and expectations. Table 14 below summarizes the results of question 5.

Table 14 : summary of the question “Communication of technical priorities”

Standard	Explanation	Impact

Tech complexities not fully understood by non-technical teams	Integration paths (e.g., API vs. messaging) affect timelines significantly	Causes misalignment and underestimated effort
Stakeholder involvement is uneven	Some engage in early refinement, others only during delivery phases	Inconsistent scoping and late feedback loops
Communication depends on individual initiative	Tools help, but personal follow-up remains essential	Increases cognitive load and time spent outside coding

Analysis question 6 – Handling requirement changes during development

Change is constant—requirements evolve mid-sprint due to missed initial details or new business needs. Tech Leads react by replanning and negotiating scope. Some teams formalize scope changes with stakeholders; others use iterative delivery as a buffer. Table 15 below summarizes the results of question 6.

Table 15 : summary of the question "Handling requirement changes during development"

Standard	**Explanation**	**Impact**
Frequent scope changes during development	Due to unclear initial specs or new business inputs	Leads to rework and delivery delays
Refinement continues during implementation	Teams adapt on the go, sometimes without full revalidation	Compromises solution integrity and planning
Formalization helps manage expectations	Changes are negotiated with stakeholders and scheduled in phases	Improves transparency and avoids conflict

Convergences and divergences between the theory and practice of the Tech Lead's role

This section analyzes the alignment between the theoretical expectations of the Tech Lead's role and the empirical findings from the interviews with professionals who perform this function in agile environments. The objective is to highlight convergences and divergences between theory and practice, focusing on the daily responsibilities, constraints, and decision-making challenges faced by Tech Leads in the context of software development.

Architectural vision and technical leadership

According to Unisinos (2024), the Tech Lead is expected to define the technologies and architectures used in a project, promoting technical alignment and ensuring the sustainability of the system. In practice, interviewees confirmed this role, often acting as key figures in technical planning. However, they also reported that architectural decisions are sometimes compromised by tight deadlines or delivery pressures, leading to technical shortcuts that may increase future maintenance efforts.

Autonomy and organizational constraints

Although Alura (2024) emphasizes that the Tech Lead should have autonomy to define technical paths aligned with strategic goals, the interviews revealed that this autonomy is often limited by internal politics, leadership preferences, or the non-approval of certain technologies. Even with experience and leadership, some Tech Leads must negotiate with other stakeholders to justify architectural choices, which transforms technical decisions into political negotiations.

Collaboration with stakeholders and POs

The role of the Tech Lead also includes strong collaboration with stakeholders, a point reinforced by Menon, Sinha, and MacDonell (2021), who highlight the importance of continuous engagement to ensure alignment of expectations. The interviewees reported that this collaboration is often informal and highly dependent on personal initiative. They play a fundamental role in "translating" technical complexities into business language, helping Product Owners and stakeholders understand the real scope and implications of architectural decisions.

Team coordination and technical mentoring

As stated by Altmann (2021), the Tech Lead acts as a mentor to the development team, encouraging knowledge sharing and best practices. This is confirmed in practice:

many participants describe themselves as both team leaders and enablers of technical growth. However, some also mentioned difficulties in dedicating time to mentoring due to urgent deliveries or the accumulation of strategic responsibilities beyond their technical role.

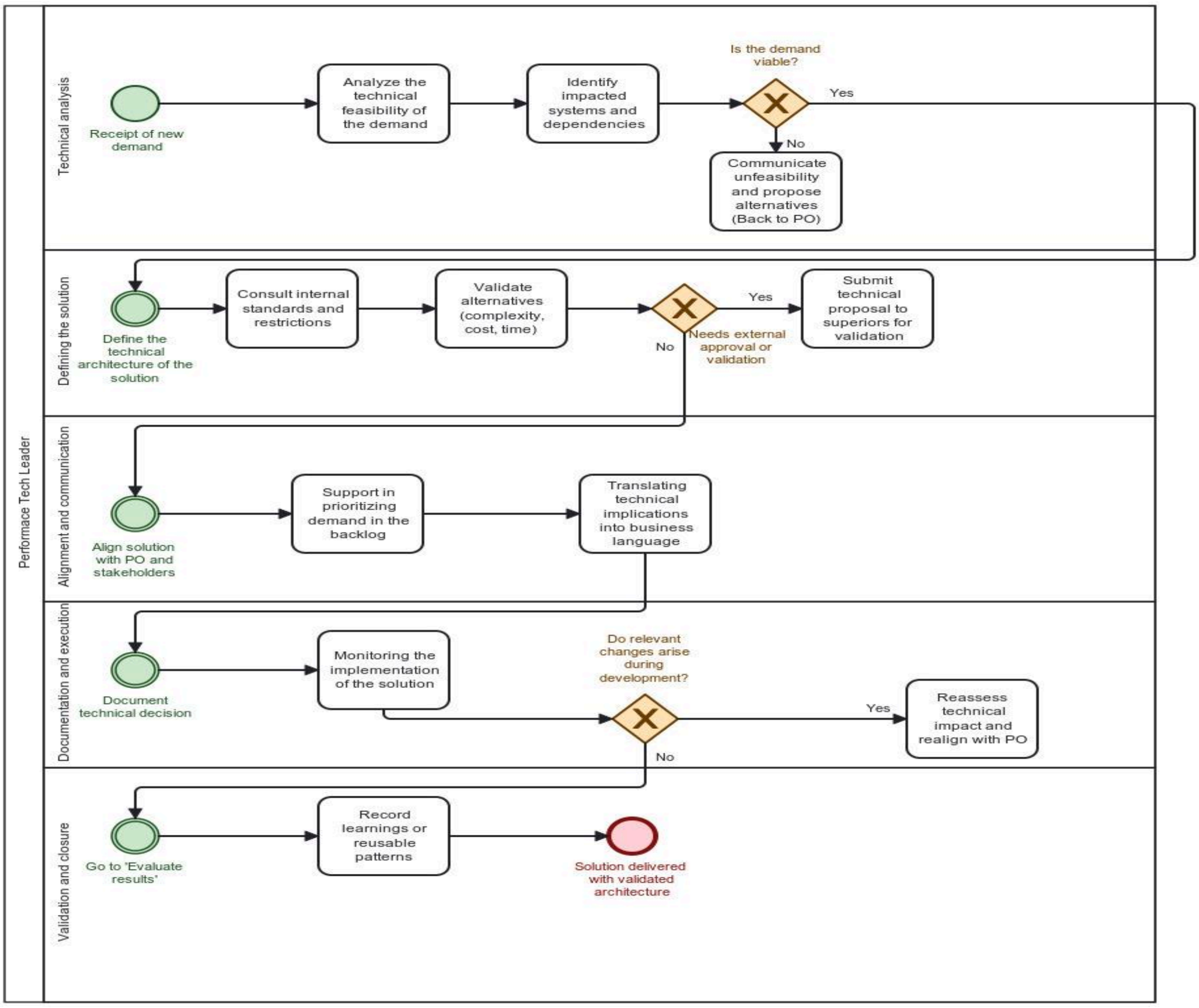
Adaptation to changing requirements

The literature points out, particularly in Arquitetocloud (2021), that the solution architect must be prepared to adapt architectures to changing business needs, often validating approaches through Proofs of Concept (PoCs). This flexibility is also observed in practice. Tech Leads reported frequent mid-sprint changes in requirements, which forces them to replan technical solutions and negotiate scope. While some teams formalize these changes, others adapt on the fly, which can compromise architectural integrity.

Macro flowchart based on research

In addition to the qualitative analysis of the interviews, it was possible to structure a macro-process that represents the flow of the Tech Lead's work based on the responsibilities reported. This flow (Figure 4), modeled in BPMN notation, describes the main stages involved from receiving the technical demand to validating the delivery, including activities such as feasibility analysis, defining the solution's architecture, aligning with stakeholders, documentation and monitoring implementation. The model seeks to represent in a visual and standardized way the recurring practices observed in the interviews, making it easier to understand the strategic and operational role of the Tech Lead in the solution development cycle.

Figure 4 – Macro flowchart based on research



Source: Authors

Overlapping responsibilities between Product Owners and Tech Leads

Although the Product Owner and Tech Lead play different roles in agile product development, the interviews revealed areas of overlap in their responsibilities. Both roles participate in refining demands, contribute to defining the scope of the solution and act as strategic interlocutors with stakeholders. Like POs, Tech Leads also participate in clarifying requirements, especially when dealing with technical restrictions and integration impacts. In many teams, Tech Leads actively contribute to discussions about the backlog, support prioritization decisions and are involved in translating business needs into technically viable solutions - responsibilities that, in theory, would be the domain of the PO. This intersection highlights the collaborative nature of demand refinement and reinforces the idea that product success depends on the synergy between commercial and technical leadership, but it can also generate a series of frictions that will be addressed later in this paper.

2.6. FELIX - Initial Structuring

The FELIX framework was built on the understanding that, although each project has its own particularities, it is essential to maintain an integrated view of the portfolio as a whole. This systemic approach allows for greater alignment and comparability between initiatives, and facilitates decision-making based on more structured data.

The following sections will detail the macro flow of the framework, its initial components and the templates and tools that support this structure, reinforcing the commitment to an agile, integrated and continuously evolving approach.

2.6.1 Recording Information

The information presented in this section was obtained through research and observational analysis of daily life at BTG Pactual. The aim is to understand how project information is currently recorded, identifying the moments that require more structured interventions and assessing which tools already in use can be maintained and adapted to the proposed framework.

This recording currently takes place across three different platforms, each with specific purposes and functionalities. The following details how these platforms are used in the current context, highlighting their role in storing and accessing information throughout the project lifecycle.

The first tool is Monday, which is used as the main platform for organizing requirements and tracking project progress. Within this tool, the “meeting notes” column is the responsibility of the PO (Product Owner), who adds information relating to meetings, decisions and important points discussed during the course of projects. This approach centralizes the communication of meetings and makes it easier to monitor the progress of demands. However, its ability to store detailed information, such as flowcharts or code, is limited, as this tool is more geared towards managing tasks and not recording technical details.

The second is an internal environment called Wiki, which is used to store more detailed and structured information, such as explanatory texts, codes, links, flowcharts and any other type of technical content that requires more in-depth documentation. This platform allows content to be easily shared via links, which facilitates collaboration between teams from different areas. Organizing and structuring the content on the Wiki is important to ensure that the information is accessible and can be found easily by all team members.

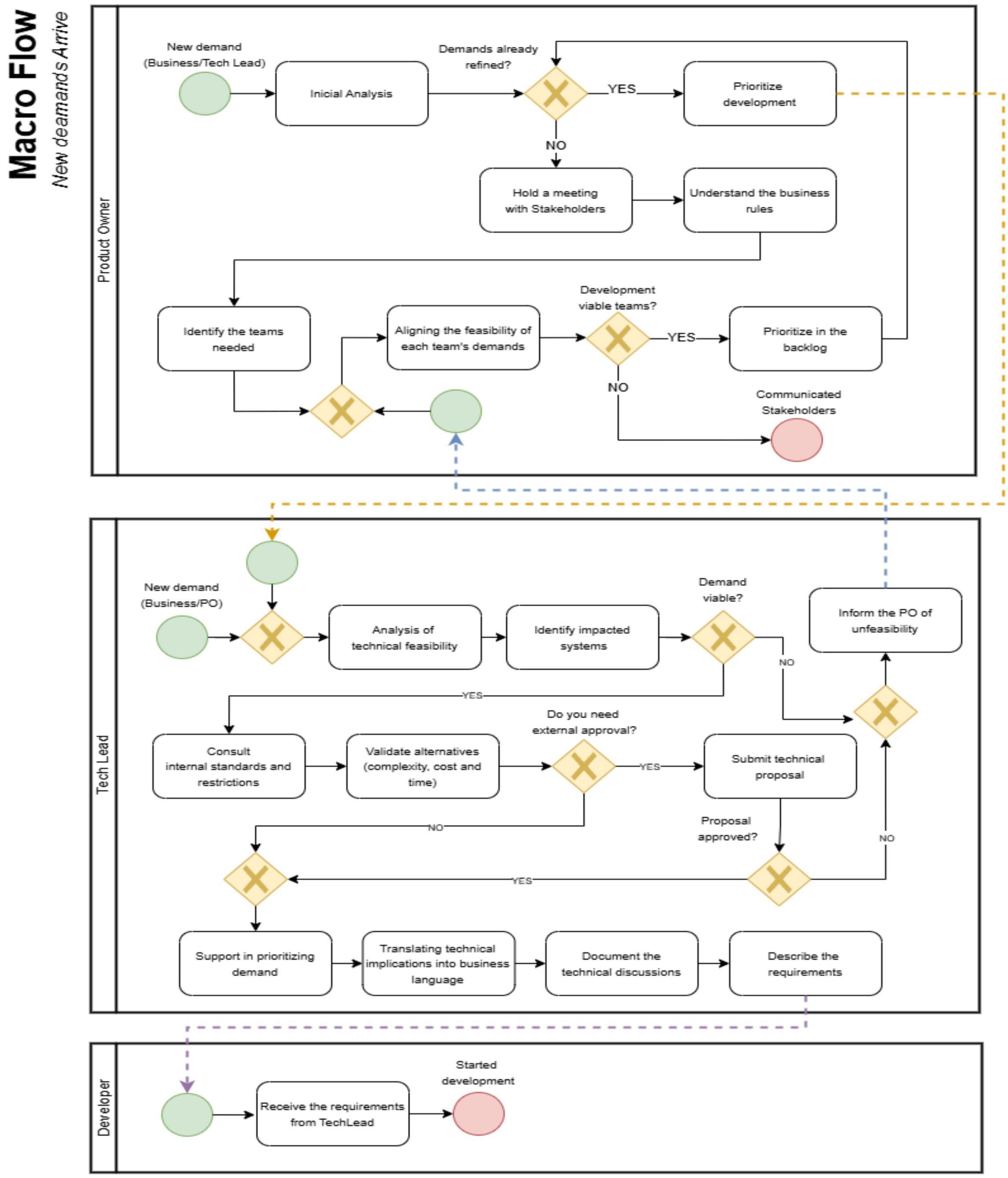
The third and final is Azure DevOps tool that is used to record smaller, more specific tasks within projects. Each task contains a detailed description with flows and steps, allowing the team to track the progress and execution of activities. Although it's a good tool for managing the progress of tasks, adding information directly to task descriptions can become a bit scattered over time, especially in larger projects. It is therefore important to be clear about the inclusion of information to avoid essential data becoming scattered or difficult to locate.

Using different platforms to record information can generate some advantages, such as specialized data storage in each tool, but it also presents challenges, such as the risk of redundancy and the dispersion of information. To ensure the effectiveness of this process, it is important that the team follows a clear plan on where and how each type of information should be recorded. In addition, integrating these platforms can improve the flow of information and facilitate access to essential data during the project lifecycle.

2.6.2 Macro Flow

Based on the initial phase of analysis, research, and bibliographic review, a macro flowchart was developed to represent the current operational structure within BTG Pactual. This flow illustrates the responsibilities currently assigned to Developers, Tech Leads, and Product Owners. The analysis revealed that the main issue in the project stems from overlapping responsibilities, indicating a lack of clarity in role definitions, as shown in Figure 5. The diagram was created using the draw.io tool.

Figure 5 – Macro flow based on research



Font: Draw.io (https://drive.google.com/file/d/13V5WKJcjIbCvSI56xIUDsRHfur-Vmu4P/view)

One of the main problems identified in the flow occurs during the handling of new commercial demands. In practice, when a technical lead receives a request directly from the business area, they often proceed with the demand without informing the product owner (PO). Usually, the PO is only informed if the technical lead determines that the demand is unfeasible. Furthermore, when the demand is highly technical, he tends to assume that the PO doesn't have the necessary knowledge to refine it properly. As a result, they assume responsibility for the refinement process independently. These behaviors contribute to a breakdown in collaboration and transparency, undermining the intended distribution of responsibilities between functions.

The proposal is that, with the construction of the framework, it will be possible to establish a new workflow that represents an ideal model according to the expected attributions for each role.

2.6.3 Foundations of the framework at the project level

A summary of the main responsibilities associated with the roles of Product Owner (PO), Tech Lead and Developer (Dev), mapped on the basis of the frameworks studied, is shown in Table 16 below. This mapping was carried out with a special focus on the role of Product Owner, which is at the heart of the proposed framework. By clearly segregating the responsibilities of the PO from the other roles, this structure will serve as the basis for defining the phases and processes that will make up the framework. The intention is to ensure that the PO's activities are well defined, actionable and properly integrated into the broader product development lifecycle.

Table 16 : Main responsibilities associated with the roles

Roles	Responsibilities	Frameworks
Product Owner	Manage backlog and prioritize value	Scrum, SAFe, LeSS, DA, DSDM
	Represent stakeholders and align vision	DSDM, Lean Inception, Scrum
	Continuous delivery of value	Lean, SAFe, DA
	Product and portfolio strategy	SAFe, DA

TechLead	Ensuring good practices and technical quality	Scrum, Kanban, Lean, SAFe
	Supporting architectural decisions	SAFe, DA, DSDM
	Facilitating technical collaboration	DA, Kanban
	Supporting integration and scaling solutions	SAFe, DA
Developers	Implement functionalities with quality	All (inference)
	Participate in ceremonies and feedback	Scrum, SAFe, LeSS
	Apply good technical practices	Lean, DA, SAFe
	Collaborate with PO and Tech Lead	LeSS, DA, Scrum

Based on the responsibilities identified for each role, we organized the data into structural components that underpin agile product development. As shown in Table 17, these components group recurring practices observed in the analyzed frameworks and interview responses. Each component connects role-specific responsibilities to a practical function within agile delivery, reflecting both theoretical foundations and real-world experience.

The 'Function' column shows the functional role of the structural component within agile delivery - for example, whether it supports strategic alignment, technical execution or continuous delivery.

Table 17 : Structural components and justifications with framework base

Role Responsability	Structural Component	Function	Frameworks Base
PO	Backlog management and prioritization	Support for continuous prioritization with a focus on value.	Scrum, SAFe, LeSS
Tech Lead	Technical governance and quality	Align architecture, technical practices and clean code	DA, SAFe, Scrum
Dev	Execution and continuous delivery	Ensure delivery flow and rapid feedback	Kanban, Lean, DA
PO	Integration with stakeholders	Continuous communication with the business and clients	DSDM, Lean Inception

All (PO, Tech Lead, Dev)	Agile collaboration and feedback	Work pace based on continuous improvement	Scrum, LeSS, DA
PO	Strategic product vision	Multi-level product scope and portfolio	SAFe, DA
Tech Lead	Evolutionary and sustainable architecture	Technical structure adaptable over time	Lean, DA, SAFe

In order to explicitly connect the role-based responsibilities (Table 16) with the structural components that support agile delivery (Table 17), Table 18 presents a direct mapping between the two. This mapping clarifies how each responsibility contributes to a broader functional structure, highlighting how individual actions reinforce collaborative and systemic agility in product development.

Table 18 - Relationship between responsibilities (Table 16) and structural components (Table 17)

Role	Responsibility (Table 16)	Mapped Structural Component (Table 17)	Connection Explanation
PO	Manage backlog and prioritize value	Backlog management and prioritization	PO responsibility feeds the continuous prioritization structure to maximize business value
PO	Represent stakeholders and align vision	Integration with stakeholders	Active role in translating needs into backlog items via communication loops
PO	Continuous delivery of value	Execution and continuous delivery	The PO ensures that what's delivered actually contributes to value, supporting the delivery rhythm
PO	Product and portfolio strategy	Strategic product vision	Ensures alignment between backlog content and long-term strategic goals
Tech Lead	Ensuring good practices and technical quality	Technical governance and quality	Guides teams to follow architectural and coding standards for quality assurance
Tech Lead	Supporting architectural decisions	Evolutionary and sustainable architecture	Helps define adaptable structures for current and future needs
Tech Lead	Facilitating technical collaboration	Agile collaboration and feedback	Enables smoother communication among developers and other technical actors

Tech Lead	Supporting integration and scaling solutions	Technical governance and quality + Strategic product vision	Ensures scalable, maintainable technical foundation aligned with business direction
Dev	Implement functionalities with quality	Execution and continuous delivery	Converts backlog items into working software with good practices
Dev	Participate in ceremonies and feedback	Agile collaboration and feedback	Engages in feedback loops and improves incrementally with the team
Dev	Apply good technical practices	Technical governance and quality	Delivers code aligned with standards defined by Tech Leads
Dev	Collaborate with PO and Tech Lead	Agile collaboration and feedback	Supports cross-functional alignment and shared ownership

Backlog management and prioritization (Responsible: PO)

It is the practice of continuously maintaining and refining a prioritized list of work items (backlog), ensuring alignment with business objectives and a focus on delivering value.

- **Scrum** works with the Product Backlog, prioritized by the Product Owner based on value. Ceremonies such as Sprint Planning and Backlog Refinement organize this process.
- **SAFe** introduces the Program Backlog and uses WSJF (Weighted Shortest Job First) as a prioritization criterion, supported by the PI Planning event for strategic alignment.
- **LeSS** centralizes the single product backlog, with a Product Owner who prioritizes based on the impact on the customer, favoring integrated decisions between teams.

Technical governance and quality (Responsible: Tech Lead)

This involves defining and monitoring technical practices, quality standards and architectural guidelines.

- **DA** suggests the role of Architecture Owner, Technical Debt Management techniques and the use of Goal Diagrams to guide technical decisions.
- **SAFe** structures governance through the System Architect and the Architectural Runway concept, guaranteeing technical preparation for the solution's growth.
- **Scrum**, although not prescriptive, reinforces the importance of Definition of Done, as well as encouraging Code Reviews and practices such as TDD and test automation.

Execution and continuous delivery (Responsible: Dev)

This refers to the ability to implement, test and deliver functional software in a continuous and fluid manner, reducing cycle time and increasing the frequency of feedback.

- **Kanban** contributes with flow visualization (Kanban Board), WIP limits and metrics such as Lead Time and Throughput.
- **Lean** focuses on eliminating waste, promoting a continuous flow of value and incremental deliveries with short cycles.
- **DA** integrates DevOps practices, CI/CD pipelines and the Continuous Delivery lifecycle, adjusting practices to the team's reality.

Integration with stakeholders (Responsible: PO)

This deals with continuous communication with clients and business areas, to ensure alignment of expectations and decisions based on real perceived value.

- **DSDM** promotes strong stakeholder involvement through collaborative workshops and regular reviews with the client.

- **Lean** Inception proposes structured sessions - such as Product Objectives, Personas and MVP Canvas - that align everyone involved from the very beginning of product conception.

Agile collaboration and feedback (Shared responsibility: PO, Tech Lead, Dev)

Represents collaborative work, in short cycles, with a focus on continuous improvement, learning and constant adaptation of processes and the product.

- **Scrum** offers structure with Daily Scrum, Sprint Review and Retrospective, creating short cycles of inspection and adaptation.
- **LeSS** extends these practices with Team Retrospectives and General Product Retrospectives, promoting collective learning.
- **DA** proposes various collaboration practices, such as Lean Coffee, Kaizen Events and Feedback Loops, leaving the choice up to the team, depending on the context.

Strategic product vision (Responsible: PO)

This involves defining the product's long-term vision, aligning strategic business objectives with the team's roadmap and backlog.

- **SAFe** offers mechanisms such as Portfolio Kanban, Epic Hypotheses and Solution Intent to connect strategic and operational levels.
- **DA** enables vision and portfolio management with a focus on adaptability, using strategic goals, capability maps and continuous evaluation of the product's direction.

Evolutionary and sustainable architecture (Responsible: Tech Lead)

This is the ability to keep the system's architecture adaptable over time, responding to business changes without compromising technical sustainability.

- **Lean** advocates simple, evolving architectures, based on real needs and with continuous delivery.

- **DA** introduces practices such as Intentional Architecture and the role of Architecture Owner, promoting a balance between emerging and planned architecture.
- **SAFe** adopts the concept of Architectural Runway, allowing technical evolution in line with delivery planning and organizational growth.

In order to operationalize the structural components presented, it is important to consider the artifacts, templates and tools that the frameworks make available to facilitate their practical application. Below is a brief overview of these tools, now organized on the basis of the structural components defined above.

2.6.4 Templates and supporting tools for agile frameworks

The use of templates, artifacts, and practical tools facilitates the application of agile practices in the daily routine of teams. These resources contribute to standardization, decision-making, and communication between team members, especially in scaled or multidisciplinary contexts.

After identifying the structural components and their respective functions (Table 19), it is important to understand what practical support the agile frameworks offer for each of these components. The following table presents the templates and tools recommended or provided by each framework, organized according to the structural components they support. This structure ensures a clearer connection between the responsibilities mapped, the practices proposed by each framework, and the artifacts that enable their practical implementation.

Table 19: Templates and tools organized by structural component and framework

Structural Component	Frameworks	Templates and Tools
Backlog management and prioritization	Scrum	Product Backlog, Sprint Backlog; Tools: Jira, Trello, Azure DevOps
	SAFe	Portfolio Kanban, Program Backlog, Team Backlog (WSJF); PI Planning templates

	LeSS	Shared Product Backlog guidelines
	DA	Agile Lifecycle templates; Decision process diagrams
	DSDM	Product Requirements Document (PRD)
	Lean Inception	MVP Canvas, Feature Brainstorming tools (Miro, Mural)
Technical governance and quality	SAFe	Architectural Runway Canvas, Solution Intent Document
	DA	Architecture Owner Checklists, Technical Debt Logs, Governance Milestones
	Scrum	Definition of Done
	DSDM	Quality Log
Execution and continuous delivery	Kanban	Workflow models with WIP Limits, Cumulative Flow Diagram, Lead Time Charts; Tools: Kanbanize, Trello, Jira
	Lean	Value Stream Mapping (VSM), A3 Report; Tools: Lucidchart, Miro
	DA	Continuous Delivery Lifecycle templates, DevOps toolchain integration
Integration with stakeholders	DSDM	Business Case Template, Risk Log, Timebox Plan
	Lean Inception	Product Vision Canvas, Personas Canvas, User Journey; Tools: FigJam, Miro, Mural
Agile collaboration and feedback	Scrum	Sprint Review, Retrospective; Tools: Jira, Confluence
	LeSS	Large-scale Sprint Review models, General Product Retrospectives
	DA	Kaizen Templates, Lean Coffee formats, Feedback Loop diagrams
Strategic product vision	SAFe	Lean Business Case, Epic Hypothesis Statement, Portfolio Kanban
	DA	Capability Maps, Strategic Goals Templates
	Lean Inception	Product Vision Canvas

Evolutionary and sustainable architecture	SAFe	Architectural Runway Canvas
	DA	Intentional Architecture templates, Architecture Owner role artifacts
	Lean	VSM and A3 Report adapted for architectural mapping

The organization of templates and tools by structural component reinforces the practical in this work. By mapping these resources to specific responsibilities and functions, it becomes easier for Product Owners and agile teams to adopt practices that are aligned with their context and goals. This structured view also supports decision-making when selecting and adapting frameworks, helping teams maintain consistency, quality, and agility as they scale or evolve their product development approach.

2.7. FELIX - The Framework

The FELIX Framework (Figure 6) was developed in response to a recurring challenge identified during an internship at BTG Pactual: the excessive dependence of Product Owners (POs) on Tech Leaders for defining solution architecture. This centralization of technical decision-making creates bottlenecks, compromises deadlines, reduces delivery predictability, and hinders alignment between business needs and technical feasibility.

With the development of this framework, we aim to improve project dynamics from their inception through their evolution within the team, with an emphasis on the active role of the Product Owner. The objective is to strengthen their position as the main party responsible for business information and decision-making, promoting greater autonomy and leadership in driving initiatives.

To this end, we propose a new project intake structure composed of a continuous flow divided into eight stages. The following section presents an overview of each of these stages, highlighting their objectives and contributions to the overall process. Notably, stage 5 constitutes the investigative core of this study, where the PO receives direct support to reflect on the architecture of the proposed solution. This stage is central to the framework's proposition, as it seeks to reduce exclusive reliance on the Tech Leader and promote better alignment between the project's technical and strategic dimensions.

In the following chapters, this work will further detail the flow using BPMN notation, demonstrating how projects are incorporated into the production pipeline — from initial documentation, through the project development lifecycle, to final delivery. It is important to note that the focus of this analysis is on the role of the Product Owner throughout the process, rather than the technical activities carried out by developers or Tech Leaders.

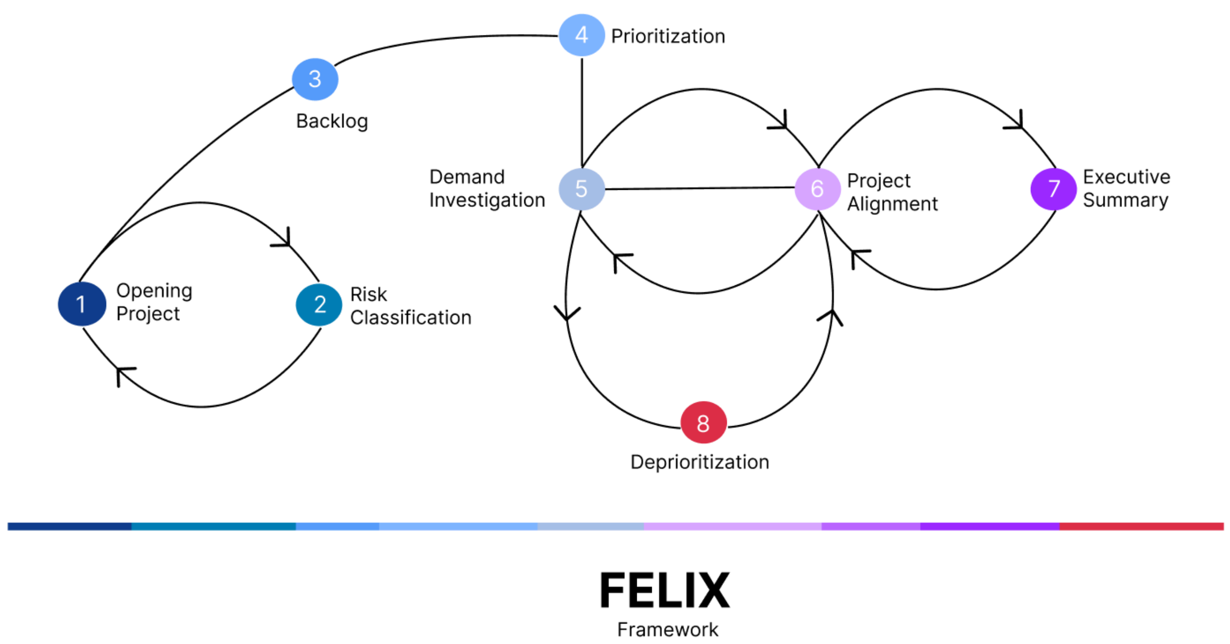
In this chapter, with the aim of addressing the identified challenges, the framework is presented as a structured model that provides greater autonomy to POs, guiding both the maturation of requirements and the construction of solution architecture in a strategic and collaborative manner. By enabling the PO's active participation from the early stages, FELIX fosters more effective communication among stakeholders, anticipates technical risks, and contributes to greater predictability within the delivery cycle.

The framework's eight stages are organized sequentially and continuously, facilitating the monitoring and management of requirements from the moment they are introduced until they are prioritized or deprioritized. A high-level explanation of each stage is presented in the following section as a foundation for understanding the complete flow, which will later be formalized through the BPMN model.

It is important to note that the FELIX Framework adopts two complementary perspectives for requirements management: one focused on monitoring a single project at a time, and another aimed at overseeing multiple projects simultaneously. Some stages — such as Project Opening, Risk Classification, Demand Investigation, and Project Alignment — are designed to analyze each project individually, allowing for a more in-depth and context-specific evaluation of the proposed solution. On the other hand, stages like Backlog, Prioritization, Executive Summary, and Deprioritization consider the broader set of ongoing projects, with the goal of ensuring strategic alignment, optimizing resource allocation, and supporting more effective decision-making.

This dual perspective reflects a clear concern with both the individual dynamics of each initiative and the integrated view of the portfolio — a balance that is essential for managing complexity and maximizing value. The proposal to implement a standardized template for capturing project information across all initiatives reinforces this mindset. After all, such a template would not be necessary if the intention were to look at each project in isolation. By standardizing how information is recorded, the framework builds the foundation for consistent evaluation, comparison, and integration of projects — an approach inspired by the TOGAF methodology, which also advocates for managing requirements from both individual and collective perspectives.

Figure 6 – High-level visualization of the FELIX Framework



Source: Prepared by the author

Stages of the FELIX Framework

- **Project Opening:** Creation and formalization of the Project Initiation Document request

- **Risk Classification:** Mandatory step within the company for the categorization of demands by classification (Efficiency Gain in Hours, Regulatory Risk, and Operational Risk), considering impact and urgency.

- **Backlog:** Organizing all the demands received into a rough backlog, without any major prioritization, serving as an initial repository of opportunities.

- **Prioritization:** Defining which projects or demands will go through the maturation cycle based on value, operational viability and strategic alignment.

- **Demand Investigation:** Scheduling meetings with the requester to detail and validate the business rules, objectives and requirements of the problem.

- **Project Alignment:** Meeting between PO, Tech Lead and Requestor to map project dependencies and discuss priorities.

- **Executive Summary:** Record of current status and next steps in a standardized format (template + e-mail), ensuring transparency with requesters.

- **Deprioritization:** Reassessment of previously prioritized demands that, due to changes in context, value, or feasibility, should be paused, reclassified, or removed from the maturation cycle.

FELIX was conceived based on interviews with Product Owners and Tech Leaders, as well as comparative studies with practices adopted in other companies and market-standard best practices, resulting in a structure that can be applied in an adaptable way to different organizational contexts. Ultimately, it aims to increase the quality of strategic decisions at the start of projects, promoting a more collaborative, efficient environment aligned with the product and technology vision.

Each of the stages of the FELIX Framework will be explained in detail below. For each stage, its objective will be presented, as well as how it can be applied in practice and the templates that make it up, offering a clear and operational view of the process. The aim is to ensure that POs and others involved have a structured and replicable guide to support the maturing of demands and the collaborative construction of solutions.

2.7.1 Stage 1 - Opening Project

This project opening form (Figure 7) is the first stage in the process of receiving new requests at BTG. It aims to centralize, standardize and qualify requests before they enter the Discovery pipeline, facilitating the prioritization and correct targeting of initiatives.

The template gathers information to assess the context of the request, including:

- Name of the area and project, channel of origin (optional) and description of the problem;
- Details of the demand, with open fields to explain what you want to solve or build;
- Strategic information such as: priority assigned (0 to 10), desired delivery quarter [1], and whether there is a need for UX involvement or APP development;
- Business Driver classifications (e.g. customer base, revenue, cost reduction) and Demand Type (technical activity, new product, feature, improvement)[2];
- Space for attachments and complementary links, such as supporting documents, wireframes or presentations.

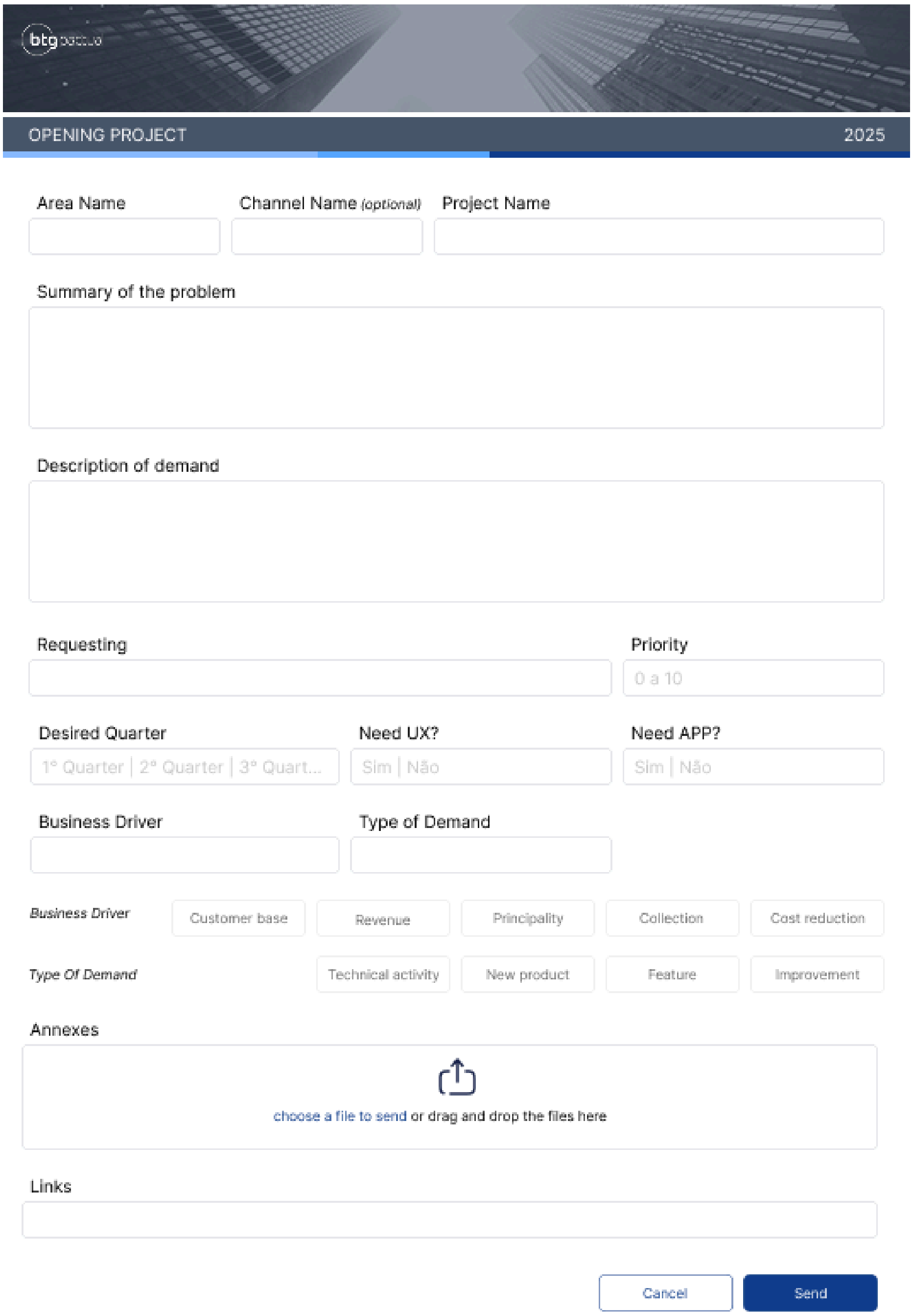
[1] Although these fields — such as assigned priority, delivery quarter, and the need for UX or APP involvement — are currently used mainly operationally to categorize requests, they also provide important input for leadership. They allow for evaluating the alignment of initiatives with broader strategic objectives and can serve as a basis for future analysis and decision-making. For now, we focus on the operational application of the framework, leaving the strategic exploration for subsequent stages.
[2] Similarly to Note 1, these Business Driver classifications are currently used operationally to categorize projects. However, they also provide relevant input for leadership, enabling identification of which strategic drivers are being addressed. The exploration of the strategic potential of this data will be considered in future stages, after the operational use of the framework has been consolidated.

This form can be easily adapted for Monday.com, the tool currently used by BTG, using Monday's Forms functionality, which automatically converts each submission into a new trackable item, already sorted into columns by priority, business driver, type of demand and attachments.

Other tools that also allow similar use:

- Typeform or Google Forms (with integration via Zapier/Integromat to feed spreadsheets or boards);
- Airtable Forms, which offers a modern look and internal database control;
- Jotform, with a corporate look and integration with workflow tools;
- Notion Forms (via third-party integration), for organizations that already use Notion.

Figure 7 – Opening Project Template



Source: Prepared by the author

Table 20 below explains the fields in the Opening Project Template presented above, detailing the meaning and application of each one to make it easier to use.

Table 20 - Opening Project Template fields

Field	What it is	Purpose
Area Name	Name of the business area responsible for the project (e.g., Investments, Partners).	Identify the business domain submitting the request.
Channel Name *(optional)*	Name of the related channel (e.g., PME, Digital, FIDC).	Specify the channel impacted by or originating the request.
Project Name	Identifier for the project or initiative.	Facilitate internal tracking and communication.
Summary of the problem	Brief description of the problem identified.	Provide quick context about the need or issue.
Description of demand	Detailed explanation of the need or proposed solution.	Clarify what is being requested for proper analysis and development.
Requesting	Name of the person or area submitting the request.	Identify the request owner or main contact.
Priority	Urgency level (scale from 0 to 10).	Support prioritization based on business perception of urgency.
Desired Quarter	Preferred quarter for delivery or project start.	Align business expectations with development planning.
Need UX?	Indicates whether UX (user experience) support is required.	Ensure UX team involvement when needed.
Need APP?	Indicates whether mobile app development or changes are required.	Direct the request to the appropriate mobile development teams.
Business Driver	Strategic justification for the demand.	Show the business rationale behind the request (e.g., Revenue, Customer base, Cost reduction).
Type of Demand	Classification of the type of request.	Indicate if it’s a technical activity, feature, improvement, or new product.
Annexes	Field for attaching additional files.	Support understanding with complementary documentation.
Links	Field for adding relevant URLs.	Reference documents, dashboards, presentations, etc.

How to apply the template

This form should be shared with all requesting areas (business, operations, product, etc.) as the official channel for new requests. Filling it in ensures that:

- All requests arrive with standardized minimum information, facilitating triage and prioritization;
- Technology and product teams can assess impact, effort and preliminary risks before starting any work;
- The input flow is documented and transparent, reducing rework, misalignment and noise with requesting areas.

Once submitted, the information can feed directly into a backlog board on Monday, starting the Discovery flow with more context and predictability.

Report Components and Agile Inspirations

The project opening form was inspired by frameworks aimed at initial demand management and scope definition. The structure of the form is based on the Product Requirements Document (PRD) from the DSDM framework, which guides the initial formalization of business needs and alignment with strategic objectives.

In addition, elements such as problem description, priority, business drivers and type of demand were organized following the guidelines of Lean Inception's Product Vision Canvas to support a clear and objective vision of the initiative's purpose from the outset.

This combination allows essential information to be captured in a standardized way, promoting alignment between requesting areas, agile teams and stakeholders, as well as facilitating prioritization in the backlog and decision-making in governance forums.

2.7.2 Stage 2 - Risk Classification

BTG Pactual already has a consolidated structure for classifying project risks, covering financial, operational, legal/regulatory and reputational aspects. In order to maintain consistency with the company's standards and ensure alignment with internal processes, we chose to reuse this same risk classification in the project status report template we propose (Figure 8 e 9). The only adaptation made was to include this template within the visual monitoring format, ensuring uniformity and ease of understanding for all the stakeholders involved.

Figure 8 – Risk Classification Template

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RISK CLASSIFICATION 2025

Financial

Name	Comment
LOW	• A low risk is considered when the financial value is less than R$100,000
MEDIUM	• A medium risk is considered when the financial value is between R$100,000 and R$400,000.
HIGH	• Considered a high risk when the financial value exceeds R$400,000

Operational

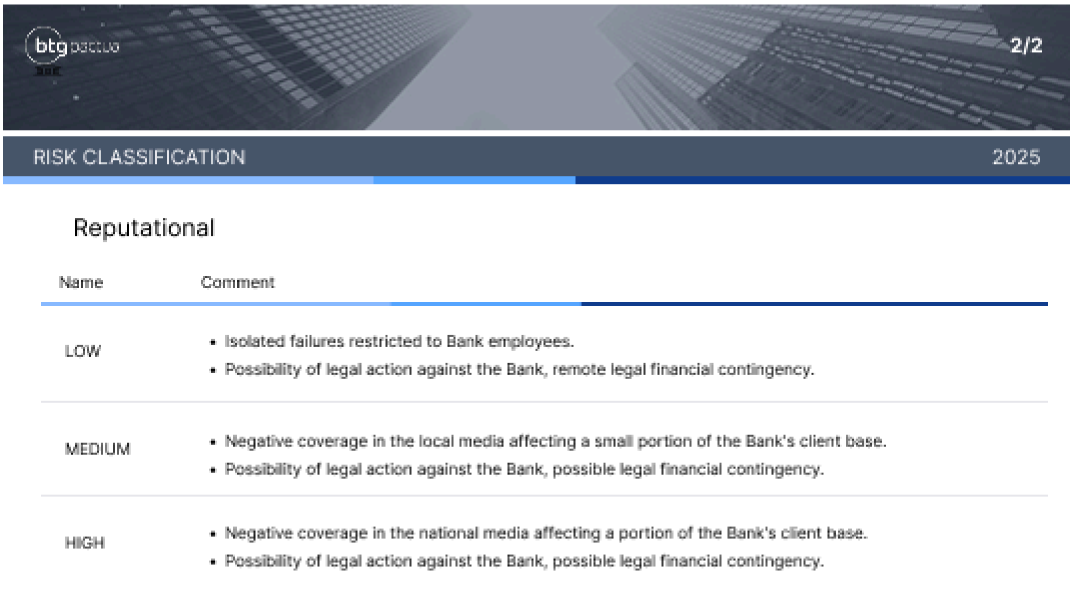
Name	Comment
LOW	• Error or unavailability of the process generates a slight impact on business processes with zero impact on the achievement of the area's objectives.
MEDIUM	• Error or unavailability of the process generates additional work, impacting one or more areas. • Moderate impact on the work schedule, generating rescheduling to normalize the flow.
HIGH	• Error or unavailability of the process generates additional work, with a high investment of working hours, impacting one or more areas. • Significant delay in the execution of tasks due to the accumulation of workload. The situation should be formalized or third parties asked to help normalize the flow.

Legal/Regulatory

Name	Comment
LOW	• No regulatory impact or action by any regulatory body. • Breach of contract at no cost.
MEDIUM	• Informal enforcement action, no warnings or supervisory procedures. • Breach of contract with some costs or censure.
HIGH	• Regulatory body with formal warnings, possible fines and enforcement action. • Breach of contract resulting in possible major litigation, fines or censure.

Source: Prepared by the author

Figure 9 – Risk Classification Template



Name	Comment
LOW	• Isolated failures restricted to Bank employees. • Possibility of legal action against the Bank, remote legal financial contingency.
MEDIUM	• Negative coverage in the local media affecting a small portion of the Bank's client base. • Possibility of legal action against the Bank, possible legal financial contingency.
HIGH	• Negative coverage in the national media affecting a portion of the Bank's client base. • Possibility of legal action against the Bank, possible legal financial contingency.

Source: Prepared by the author

2.7.3 Stage 3 - Backlog

This Backlog document (Figure 10) is the continuation of the flow started by the project opening form. It organizes all the demands received, consolidating the essential information for planning and prioritization by quarter.

The structure is divided into quarters (1st to 4th), and each line represents a new validated demand, containing the main fields captured in the previous form:

- Source information: Area, Channel, Project Name;
- Context of the problem and description of the desired solution;
- Indicators such as: priority, need for UX or APP, business driver and type of demand;
- Operational fields for the technical and product teams: attachments, expected timeline, current status, expected efficiency (in hours/month) and expected delivery date.

This backlog allows the product and technology team to:

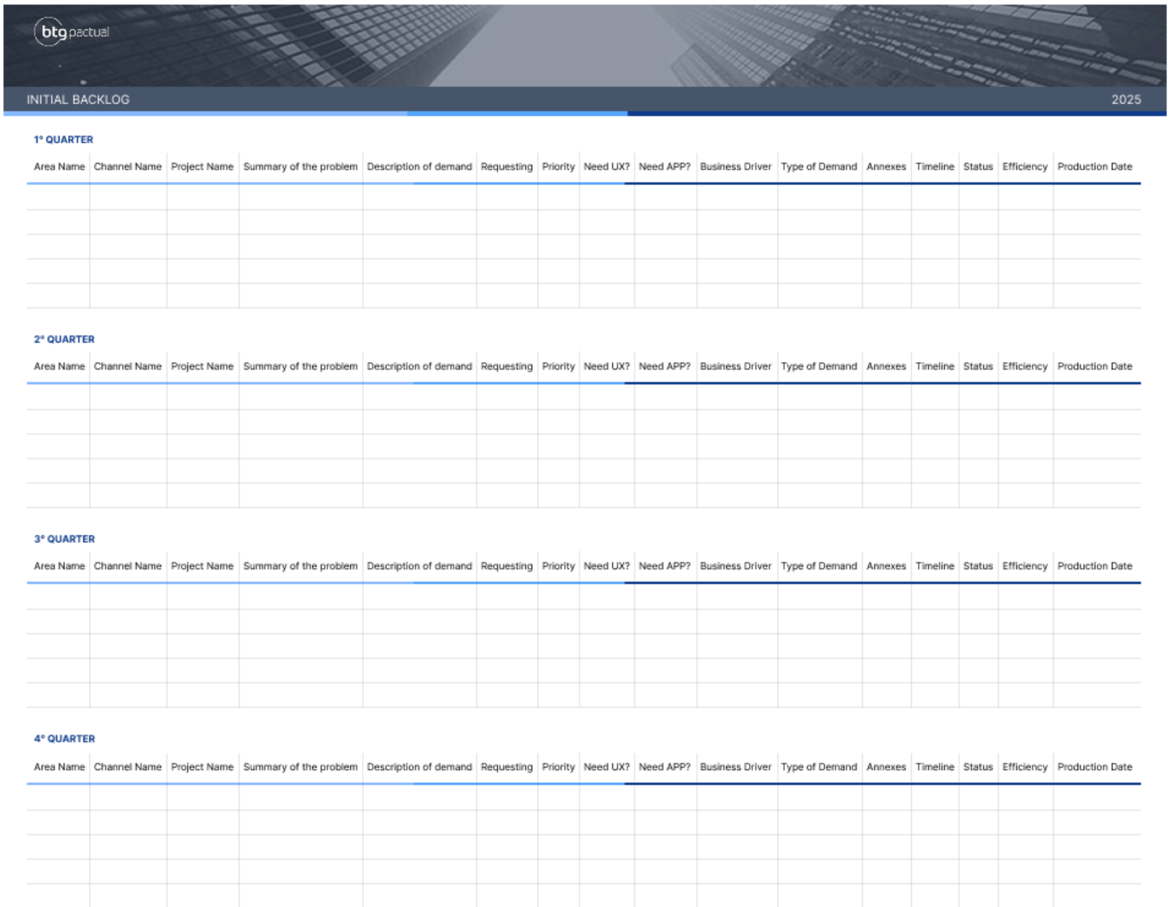
- Centralize and order the queue of initiatives;
- View the volume of demands by quarter;
- Cross-reference decision factors such as effort, impact, risk and dependencies.

At BTG Pactual, this backlog can be adapted directly in Monday.com, by means of a board with a table view, grouped by quarter, with columns representing the same fields listed in the template. Monday's automation also allows the demand to be automatically added to this board when the opening form is submitted, with all the information filled in and risk ratings applied.

Other tools that allow similar operation:

- Airtable: enables table, kanban and calendar views, as well as dynamic filters by quarter;
- Smartsheet: focused on schedules and executive portfolio management;
- Google Sheets + Forms, with automation scripts;
- Notion Database, with a user-friendly look and customization by quarter and status.

Figure 10 – Backlog Template



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INITIAL BACKLOG 2025

1º QUARTER

Area Name	Channel Name	Project Name	Summary of the problem	Description of demand	Requesting	Priority	Need UX?	Need APP?	Business Driver	Type of Demand	Annexes	Timeline	Status	Efficiency	Production Date

2º QUARTER

Area Name	Channel Name	Project Name	Summary of the problem	Description of demand	Requesting	Priority	Need UX?	Need APP?	Business Driver	Type of Demand	Annexes	Timeline	Status	Efficiency	Production Date

3º QUARTER

Area Name	Channel Name	Project Name	Summary of the problem	Description of demand	Requesting	Priority	Need UX?	Need APP?	Business Driver	Type of Demand	Annexes	Timeline	Status	Efficiency	Production Date

4º QUARTER

Area Name	Channel Name	Project Name	Summary of the problem	Description of demand	Requesting	Priority	Need UX?	Need APP?	Business Driver	Type of Demand	Annexes	Timeline	Status	Efficiency	Production Date

Source: Prepared by the author

How to apply the template

The new demand should be automatically or manually directed to this backlog. The aim is to enable product managers and analysts to have:

- A complete and orderly view of what has been requested and validated;
- Ease of planning deliveries by quarter, based on effort, value generated, risks and team capacity;
- Support material for prioritization meetings with stakeholders and technical leaders.

Using this backlog improves governance over the portfolio, helps identify bottlenecks and future overloads and strengthens transparency between the requesting areas and the execution teams.

Report components and agile inspirations

1. Segmentation by quarter (1st to 4th Quarter)

The segmentation by quarters (1st to 4th Quarter) is inspired by SAFe, Disciplined Agile (DA) and LeSS. It uses templates such as SAFe's PI Planning and Portfolio Kanban, as well as DA's Agile Lifecycle Templates.

The justification for this approach lies in the division of the backlog into quarters, which reinforces alignment with SAFe's incremental planning practices, organizing deliveries into quarterly Program Increments. This segmentation also provides a strategic view in short cycles, in line with the DA approach. In the context of LeSS, the division by time cycles facilitates coordinated and synchronized reviews between multiple teams.

2. Description fields and demand context

The demand description and context fields - such as “Summary of the problem”, “Description of demand”, “Project Name”, “Channel Name” and “Requesting” - are inspired by Scrum, LeSS and Lean Inception. They use templates such as Scrum's Product Backlog, LeSS's shared backlog guidelines and tools such as Lean Inception's Product Vision Canvas and MVP Canvas.

The justification for this approach lies in providing a rich context for understanding demand, a common practice in the Scrum Product Backlog and essential for shared backlog management in LeSS. In addition, the focus on aspects such as problem, channel and requester reflects Lean Inception's user- and stakeholder-centered approach, which uses tools such as User Journey and Personas Canvas to deepen the understanding of needs.

3. Fields of prioritization and technical need

The prioritization and technical requirement fields - such as “Priority”, “Need UX?”, “Need APP?”, “Business Driver” and “Type of Demand” - are inspired by SAFe, Disciplined Agile (DA) and DSDM. They use templates such as SAFe's WSJF, DA's decision diagrams and DSDM's Product Requirements Document (PRD).

The justification for this structure lies in the possibility of carrying out an organized assessment of business value and technical feasibility. Fields such as “Need UX?” and “Need APP?” contribute to architectural and functional scope decisions already in the initial phases, a practice present in DA checklists. The classification of the type of demand, in turn, is common in documents such as the PRD in DSDM, offering a clear and categorized view of the project's needs.

4. Governance, Traceability and Metrics

The fields related to governance, traceability and metrics - such as “Annexes”, “Timeline”, “Status”, “Efficiency” and “Production Date” - are inspired by Disciplined Agile (DA), Lean and SAFe. They use templates such as Lean's Value Stream Mapping, DA's Continuous Delivery Lifecycle and SAFe's Portfolio Kanban.

The justification for this approach lies in the presence of fields such as Timeline and Production Date, which reinforce traceability throughout the demand lifecycle, a practice encouraged by the DA templates. The Efficiency field, representing possible time savings, is strongly inspired by Lean Value Stream Mapping. The status and attachments fields contribute to technical governance and proper documentation, in line with the structured practices promoted by SAFe.

5. Visual organization in a structured table

The visual organization in a structured table is inspired by SAFe, DSDM and Disciplined Agile (DA). It uses templates such as SAFe's Portfolio Kanban, DSDM's Risk Log and DA's architecture governance artifacts.

The justification for this approach lies in the tabular layout with multiple fields, which allows future integration with platforms such as Monday.com and reflects a governance-oriented corporate backlog model. This structure is highly compatible with the risk, architecture and governance artifacts provided for in the DA and DSDM, as well as offering the potential to apply filters, automations and dashboard visualizations.

2.7.4 Stage 4 - Priorization

The stage of defining which projects or demands will move on to the maturity cycle, based on criteria such as value generated, operational viability, and strategic alignment, is informed by concepts present in frameworks such as SAFe, DA, and DSDM, which emphasize structured prioritization and business impact analysis. Elements like the WSJF model (SAFe), decision diagrams (DA), and business case templates (DSDM) provide objective methods for prioritization. However, in practice at BTG, decisions on what advances or is deprioritized rely heavily on the knowledge and experience of the business teams. While these individuals naturally apply concepts aligned with these frameworks, the process remains largely intuitive and discussion-based rather than formally structured or systematically documented.

Despite this conceptual basis, this phase doesn't have a mandatory template - each Product Owner can adapt it according to the reality of their organization. At BTG Pactual, for example, this prioritization takes place directly in the backlog already set up within Monday.com, by filling in the tool's columns. Other platforms such as Jira, Trello, ClickUp and Azure DevOps also support backlog management and can be used according to the team's context. At the end of this stage, it is hoped to have a prioritized list of initiatives that will be matured, with clear visibility of the demands that bring the greatest value to the business and that are suitable for moving on to the discovery and technical planning phases.

2.7.5 Stage 5 - Demand Investigation

The Demand Investigation stage (Figures 11 and 12) marks the beginning of the refinement of the demand after its inclusion in the backlog. This is when the Product

Owner (PO) gathers and structures all the information needed to enable the request to be developed. The aim is to deepen the understanding of the problem or need, detailing the business context, the rules involved, the dependencies identified and any technical or functional doubts.

During this phase, the PO conducts alignments with the desk team to clarify outstanding points, validate assumptions and identify risks. They are also responsible for designing the solution's architecture - which comprises the complete flow that the system user will have to follow in order to execute the demand. This flow serves as the basis for the solution design and development stage, ensuring alignment between the business vision and the technical implementation.

The structure of the template is divided into six main sections. Currently, however, none of these sections is formally established or used in a standardized way in the company. Each PO is free to conduct the investigation as they see fit, which varies greatly depending on the area involved. The purpose of these sections is precisely to help standardize and record information about demands, promoting a more consistent, traceable and aligned process between teams.

Solution Details

This section is dedicated to documenting general information about the request. The "Demand name" field identifies the title of the request, while "Request date" records the time of the request. The "Type of request" field classifies the nature of the request, which may be a new product, functionality, improvement or technical activity.

The description of the current situation is recorded in the "Problem description" field, which should objectively present the need faced. The proposed solution is detailed in "Solution description", representing the initial vision of what is expected to be developed.

Contextual information, such as "Requestor" and "Requesting area", identifies the origin of the demand within the organization. Strategic categorization is done through the "Classification" field, which allows the demand to be related to topics such as customer base, revenue, principality, collection or cost reduction.

Operational aspects are also considered: the "Frequency of manual intervention" field describes whether the activity is currently carried out manually, and how often. The "Efficiency gain" field estimates the savings in time or resources that could be achieved. Finally, the "Monday link" field centralizes access to the demand card in the management tool.

Process Information

This section focuses on the technical, operational and regulatory aspects that may impact the development and delivery of the demand. The "Operational risk" field assesses the potential impact on the operation, with a scale of 0 to 3. The same logic applies to the "Regulatory risk" field, which considers possible legal or regulatory implications.

The urgency of the demand is recorded under "Priority of demand", with a score from 0 to 10. The "Systems involved" and "Teams involved" fields make it possible to map technological dependencies and support areas. The "IT Manager" field defines the person technically responsible for the initiative, while "Involves external client?" indicates whether the solution will have a direct impact on the organization's clients.

Overview

The purpose of the "Overview" section is to provide an introductory summary of the request. In this field, you should describe the current situation faced by users, the activities carried out and how the proposed solution will contribute to carrying out these tasks.

When constructing this overview, it is important to consider the business domain, the operating rules, the services and tools used, as well as the desired characteristics of the solution and the expected performance.

Highlights

This section concentrates the most relevant elements identified in the investigation. The "Key information" field should compile the most relevant data collected during the analysis. The "Key opportunities" field is used to highlight the potential gains associated with the delivery, such as productivity improvements, strategic impact, cost reduction or increased revenue.

Points of Attention

This section is dedicated to identifying critical aspects or those that require special care. The "Aspects with the greatest impact" field should list the points that pose the greatest risk or represent decisive factors for the project's success. The "Dependence on validation" field informs you of any approval or prior analysis requirements that may affect the progress of the initiative.

Risks and Problems

The final section of the template aims to record risks and obstacles that have already been identified. The "Main risks" field describes potential threats to the project, classifying their severity or impact. The "Main problems" field should point out technical, operational or organizational limitations that could compromise the execution of the demand.

Figure 11 – Demand Investigation Template

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☐ Solution details

Questions	Description
Name of demand	
Date of request	
Type of demand	New Product, Feature, Improvement or Technical Activity
Description of the problem	
Solution description	
Requesting	
Requesting Area	
Classification	Customer base, Revenue, Principality, Collection or Cost Reduction
Frequency of manual intervention	
Efficiency Gain	
Monday link	

☐ Process Information

Questions	Description
Operational Risk | Impact	0-NA | 1-Low | 2-Medium | 3-High
Regulatory Risk | Impact	0-NA | 1-Low | 2-Medium | 3-High
Priority of demand	0 to 10
Systems Involved	
Teams Involved	
IT Manager	
Does it involve an external client?	

Source: Prepared by the author

Figure 12 – Demand Investigation Template

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☐ Overview

In this specification stage, an introductory text should be drawn up explaining what the request is about, what tasks users currently perform and how a new system can contribute to carrying out these activities.

During the information gathering process, it is important to understand the business domain and rules, the activities involved, the services and tools used, as well as the desired characteristics of the solution and the expected performance.

☐ Highlights

Questions	Description
Main information	
Main opportunities	

☐ Points of attention

Questions	Description
Points with the greatest impact	
Dependence on validation	

☐ Risks and problems

Questions	Description
Main risks | Impact	
Main problems | Impact	

Source: Prepared by the author

The Demand Investigation stage is strongly inspired by good practices from Lean Inception, Design Thinking and the continuous refinement of the backlog in Scrum. It is also aligned with the Discovery concept of the Dual-Track Agile model, where the discovery of value precedes the execution of the delivery.

To ensure that the investigation is both comprehensive and adaptable, the template draws on a blend of methodologies that emphasize collaboration, clarity and early risk assessment. Each of these frameworks contributes specific strengths that shape the structure and purpose of this stage:

- **Lean Inception and Design Thinking:** the modular structure and open questions favor collaborative construction and alignment between different visions.
- **Scrum (Backlog Refinement):** promotes constant conversations and iterative clarification of requirements.
- **Dual-Track Agile:** reinforces the separation between discovery and execution, prioritizing the validation of value before the prioritization of effort.
- **Disciplined Agile Delivery (DAD) and SAFe:** provide guidelines for analyzing risks and regulatory impacts at the earliest stages.

How to apply the template

The Demand Investigation template should be used as soon as a request has been prioritized for analysis. It can be filled in during or immediately after the meeting with the requester, serving as formal documentation of the initial understanding of the demand.The answers to the questions below will compose the overview section of the template.

To ensure consistency in conversations, we recommend this basic guide to questions during meetings, which serves as a starting point for the investigation:

- Can you give me an overview of the project? What is the main objective of this initiative?
- What is the initial time estimate for implementation[3]?
- Are there any risks already identified related to the project?
- Which systems will be impacted?[4] Who is responsible for these systems?
- What is the expected result of delivering this project?
- In the event of failures or errors, how do you expect the system to react?
- What is the expected volume (e.g. number of users, transactions, accesses, etc.)?
- Are there any other projects being planned, developed or implemented that could impact or be impacted by this demand? If so, what are they and how are they related?

These questions are intended to guide the development of the architecture, ensuring that the essential information is gathered for a proper technical and strategic analysis. They act as a roadmap that directs the collection of relevant data, ensuring that all important aspects of the demand are considered from the outset, which contributes to more precise and effective planning.

Report Components and Agile Inspirations

The Demand Investigation template not only structures the analysis of new demands, but also integrates key principles from several agile methodologies to ensure a comprehensive and collaborative approach. Its format mirrors the logic of modular

[3] Although this field is currently used to record the initial implementation estimate operationally, it also has the potential to feed dashboards for tracking the project lifecycle, providing strategic visibility for leadership. At the current stage of the framework, the focus is on documenting the initial understanding of the demand; leveraging this data for strategic purposes may be considered in later phases.

[4] Currently, this information is recorded primarily operationally, identifying the impacted systems and their responsible parties. However, it also has the potential to support dashboards that provide strategic visibility of the interface between projects and systems, helping leadership monitor the project lifecycle. Leveraging this data for strategic purposes may be considered in later phases.

documentation seen in Lean Inception and Design Thinking, where information is captured incrementally and refined through ongoing dialogue.

The six-part structure helps facilitate traceability and transparency, ensuring that each dimension of the demand, from business motivation to operational risks, is addressed systematically. By incorporating backlog refinement practices from Scrum and the discovery track from Dual-Track Agile, the report fosters early validation of value and reduces uncertainty.

Additionally, the attention to risks, regulatory constraints and dependencies reflects influences from more structured frameworks like Disciplined Agile Delivery (DAD) and SAFe, enabling alignment with enterprise-level governance.

BPMN Guide for Demand Structuring

To make it easier to draw up BPMN diagrams and better structure demands, a practical guide has been created that brings together the main components, good practices and useful guidelines for mapping processes. This material serves as support for Product Owners, helping them to understand workflows in depth, to document processes clearly and objectively, and to align the teams involved. In addition, the guide promotes the standardization of diagrams and helps to avoid ambiguities, which is fundamental for effective communication and successful deliveries.

Using this guide enables POs to conduct their analyses and technical discussions with greater confidence and precision, facilitating dialogue with stakeholders, developers and other impacted areas. It thus contributes to building solutions that are more in line with business objectives and user needs.

Product Owners can access the complete guide via the link below, which will always be available as a reference for their analysis and technical discussions: BPMN Guide for Product Owners

2.7.6 Stage 6 - Project Alignment

The main purpose of the meeting between the Product Owner, Tech Lead and the requester is to map out the project's dependencies, align priorities and ensure a shared vision of the next steps. This practice is inspired by the SAFe, DA and Lean Inception frameworks, which encourage continuous alignment between technical and business stakeholders for informed decision-making. SAFe contributes with practices such as PI Planning and Solution Intent to ensure coherence between strategy and execution, while DA provides checklists and technical governance processes, and Lean Inception values collaborative mapping of vision and scope through visual dynamics and facilitated discussions.

At BTG Pactual, this meeting takes place via Microsoft Teams, the official platform used for internal communications. Other organizations should adapt this practice to the meeting environment they use, such as Google Meet, Zoom or Slack Huddles. There is no fixed template for this stage, as the focus is on discussion oriented towards the materials already completed - especially the backlog, and the technical refinement template that includes business rules and the solution's architecture. At the end of this stage, all those involved are expected to be aligned on the context, technical needs and critical factors of the demand, allowing for more agile and integrated decisions in the following stages of the project's life cycle.

2.7.7 Stage 7 - Executive Summary

This weekly project status template (Figure 13) has been developed to provide a clear and objective executive view of the progress of ongoing initiatives. Its main purpose is to communicate progress, identify blockages and highlight relevant points to stakeholders at different levels, from executing areas to senior management.

The structure is divided into:

A monitoring panel with status indicators (Total, In Progress, Under Approval, Blocked, Completed), allowing a quick overview of the distribution and efficiency of projects (in hours/month).

- A table with details per project, including current status, schedule, estimated efficiency and delivery forecast.
- A highlights section, to record deliveries, important milestones or alerts.
- A space dedicated to blocking reasons, making it easier to deal with impediments proactively.

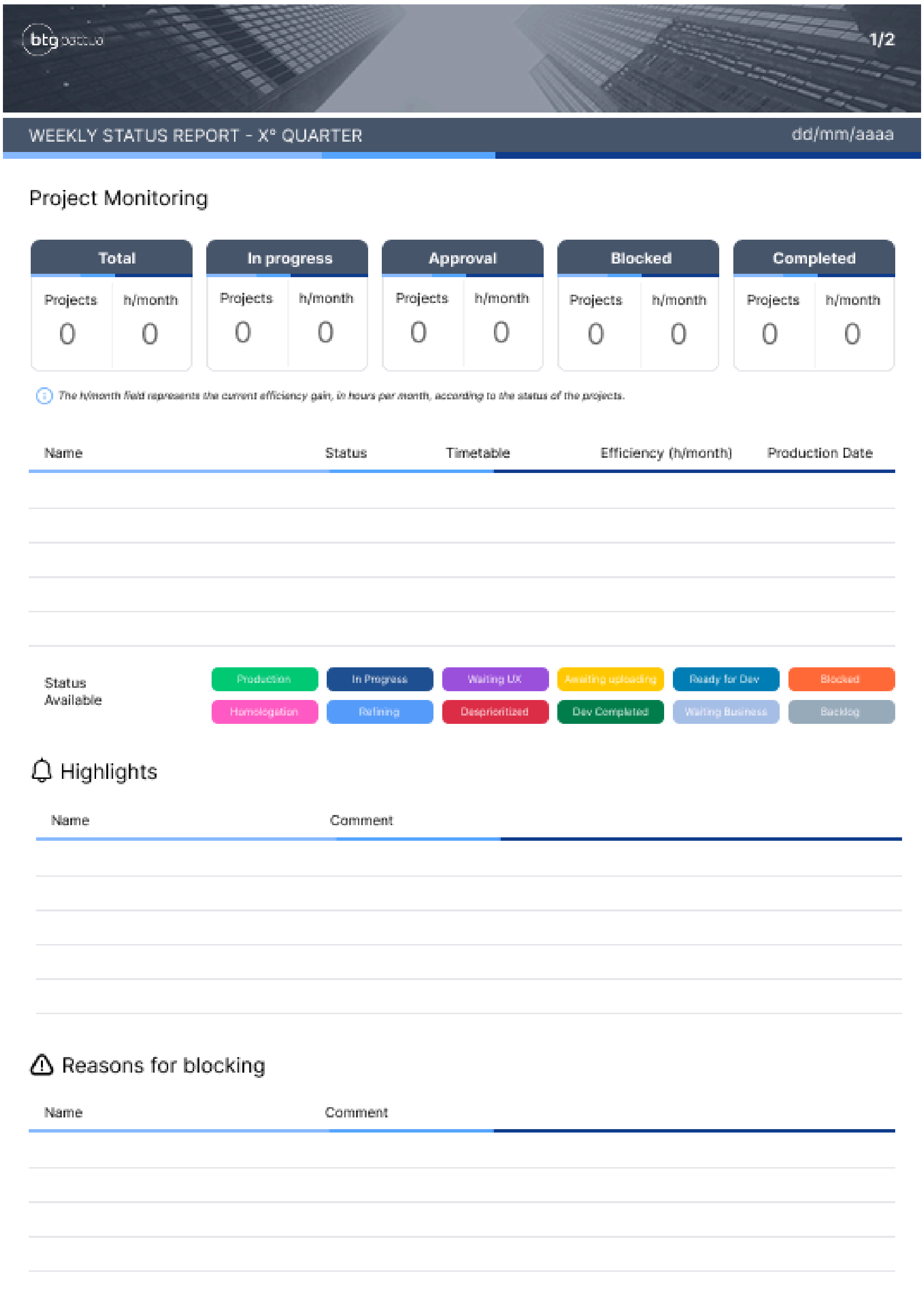
BTG Pactual currently uses Monday.com as its main tool for monitoring project portfolios. This is a paid platform that allows you to set up visual dashboards, apply automations and centralize the updating of information in an efficient and collaborative way.

If the use of Monday is not available to any team, the template can be easily adapted to other tools such as:

- Google Sheets or Excel Online, which are free and allow simple automation with formulas.
- Trello, which allows visual organization in lists, ideal for small teams.
- Jira + Confluence, already used by technology teams, great for agile teams, although it is also paid for.
- Notion, which in its free version already offers a good experience for small teams, with good visual presentation and sharing.

These alternatives make it possible to apply the template without compromising the central objective of executive visibility, even in teams that don't have access to full Monday licensing.

Figure 13 – Weekly Status Report Template



Source: Prepared by the author

How to apply the template

Filling in the template should be done weekly, preferably by the end of business on Fridays, based on the most recent updates from those responsible for each initiative. When filling in the template, you are expected to:

- Update the status according to the standardized labels;
- Indicate efficiency gains (in hours/month), considering the positive impacts of execution;
- Record milestones, deliveries or points of attention, contributing to alignment with leadership;
- Signaling existing blockages in a clear and objective way, to facilitate the support needed to resolve them.

With this, the report becomes a tool for increasing transparency, ensuring governance and speeding up decision-making among those involved in projects.

Report components and agile inspirations - Template 8 - Part I

1. Project monitoring by status

Project monitoring is inspired by practices found in Kanban, SAFe, Scrum and Disciplined Agile (DA). It uses templates such as SAFe's Portfolio Kanban, workflow models with Kanban's WIP Limits and Scrum's product or sprint backlogs.

The justification for this approach lies in the segmentation of projects into columns such as “In progress”, “Approval”, “Blocked” and “Completed”, which is a typical Kanban practice and also present in SAFe backlog boards (Program/Team Backlog). This visualization follows the logic of a continuous flow of work, making it possible to monitor the progress of projects and identify possible bottlenecks.

2. Efficiency metric (Efficiency h/month)

The Efficiency h/month metric is inspired by Disciplined Agile (DA) and Lean. It uses templates such as DA's Continuous Delivery Lifecycle and Lean's Value Stream Mapping.

The justification for this metric lies in the proposal for continuous improvement and value stream monitoring, as addressed by Lean through value stream mapping (VSM) and reinforced by continuous delivery models and the metrics suggested by DA.

3. Available statuses (colored legend)

The available statuses represented by a colored legend are inspired by Scrum, SAFe and LeSS. Elements such as the Scrum Product/Sprint Backlog, the LeSS Shared Backlog and the SAFe Program Backlog are used.

The justification for this approach lies in the detailed categorization of the status of tasks, such as “Waiting UX”, “Ready for Dev” or “Dev Completed”, a common practice in Scrum and SAFe environments, in which work is continually refined and reclassified according to its progress.

4. Highlights / Reasons for blocking sections

The Highlights / Reasons for Blocking sections are inspired by Scrum, Disciplined Agile (DA) and LeSS. They use templates such as Scrum's retrospectives and DA's Kaizen models and continuous feedback loops.

The rationale for this approach is to promote visibility and transparency about the main points of the week, reinforcing the focus on continuous feedback and the removal of impediments, as practiced in Scrum retrospectives and DA continuous improvement cycles.

5. Visual structure and regularity of the report

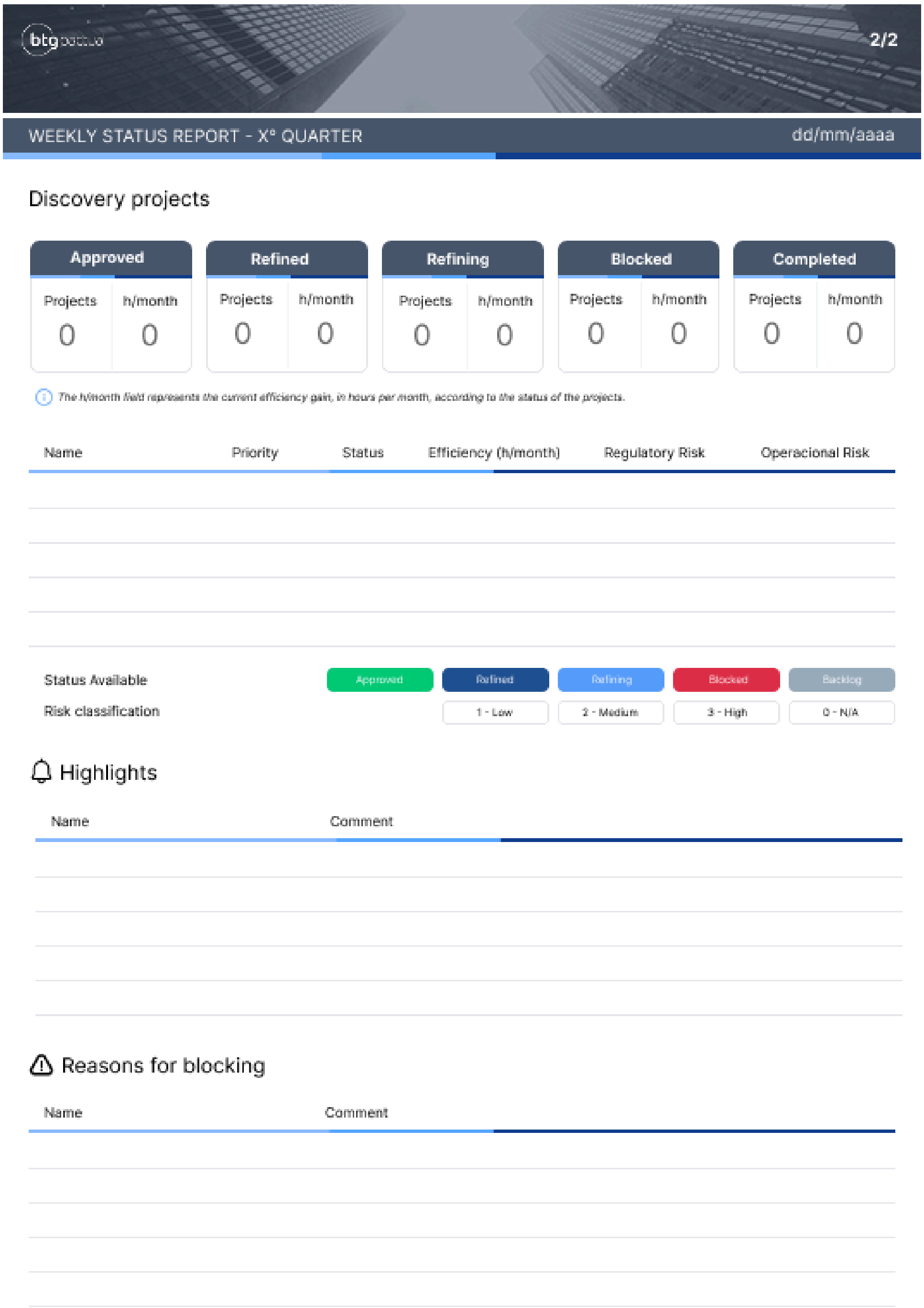
The visual structure and regularity of the report are inspired by SAFe's PI Planning reports, Scrum artifacts and tools such as Jira and Azure DevOps.

The justification for this approach lies in the weekly and objective formatting, which aligns with the cadence of the Scrum Sprints and the itineraries of the SAFe Program Increments. In addition, the use of standardized fields reflects the common practices of agile management tools.

Discovery Weekly Status Report

This template (Figure 14) for weekly project monitoring in Discovery is the second page of the executive report, aimed specifically at initiatives that are still in the design, analysis and prioritization phase. It allows you to monitor the maturity funnel of projects, even before they go live, ensuring clear visibility on risks, efficiency and anticipated bottlenecks.

Figure 14 – Weekly Status Report Template 2



Source: Prepared by the author

The structure includes:

- A dashboard with status indicators for projects in Discovery, with columns such as: Approved, Refined, Under Refinement, Blocked and Completed, along with the estimated efficiency gain per project (h/month).
- A table detailing by project: priority, current status, expected efficiency, as well as regulatory and operational risks classified into levels (1 to 3).
- Specific sections for relevant highlights and reasons for blockages, essential for management decisions.

Like the first page of the report, this template was designed for use on Monday.com, the current tool used by BTG Pactual to track projects. However, other tools can also be used if necessary, such as:

- Google Sheets/Excel, allowing simple automations for teams without access to Monday;
- Airtable, which combines spreadsheet flexibility with a modern look and interactive filters;
- Notion, which can be adapted with a clean look and dynamic statuses, even in its free version;
- ClickUp, an alternative with a robust free version and good usability for statuses and workflows.

How to apply the template

It should be filled in weekly, together with the first page of the report. The correct application of this template helps:

- Control the project pipeline before execution, anticipating risks and bottlenecks;
- Prioritize initiatives with the greatest impact and lowest risk, based on efficiency and risk rating fields;
- Giving leadership visibility of the stage of maturity of ideas and requests, before they become active demands on the development team;

- Registering and flagging impediments from the earliest stages, enabling support for resolution while still in Discovery.

Applying this model is expected to lead to more strategic portfolio management, focused on value, alignment with regulatory targets and the avoidance of waste through immature or misdirected demands.

Report Components and Agile Inspirations - Template 8 - Part II

1. Status columns: Approved, Refined, Refining, Blocked, Completed

The status columns - Approved, Refined, Refining, Blocked, Completed - are inspired by SAFe, Disciplined Agile (DA), LeSS and Scrum. They use templates such as SAFe's Portfolio Kanban and Program Backlog, LeSS's Shared Product Backlog and Scrum's Product Backlog.

The justification for this structure lies in the separation by stages of the discovery phase, which is in line with the layered organization of backlogs, as in SAFe's Team/Program Backlog and in the refinement practices of Scrum and LeSS. The use of columns reflects the flow visualization approach typical of Kanban, adapted by these frameworks.

2. Risk prioritization and classification

Risk prioritization and classification is inspired by SAFe, DSDM and DA. It uses templates such as SAFe's WSJF (Weighted Shortest Job First), DSDM's Risk Log and DA's decision diagrams. The justification lies in the presence of priority fields and regulatory or operational risks, which point to a structured analysis of value and risk. This approach is consistent with SAFe's WSJF, DSDM's risk logs and the checklists and decision-making models proposed by DA.

3. Efficiency metric (Efficiency h/month)

The efficiency metric (Efficiency h/month) is inspired by Lean and DA, using templates such as Lean's Value Stream Mapping and DA's Continuous Delivery Lifecycle. The justification for this metric lies in measuring the estimated gain in hours per month, even in the discovery phase, which reflects the Lean principle of mapping value and reducing waste from the earliest stages. This practice is also supported by DA's agile lifecycle templates.

4. Highlights / Reasons for blocking sections

The Highlights / Reasons for Blocking sections are inspired by Scrum and DA, using templates such as Scrum's Sprint Review and Retrospective, as well as DA's Kaizen models and feedback loops. The justification for this approach lies in promoting transparency, collaboration and continuous improvement, which are central principles of both Scrum and DA. Catching impediments in the discovery phase is an effective preventive risk management practice.

5. Visual structure and risk segmentation

The visual structure and risk segmentation are inspired by DSDM, SAFe and DA, using templates such as DSDM's Risk Log and SAFe's Portfolio Kanban. The justification for this practice lies in the visual classification of risks and statuses using color codes and legends, which reinforces the governance and visual communication of the project. This approach is strongly recommended in regulatory or corporate environments, as suggested by DSDM and SAFe.

2.7.8 Stage 8 - Deprioritization

The Deprioritization stage consists of re-evaluating demands that had already been prioritized but which, due to changes in context, loss of value or technical unfeasibility, no longer make sense at the present time. The aim is to keep the backlog lean, relevant and aligned with the company's strategic objectives.

This process is essential to ensure that the teams' efforts are directed only at initiatives that really generate value and have real conditions for being executed. By constantly reviewing the backlog and removing obsolete items, wasted time and rework are avoided.

The practice of deprioritization is present in frameworks such as Scrum, SAFe and Lean, which reinforce the need to keep the backlog dynamic and adapted to the current reality. It also connects to the agile principle of responding to change rather than following a plan, promoting continuous focus on what generates the greatest value at the right time.

Deprioritization depends heavily on the context in which the demand was assessed. Each decision to deprioritize involves specific factors - such as a change in strategy, technical problems, new dependencies or a reduction in the expected impact - which cannot be captured in a standardized way. For this reason, there is no fixed template for this step, as the analysis is situational and must be documented according to the specific context of each case, usually within the backlog management tools already in use.

Unlike other more structured stages, such as project opening or the initial backlog, here the process is more analytical and situational. Normally, the discussion takes place in governance forums, prioritization meetings or one-off alignments between product and technology areas. What matters is recording the decision and justification, which can be done directly in the tools used (such as Monday.com, Jira, Notion, etc.), without the need for a fixed form or template.

De-prioritization can be considered in the following cases:

- A change in the company's strategic priorities;
- A redefinition of objectives by the requesting area;
- Identification of technical unfeasibility or high risks;
- A drop in the perceived value of the delivery;
- Demands that have been stalled for a long time without progress.

Possible courses of action after de-prioritization:

- Pause: the demand remains registered, but leaves the active queue;
- Reclassification: returns to previous stages, such as investigation or reformulation of the scope;

- Removal: is filed definitively, with documentation of the reason.

This step can directly impact the backlog stage, either by removing items, reclassifying them to previous stages (e.g., Investigation), or prompting a revision of priorities. Therefore, deprioritization not only refines the current scope but may also reshape the prioritization structure of the ongoing pipeline.

2.7.9 Skills for Framework Use

This section aims to detail the competencies that a PO must develop in order to apply each stage of the framework. The idea is to go beyond simply listing knowledge and skills, presenting a structure that guides both professional development and the assessment of the maturity of POs within the organization.

To this end, the analysis is organized into three complementary dimensions:

- **Knowledge:** refers to the theoretical knowledge needed to understand and contextualize the activities at each stage of the process.
- **Skills:** refers to the practical ability to apply this knowledge on a routine basis, using tools, conducting meetings and making decisions.
- **Attitudes:** encompasses the expected behavior when faced with demands, such as a collaborative attitude, a sense of responsibility and a focus on value.

Each of these competencies is associated with the stages of the framework - from the opening of the project to the deprioritization of demands - and is accompanied by a maturity scale (levels from 1 to 4) that allows you to classify the PO's expected level of mastery in each item (Table 21):

Table 21 - Maturity levels and descriptions

Level	Description
1. Basic	Understands the concept or performs with support; requires constant supervision or guidance.

2. Intermediate	Applies independently in common situations; recognizes limitations and seeks support in complex contexts.
3. Advanced	Performs confidently, mentors peers, and acts with critical thinking and autonomy even in complex scenarios.
4. Expert	Serves as a technical or behavioral reference on the topic; drives improvements, trains others, and influences organizational decisions.

The following table 22 details, for each stage of the framework, the main elements of knowledge, skills and attitudes required, accompanied by a clear definition for each of them and the expected level of maturity for the PO. This structure aims to guide professional development and promote a more standardized performance in line with the organization's strategic objectives

Table 22 - PO Skills for Framework Use

Stage	Knowledge	Skills	Attitudes
Risk Classification	Risk management (Level 3): Understanding processes and methods to identify, assess, and mitigate risks. Types of risk (regulatory, operational, etc.) (Level 3): Knowledge of different risk categories and their specificities. Impact assessment (Level 3): Ability to evaluate potential consequences of identified risks. Risk classification (Level 3): Skills to prioritize risks based on severity and likelihood.	Responsibility (Level 3): Accountability for managing risks and communicating them clearly. Systemic vision (Level 3): Ability to see how risks impact different parts of the system and organization.	Responsibility (Level 3): Commitment to owning and managing risks effectively. Systemic vision (Level 3): A holistic approach to anticipating and addressing risks.

Backlog	Backlog structuring (Level 3): Organizing and maintaining the backlog to reflect priorities and dependencies. Types of demand and categorization criteria (Level 3): Recognizing different demand types and applying categorization frameworks. Organization of information (Level 3): Managing data and documentation clearly within tools.	Use of tools such as Monday/Airtable (Level 3): Proficiency in backlog management software and platforms. Clarity (Level 3): Clear presentation of backlog items and priorities.	Organization (Level 3): Keeping backlog and related information structured and accessible.
Prioritization	Prioritization techniques (Level 3): Familiarity with frameworks like MoSCoW, WSJF, or value-risk matrices. Company strategy (Level 3): Understanding strategic goals to guide prioritization decisions. Data-driven decision making (Level 3): Using metrics and evidence to support prioritization.	Stakeholder alignment (Level 3): Facilitating agreement and buy-in among involved parties. Critical thinking (Level 3): Analyzing trade-offs and potential impacts objectively.	Commitment to value (Level 4): Dedication to prioritizing initiatives that maximize business impact.
Demand Investigation	Business rules (Level 3): Knowledge of the policies, constraints and logic governing the business domain. BPMN (Business Process Model and Notation) (Level 2): Ability to read and create process flow diagrams to map user/system interactions. System architecture and overall technical view (Level 2): Understanding the high-level technical components and integration points.	Attention to detail (Level 3): Precision in documenting requirements and processes. Clear and structured writing (Level 3): Communicating complex information understandably and logically.	Attention to detail (Level 3): Maintaining accuracy throughout demand investigation. Focus on clarity (Level 3): Ensuring transparency and simplicity in documentation.

Project Alignment	Technical and business dependencies (Level 3): Identifying and understanding interdependencies impacting delivery. Roles and responsibilities of teams (Level 3): Knowledge of who does what across involved stakeholders.	Coordination with Tech Lead and requesters (Level 3): Facilitating collaboration and communication among teams. Negotiation (Level 3): Managing conflicts and aligning expectations.	Empathy (Level 3): Understanding perspectives and challenges of others. Collaborative spirit (Level 4): Actively fostering teamwork and shared ownership.
Executive Summary	Reporting techniques (Level 3): Ability to synthesize information in written and visual formats. Written and visual communication tools (Level 3): Proficiency in tools like PowerPoint, dashboards, or report generators.	Information synthesis (Level 3): Extracting key messages and insights. Executive writing (Level 3): Producing concise, clear, and impactful summaries.	Transparency (Level 3): Honest and clear communication of progress and challenges. Organization (Level 3): Keeping information and communications well-structured.
Deprioritization	Business value analysis (Level 3): Evaluating the impact and cost-benefit of demands to inform deprioritization. Evolving backlog management (Level 3): Adjusting priorities dynamically as new information emerges.	Contextual decision-making (Level 3): Making informed deprioritization decisions based on the bigger picture. Justification of deprioritizations (Level 3): Clearly explaining the rationale behind deprioritization choices.	Flexibility (Level 4): Willingness to adapt priorities and plans as needed. Responsibility with a focus on value delivery (Level 4): Accountability for ensuring efforts deliver real business value.

2.7.10 FELIX - New Flow

Previously, the process of receiving and managing requests was decentralized and not very standardized. Requests came in through various channels, such as chats, emails and phone calls, which made control and traceability difficult. With the changes implemented, every new demand or project now goes through stage 1 (Project

Opening) of the framework, and is registered exclusively using a standardized form. This change has brought more organization and visibility to the flow of incoming demands.

In addition, in the old process, requests could go directly to the developer or Tech Lead, which created risks of misalignment and loss of control. With the implementation of the form in stage 1 (Project Opening), this risk is mitigated, as the process centralizes the entry of demands and reinforces the role of the PO as the person responsible for receiving them and directing them appropriately - an activity that was previously often done by the Tech Lead himself.

In addition, in the old process, requests could go directly to the developer or Tech Lead, which created risks of misalignment and loss of control. With the implementation of the form in stage 1 (Project Opening), this risk is mitigated, as the process centralizes the entry of demands and reinforces the role of the PO as the person responsible for receiving them and directing them appropriately - an activity that was previously often done by the Tech Lead himself.

Before the template was standardized, requirements management meetings took place without a defined structure, which often resulted in misalignment between POs and business teams. The lack of clarity about the decisions made during the refinement of requests and the absence of a structured process made follow-up difficult and led to divergent interpretations about the status of requests. With the formalization of stages 4 (Prioritization) and 5 (Demand Investigation), the framework began to establish a clear sequence of meetings and documents to be used in each phase, promoting greater organization, transparency and alignment. In this way, everyone involved now shares a common vision of the progress of demands, which facilitates collaboration between POs, business teams and other stakeholders.

Finally, the autonomy of the PO in terms of understanding the projects and the architecture evolved. Previously, there was a strong dependence on the Tech Lead to access this information. With steps 5 (Demand Investigation) and 6 (Project Alignment) of the framework, this reality has changed: now, the documentation is built jointly by the

PO and the Stakeholder, and there is a guide that allows the PO to carry out the definition of the project's solution architecture more independently.

Table 23 summarizes the main changes made to the process, comparing the previous scenario with the current one, after the framework was created. For each phase, we have highlighted how it worked before, what has changed with the new practices and what value has been generated from these improvements.

Table 23 - Comparison of AS-IS vs. TO-BE demand management process

Framework Stage	AS-IS	TO-BE	Value Generated
1. Project Opening	Requests came through chats, emails, and phone calls.	All new requests are submitted via a standardized form.	More control, traceability, and visibility of requests.
1. Project Opening	Requests could go directly to the developer or Tech Lead.	The PO receives and directs requests through the form.	Centralized entry and clearer PO role; reduced risk of misalignment.
3. Backlog **5. Demand Investigation**	Each PO used different tools and formats.	Standardized tools and templates for documentation.	Greater consistency and organization.
4. Prioritization	Meetings lacked structure and decision clarity.	Defined sequence of meetings and required documents.	More transparent and aligned process among all stakeholders.
5. Demand Investigation	Lack of clarity on status and decisions; dependency on the Tech Lead.	PO conducts the investigation with the stakeholder; guided by a template to define the solution's design.	Greater PO autonomy, shared vision, and clarity on demands and solutions.
6. Project Alignment	Tech Lead led the solution architecture.	PO participates in defining the architecture with the support of the guide and stakeholder.	Reduced dependency; better alignment between business and solution.

2.7.10.1 - BPMN Flowchart of the Product Owner's Role in the Felix Framework

Understanding that demands and core organizational processes require prioritization by the Product Owner—and given that their actions are central to the workflow—it is appropriate to model a BPMN flowchart that focuses solely on their responsibilities. Below is the idealized process model within the scope of the Felix framework.

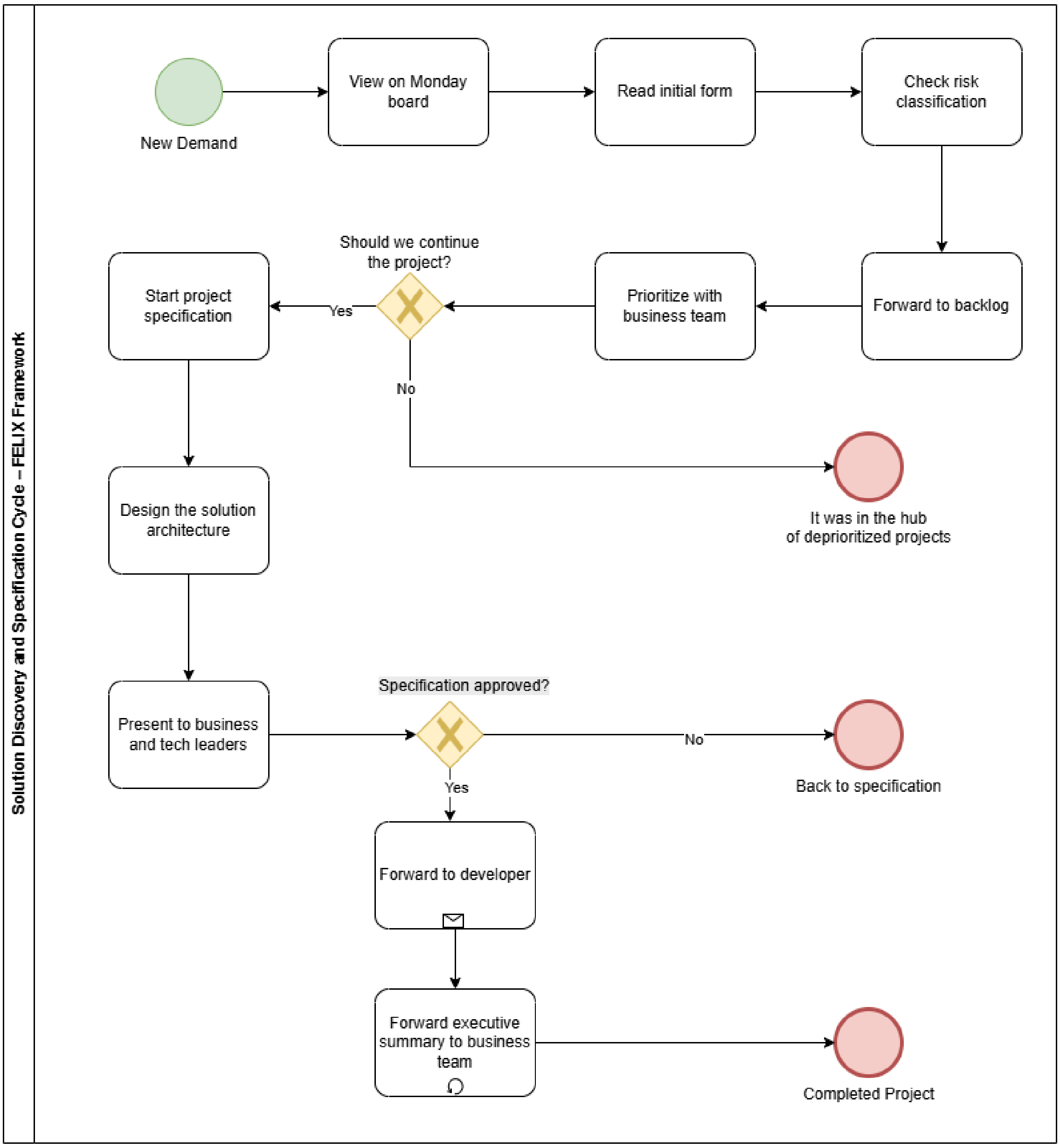
BPMN Process Description:

1. **New Demand Received:** The request is recorded and visualized on the Monday board.
2. **Initial Form Review:** The Product Owner reads the data submitted through the initial form.
3. **Risk Classification Assessment:** The level of risk associated with the demand is evaluated.
4. **Forward to Backlog:** If not classified as urgent or critical, the demand is directed to the backlog.
5. **Prioritization with Business Team:** Backlogged items are discussed and prioritized with the business stakeholders.
6. **Decision: Should the Project Continue?:** If there is no justification for continuation, the demand is placed in the deprioritized project hub.
7. **Start Project Specification:** If prioritized, the specification phase is initiated.
8. **Design of Solution Architecture:** The Product Owner drafts the proposed solution architecture based on requirements.
9. **Presentation to Business and Tech Leader:** The specification is presented to stakeholders for validation.
10. **Decision: Is the Specification Approved?**
    - If **not approved**, it returns to the specification phase.
    - If **approved**, the package is sent for development.

11. **Forward to Developer:** The specification is sent to the development team for execution.
12. **Send Executive Summary**
 The Product Owner delivers an executive summary to the business team.
13. **Project Completed**

After the executive summary has been handed over to the commercial team, the discovery and refinement cycle for that specific demand is considered complete. From this point on, the Product Owner continues to play a central role in managing the backlog, supporting the development teams and ensuring the evolution of the solution through delivery, monitoring and iteration. Below is the flow of how the Product Owner begins this discovery stage and follows up on projects (in macro form).

Figure 15 - BPMN flow of the discovery and specification cycle led by the Product Owner in the FELIX framework



Font: Draw.io (https://drive.google.com/file/d/1tXtZarG5JI8f-Zg5lvcJXHCP55qoGF-d/view?usp=sharing)

This BPMN flow focuses exclusively on the mapping of the Product Owner's activities, as it is the role most impacted by the framework's proposition. No significant changes are proposed for the responsibilities of Tech Leads or Developers, whose contributions remain aligned with existing agile practices. Among them, only the Tech Lead may participate in a supportive and organic manner—when necessary—to assist the PO in evaluating technical feasibility. However, in this model, the discovery and specification of the solution are defined as business-driven activities led by the PO. From a process perspective, this justifies modeling the PO's path as the primary focus, since it is the only role undergoing a transformation in how new projects are explored and structured.

2.8. Implementation

Monday.com was defined as the organization's official tool for managing the lifecycle of projects and demands, while the Wiki was established as the official platform for documentation. Both tools were already in use within the bank, which facilitated their adoption as the foundation for implementing the framework.

The choice ensured that the framework would be integrated into the existing ecosystem, providing consistency and reducing the learning curve for users. Monday enables a high level of workflow customization, the creation of specific templates for each defined form, and automations that facilitate integration between stages. Meanwhile, the Wiki centralizes all documentation in a single environment, offering features such as folder structuring, standardized formatting, and embedded tools like Excalidraw for diagramming.

Together, these platforms made it possible to adapt the framework to the organization's reality without the need for additional tools, ensuring standardization, visibility, traceability, and proper documentation throughout the process.

2.8.1 Customizations and templates implemented

After choosing the tools, Monday and Wiki, the framework was translated into templates and customizations that support the main stages of the process:

- Form templates (WorkForms): Allow stakeholders to register demands directly, with fields configured to capture essential information.

- Standardized boards: each request is recorded on a structured board with predefined columns, ensuring consistency in monitoring.

- Automations: Rules configured to automatically change status, notify those responsible, and create links between boards, reducing rework and ensuring fluidity in the process.

- Executive dashboards: Consolidated view for monitoring indicators, prioritizing backlog, and monitoring project progress.

- Standardized documentation: Creation of specific templates to support Product Owners in organizing and recording information, ensuring consistency and clarity in communication between areas and in the evolution of demands.

These customizations ensured that the framework was not just a set of conceptual guidelines, but rather an operationalized process, accessible to all involved.

2.8.2 Practical example: creating a board on Monday

To better illustrate how these customizations were operationalized in practice, the following subsection presents a step-by-step example of creating a board in Monday.com. This example demonstrates how the predefined columns and configurations can be applied to structure the demand intake process in alignment with the framework.

1) **Board with the requested columns**

To create the board on monday.com with the requested columns, click + Add (or New) and select Board, then choose the From scratch option and define whether it will be Main or Private, as you prefer, then give the board a name. As a tip, rename the mandatory Pulse name/Item name column to something useful, remembering that it does not need to be included in the Form.

2) **Add the columns (exact name and type from Monday)**

For each row below, click + to the right of the columns, select More columns, choose the type, and then rename it exactly as indicated in the table 24 below:

Table 24 - Description of columns and types in the table

Column Name	**Type**
Area Name	Status
Channel Name	Status
Requester	Text
Request Date	Date
Business Driver	Status
Risk	Status
Risk Rating	Status
Demand Type	Status
Problem Summary	Long Text
Demand Description	Long Text
Quarter	Status
Desired Quarter	Status
Priority	Number
Discovery Status	Status
Wiki Link	Link
Status	Status

Timeline	Timeline
PROD Date	Date
Ux	Status
App	Status

3) Useful column settings (optional, but recommended)

For Status columns (labels/colors), click on the column header, select Edit labels, and set the desired options; repeat this for each “Status” type column according to your needs. For the Wiki Link column, you can set Preview to open in a new tab if preferred.

4) Form on the same board

To create a Form (WorkForms) using the fields, go to the top of the board, click + Add view > Form (or Open in WorkForms), and select Use board columns to convert the columns into questions.

5) Fields that must be in the Form

Keep only the following fields in the form, removing or hiding all others, indicated in the list below:

- Area Name;
- Channel Name;
- Requester;
- Request Date;
- Business Driver;
- Risk;
- Risk Classification;
- Demand Type;
- Problem Summary;

- Demand Description;
- Desired Quarter;
- UX; and
- App.

You can drag fields to reorder them, and click on each one to make it Required, add help text/placeholders, and adjust labels or descriptions as needed.

6) Publish and use the Form

Click Share/Publish to obtain the Form link, then in Submission settings, select the board group where the items will be created and, if desired, enable options such as Send attachment, confirmation, and other preferences.

7) Create dashboard

Create a separate Dashboard (the “blank board”) and set up the widget layout, which can pull data from the board created earlier. Click + Add > Dashboard, choose Blank dashboard, give it a name, and then connect the data source by adding your board (or multiple boards, if needed).

8) Set up the layout (order and widths)

On the dashboard, each widget can be resized: hover over the corner and drag until it occupies 100% of the width (entire page) or just a fraction (side by side). The exact sequence is illustrated in Table 25 below:

Table 25 - Description of the executive summary's sections, widgets, and layouts

Line	Widget Type	Layout / Notes
1	Text	Full width (Panel title/description)
2	Numbers	5 side by side (e.g., Total requests, Under review, Approved, In development, Completed — each filtering a status)

3	Text	Full width
4	Table	Full width (pulls items from the board; configure visible columns and filters)
5	Text	Full width
6	Text	Full width
7	Text	Full width
8	Numbers	4 side by side
9	Table	Full width
10	Text	Full width
11	Text	Full width

2.8.3 Practical example: creating a board on Wiki

To better illustrate how documentation standards were incorporated into the framework, the following subsection presents the template available in the Wiki for demand refinement. This template was designed to support Product Owners in structuring the problem statement and acceptance criteria, ensuring consistency and traceability across all documented demands.

1) First steps on the wiki

- Access the Wiki: Open your browser and navigate to the following address: https://wiki.quickfin.com.br/ (Intern Link)

- Navigate to the desired location: On the home page, find and click on the add page link in the top right menu. This will open the Wiki folder structure, allowing you to choose where your new page will be stored.

- Choose the correct location: The location of your page will depend on the type of content you want to add. The path depends on the project being refined.

Example: To create a page on “Automatic Order Settlement”, you need to navigate to: Products/Trade Finance/Project. Click on “Select” to confirm the route of your new page.

- Define the name of the page (Route): At the bottom of the screen, you'll see the route of the folder you're in. After the last folder displayed (in our example, “Projects”), add a slash / and then type in the name you want for your page.

- Select the type of page: A new window will open asking you to select the page type. For our organization, ALWAYS select the “markdown” option.

Important note: *The page name must not contain spaces or special characters. Use lowercase letters and, if necessary, separate words with a hyphen (-) or underscore (_).*

- Edit the page properties: The page properties screen will be displayed. In this step, only change the “Title” field.

- Create the page: Finally, to finish creating your page, click on “Create” in the top right-hand menu.

2) How to add a diagram to a page

- Go to page editing: Navigate to the page on which you want to add the diagram. Click on the "Edit" icon located on the page.

- Open excalidraw: In the left-hand side menu that opens, find and select the Excalidraw icon. Clicking on this icon will open the Excalidraw editor directly on your page.

- Create and save the diagram: Use Excalidraw's tools to draw your diagram. You can add shapes, text, arrows and other elements as needed.

Important note: *Once you have finished creating the diagram, click on the "Save" button located in the top right menu of the Excalidraw editor. A pop-up confirmation window may appear. Click "OK" to confirm saving the diagram.*

- Finish editing the page: The diagram you created in Excalidraw will now be inserted into your page. To save the changes to the page as a whole, click on the "Save" button located in the top right menu of the page. If you want to close the page editing mode, click on the "Close" button also located in the top right menu.

3) Implementing the template

After creating the Wiki page for the demand refinement template, click on “Edit” and enter the template content. It shows the titles of the sections and guidance on the information that should be filled in for each one, ensuring the standardization of demand records.

[Refining] Demand X

Field	**Description**
Project Name	
Problem Description	
Solution Description	
Systems Used	
Teams Involved	

Fill in the necessary demand information in this refinement phase using the topics listed below. Remember to delete the guide items after filling them in.

🌐Overview

In general terms, the context of the demand needs to be highlighted.

The person responsible for drawing up the specification should develop an introductory text that conveys what the demand is about, understand the tasks that users perform, and try to understand how they would use a new system to help them carry out these tasks.

🎯Highlights

- Highlight key information.
- Highlight the main opportunities.

⚠ Points of attention

- Highlight the main points of attention.

⁉ Risks and Problems

- Highlight the main risks that the demand could generate.
- Highlight the main problems encountered during project specification / development.

📌 Business rules

During the information gathering, you will need to know about the domain and the business rules, what activities should be involved, the services, the tools, the characteristics of the application they want, as well as the expected performance.

🧩 Flowchart

In addition to the brief description, it will be necessary to develop a drawing of the current flow carried out by the requester, including tasks that are performed, systems involved, receipt of information, etc...

Note: *Once the flowchart prototype has been finalized, pass it on to the person responsible for the request for any questions, corrections, etc.*

📘 Stages

Create bullet points for the phasing of deliveries.

🔗 System integrations

Integration of internal and external systems related to the project[5].

2.9. Practical Validation of the Solution

2.9.1 Business Presentation

The presentation took place on August 4 with the aim of providing context to Tech Leaders potentially involved in the Felix product, i.e., the new methodology proposal for refining demands to be implemented at Banco BTG Pactual.

The meeting was structured into topics, covering: project introduction; objectives; overview of the framework structure; summary description of each stage; individualized

[5] We intended to include, at this point, an illustrative image to complement the explanation and provide a more practical demonstration of the process. However, due to information security restrictions, it is not possible to include direct screenshots of the internal system (Wiki). Therefore, we opted to present only the textual description or to use anonymized versions with redactions, as previously adopted in other sections of this study.

presentation of each phase with its respective template; monitoring metrics; implementation schedule; and proposal of relevant projects for testing.

The main feedback received focused on the first stage of the framework, the project opening. The concern raised was the excess of information to be filled in by the business team, including data that, in some cases, they may not be familiar with, requiring interaction with the IT team for clarification, especially with regard to risk classification.

In general, participants showed interest in the initiative and suggested projects from different areas to serve as pilots, which will be detailed in the following topic. The proposal is that these tests be conducted using the bank's own official tools and systems currently in use, ensuring greater adherence and feasibility in the implementation phase.

2.9.2 Metrics for evaluating the effectiveness

We conducted a brainstorm to identify the metrics that would be feasible to monitor, considering the resources available in the chosen tool, Monday. Based on this, we selected the most relevant metrics aligned with FELIX's strategic objectives, with a focus on measuring concrete results and supporting decision-making.

Next, we will move on to drawing up a structured test plan, which will define in detail how and when the metrics will be applied. In addition, we will select the projects that adhere most closely to the context and objectives of the framework, ensuring that the application of the metrics is representative and offers real insights into the effectiveness of FELIX in different scenarios.

These metrics will allow us to compare performance before and after implementing the framework, focusing on the following aspects:

1. Average demand cycle time (lead time to "Ready for IT")

Related Stages:

- **Stage 3 - Backlog**: This is the entry point for all new demands into the structured workflow. At this stage, items are cataloged but not yet prioritized or matured.
- **Stage 4 - Prioritization**: Determines which demands will enter the refinement pipeline, impacting when the cycle effectively begins.
- **Stage 5 - Demand investigation** and **Stage 6 - Project alignment**: These are the core stages for requirement refinement and architectural validation. The handoff to development — or the "ready for IT" milestone — occurs at the end of Stage 6.

Metric: Average number of working days between the entry of a demand into the backlog and its readiness for development.

Objective: To assess whether FELIX improves the efficiency of the refinement process, accelerating the maturation of demands through clearer structure and decision flows.

How to measure: Record the date on which the demand enters the backlog (step 3) and the date it is marked as "ready for IT" (i.e., post-alignment and refinement). Calculate the average time over a given period, and consider segmentation by type of demand (feature, improvement, technical activity, etc.).

2. Clarity and Quality of Refinements

Related Stages:

- **Stage 5 - Demand investigation:** At this point, the PO conducts meetings with the requestor to understand the business context, collect detailed requirements, and document the rules that guide the solution.
- **Stage 6 - Project Alignment:** Brings together PO, Tech Lead, and stakeholders to validate the proposed approach, surface dependencies, and ensure mutual understanding before development.

These two stages are responsible for the quality and clarity of documentation and alignment — which directly impact how well the development team understands what needs to be built.

Metric: Developers' and stakeholders' perception of the clarity of the requirements when they are handed over to the technical team.

Objective: To measure whether the investigation and alignment stages (stages 5 and 6) are in fact generating clearer and more complete documentation, reducing rework and dependence on clarifications during development.

How to measure: Application of semi-structured interviews and/or quarterly surveys with the following audiences:

- **Developers:** ask if the requirements were clear, with defined acceptance criteria and few points of doubt.
- **Stakeholders:** ask if they feel greater alignment and visibility of what will be delivered.

Suggested scale: 1 (very dissatisfied) to 5 (very satisfied).

3. Rework Index (demands returning to the backlog)

Related Stages:

- **Stage 5 - Demand investigation and Stage 6 - Project alignment:** These stages aim to ensure that the demand is well understood and that technical feasibility and business alignment are achieved before development starts. If demands return after these points due to lack of clarity, missed dependencies, or changing scope, this indicates a breakdown in these critical refinement stages.

Monitoring returns to backlog from post-alignment stages provides insight into whether FELIX is effectively preparing demands for development or if gaps remain in the investigative process.

Metric: Percentage of demands that, after progressing to development, need to return to the backlog or alignment stage.

Objective: To assess whether there has been an improvement in the initial definition of the solution, indicating maturity in the initial phases of the framework.

How to measure: Monitor the number of demands that, after passing through the investigation and alignment stages, return for reassessment before delivery. A return can occur due to

- Significant change in scope
- Lack of technical understanding
- Unexpected priority adjustment

This metric can be expressed as: % rework = (No. of demands that returned / Total demands delivered in the period) x 100

2.9.3 Pilot projects

In order to assess the applicability of the proposed framework, four projects were selected to serve as pilots. The choice was made considering different areas of activity, levels of complexity, and user profiles, in order to enable a comprehensive analysis of the model's performance in various scenarios.

Although these projects represent the initial phase of practical application, they will be conducted exclusively following the framework model, without other current processes or methodologies being applied in parallel.

Project 1 - Automatic Order Settlement

The Automatic Order Settlement project will be implemented to optimize the conversion and receipt of funds from abroad, reducing bureaucracy and operating costs. Aimed at content creators, the initiative will be carried out exclusively for a select group of companies accredited with the bank.

Currently, the process faces long delays, high costs, and fragmented procedures, which negatively affects predictability and customer satisfaction. The proposal is to develop an automated system that processes foreign exchange orders and performs automatic settlement and direct disbursement to the customer's account, ensuring regulatory compliance, operational security, and transparency in fees.

Project 2 - Limit Reallocation

The project focuses on implementing a global limit shared across specific bank products, allowing customers to freely allocate and reallocate this limit among the contracted products. This change aims to replace the current fixed limit management, eliminating the need for manual requests and enabling automatic reallocations directly through the bank's digital channels.

Project 3 – Full Assignor – Drawee Risk

The project aims to improve the process of contracting drawee risk products through biometric authentication, ensuring greater security and reliability in the validation of companies' legal representatives. The proposal is to implement biometric collection integrated with internal and external systems, with features that allow tracking the validation status, sending notifications, and automatic resends in case of failures. In addition, the process will maintain the feature of not requiring the opening of a corporate account, streamlining the contracting process.

Project 4 - Credit Taker - Solar

The project aims to develop a new contracting flow for solar credit borrowers, focusing on establishing relationships without the need to open a corporate account, a feature that does not currently exist. In addition to incorporating biometric authentication to securely validate legal representatives, the initiative includes adjustments to the contracting process to align with this new model, relying on the synergy of the areas involved.

2.9.4. Validation Steps

The test plan for evaluating the effectiveness of the framework was drawn up for a ten-week cycle. The main aim of this plan is to validate the framework's impact on the organization and the maturity of the process of refining and aligning demands up to the “Ready for IT” milestone.

The structure includes success and failure criteria, objective metrics and qualitative assessments with the main players involved - Product Owners, Tech Leads and stakeholders. The plan also describes the stages week by week, from the selection of pilot projects to the consolidation and final analysis of the results, guaranteeing complete traceability of the application in real contexts.

The full document detailing the test plan is presented below. This material serves as a formal reference for future validations, adaptations, and possible scalability of the framework in other organizational contexts.

2.9.4.1 Test Plan

This document describes the test plan for evaluating the effectiveness of the FELIX framework, with the aim of proving improvements in the process of refining and aligning demands to the point of “Ready for IT”. The tests will be applied to selected projects that represent the context of the framework.

Scope

The test will cover the following aspects and stages of the workflow:

- Stage 3 - Backlog
- Stage 4 - Prioritization
- Stage 5 - Demand Investigation
- Stage 6 - Project Alignment

Focusing on the following metrics:

1. Average time from demand cycle to "Ready for IT"
2. Clarity and quality of refinements (perception of developers and stakeholders)
3. Rework rate (demands returned to spec)

Project selection

To ensure a representative and consistent analysis, three to five pilot projects will be selected. The choice will primarily consider initiatives that have not yet used the framework, allowing for a clear comparison between before and after its application. In addition, the projects must have stable teams, which ensures greater reliability in data collection and interpretation throughout the testing period.

Metrics and evaluation criteria

To validate the effectiveness of the FELIX framework in real project contexts, a structured ten-week test plan was drawn up. The plan includes the progressive application of the FELIX stages, the use of specific success and failure criteria and the monitoring of relevant metrics. The aim is to assess whether the structure improves the organization of demand, promotes greater clarity in specifications, reduces rework and allows the Product Owner to act with greater autonomy and assertiveness. The table below summarizes the weekly testing strategy, the evaluation criteria and the expected results for each phase of the validation cycle.

Weekly	Step	Strategy	Success Criterion	Failure Criterion	Observations
1	Selection of Pilot Projects	Choose 3 to 5 projects with different profiles (low, medium and high complexity)	Projects that are representative of the team's real scenario	Generic projects, without characteristics that validate the framework	Selection in progress

2	Application of FELIX up to stage 4	Use stages 1 to 4 of FELIX (reception, backlog, classification, prioritization)	Clearly organized demands and transparent prioritization	Demands without clear criteria, or unorganized backlog	Measure: average time from entry to prioritization
3–4	Carry out steps 5 and 6 in real time with Product Owners	Product Owner conducts discovery and alignment sessions with stakeholders	Business rules recorded, requirements clear, dependencies mapped	Meetings inconclusive, documentation missing or vague	Number of meetings rescheduled
5	Application of the "Clarity of Refinements" metric	Check the number of meetings to specify the demand	Complete documentation with fewer recurring meetings	Incomplete documentation with recurring refinement meetings	Ratio between the number of refinement meetings and the number of items completed in the specification document
6	Metric: "Rework rate"	Check the number of demands that come back for re-specification after stage 6	Maintain rate ≤ 10% after complete refinement	More than 10% of demands come back after stage 6	Monitor via projects that come back for specification
7	Application of Step 7 + submission of the Executive Summary	Validate that the Product Owner can consolidate information and communicate status clearly	Stakeholders understand the plan and validate what will be delivered	Confusions, doubts or requests to rework the content of the summary	Direct feedback via form/meeting
8 **(Module 4)**	Consolidation and analysis of results	Collection of quantitative and qualitative data from the previous stages	Clear conclusion on the impact of the framework, with evidence	Lack of clear empirical evidence to support improvement	Produce final report

At the end of this ten-week cycle, the data collected will provide a solid basis for evaluating the impact of the FELIX framework in the real world. By systematically applying the methodology to different types of projects and analyzing process indicators and user perceptions, it will be possible to determine whether the proposed improvements effectively increase maturity in the early phases of demand management. The findings will support future iterations of the framework, guiding its evolution and possible adoption at scale in agile environments.

Test schedule

The schedule of activities planned for the execution of the framework test plan is detailed below. The actions were spread over 10 weeks.

Period	Activity
Week 1	Selection of pilot projects and configuration of the board with the necessary fields and automations.
Week 2	Integration of participants (technical leaders, stakeholders) on the use of the framework and models.
Weeks 3-8	Practical application of the framework to selected demands and data collection.
Week 9	Application of perception surveys (clarity, alignment, satisfaction).
Week 10	Consolidation of quantitative and qualitative data.

The above timetable guarantees an adequate period for the application to mature, allowing for the structured collection of data and feedback that will inform the evaluation of its effectiveness.

2.9.5 Execution - Part One (Steps 1, 2, 3 e 4)

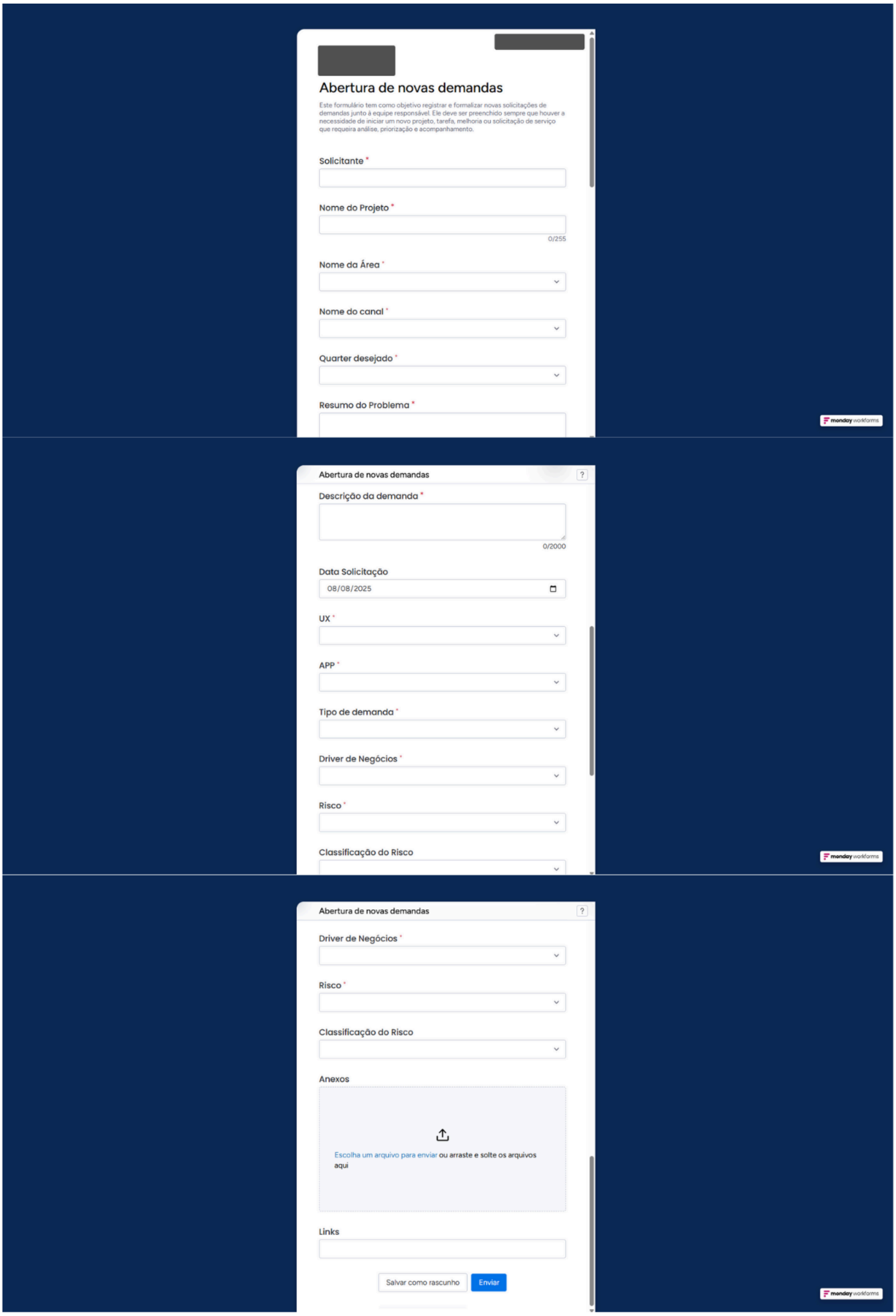
The execution of the structure began with the creation of templates on the Monday platform and the bank's Wiki. Most of these templates involved forms and tables — all native features of Monday. The adaptation process consisted of reviewing the topics and structure defined in the conceptualization phase and manually building this first version (V1) within Monday. For the discovery research phase, a corresponding format was also created on the Wiki to centralize and standardize all related information.

To translate the initial concept of the Project Charter into a practical and collaborative workflow, we chose Monday.com as the implementation platform. This decision was based on two main factors: (1) the tool was already part of the team's daily routine, ensuring zero friction in integration; and (2) its flexibility allowed the creation of structured models with native features such as forms, tables, dashboards, and automation rules.

The original fields on the forms were broken down into columns in the table, each representing a key element for classification and decision-making. These included Priority Level, Risk Type (Regulatory, Operational, or Efficiency Gain), Estimated Time, Requesting Area, Affected Systems, and Current Stage in the FELIX workflow. These columns served not only as storage for project data, but also as parameters for filtering, grouping, and classifying items in real time.

For Stage 1 – Project Opening, the information collected via the Monday.com form is automatically populated into a new board item, streamlining the process and eliminating manual data entry. This ensures that all required fields from the project charter are consistently captured and immediately available for use in the framework's implementation.

Figure 15 – Monday.com board for Stage 1 – Project Opening

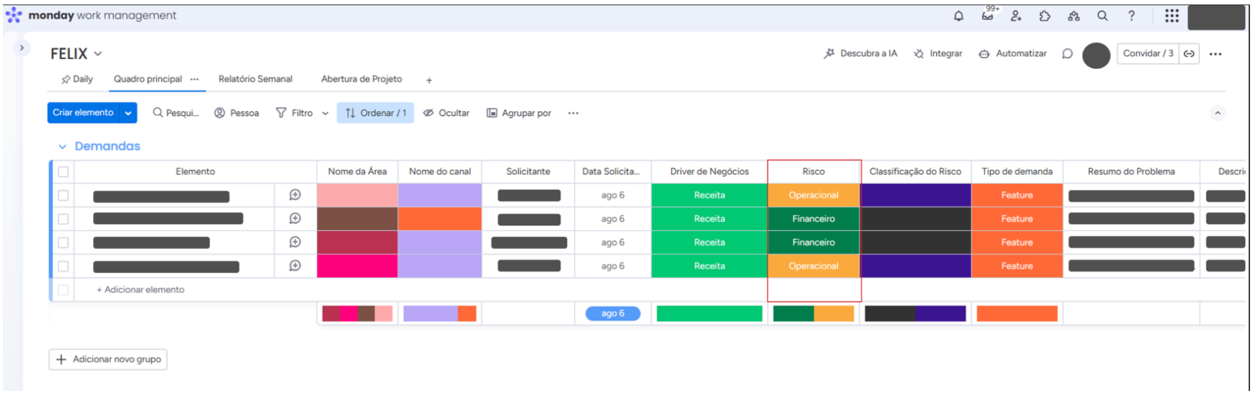


Source: Prepared by the author. Based on the conceptual template presented in Figure 7, Chapter 8.

In Phase 2 – Risk Classification, a column was configured to classify the type of risk, which is filled in manually by the Product Owner who is refining the demand, so that they can better understand the origin of the demand.

This classification, stage 2, was carried out jointly by the PO, Tech Lead, and business team, ensuring multidisciplinary alignment.

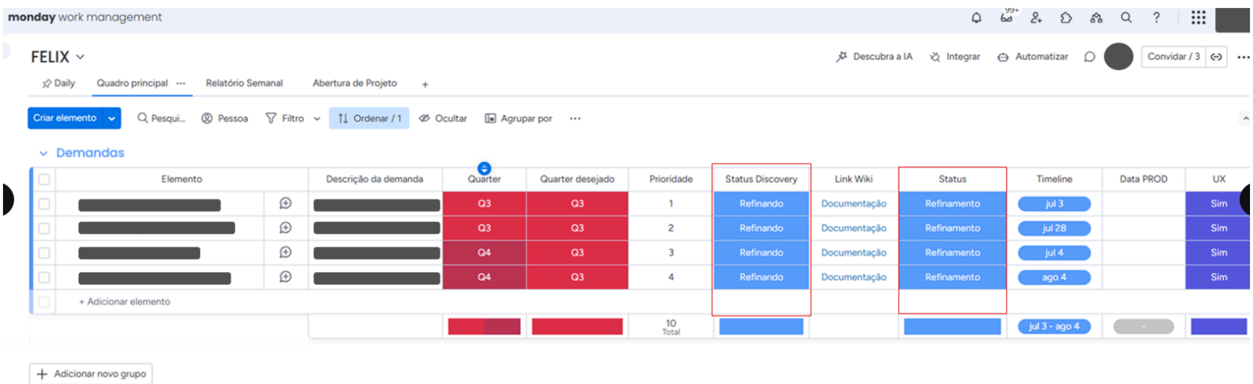
Figure 16 – Monday.com board for Stage 2 – Risk Classification



Source: Prepared by the author. Based on conceptual template presented in Figures 8 and 9, Chapter 8.

With the projects properly entered and all columns filled in, the third stage was completed: the formation of the backlog. In this phase, the table grouping feature was used to segment projects by priority and complexity. A native Monday function was also applied for prioritization, in which the first item on the list is the most important, and the priority column records this order numerically.

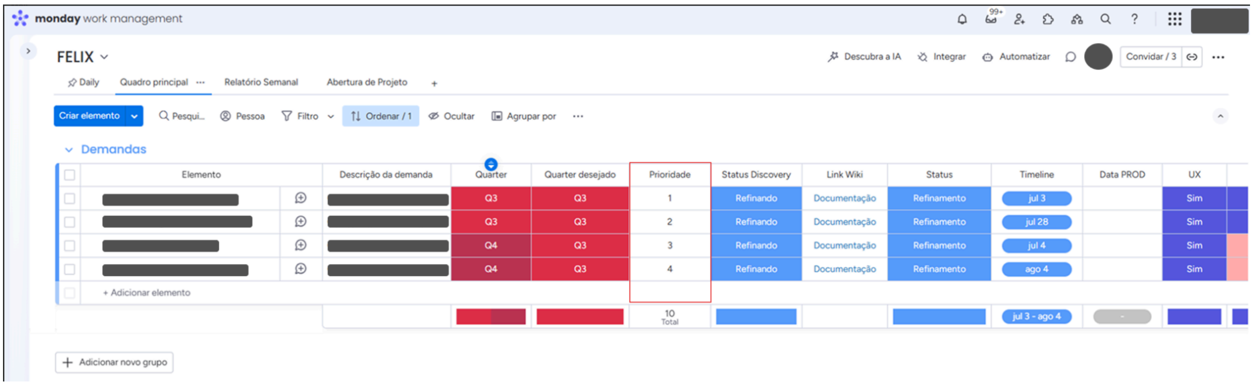
Figure 17 – Monday.com board for Stage 3 – Backlog



Source: Prepared by the author. Based on conceptual template presented in Figure 10, Chapter 8.

For the last stage planned in the first sprint (August 4 to August 15), an alignment was carried out with the areas responsible for each project in order to understand their priorities. Based on this analysis, the start and delivery order was defined, concluding the fourth stage.

Figure 18 – Monday.com board for Stage 4 – Prioritization



Source: Prepared by the author. Based on conceptual process described in Section 8.4, Chapter 8.

Once these points have been completed, the next steps will involve working with the project team to develop the Executive Summary. This document will bring together consolidated dashboards to present strategic metrics such as: total number of projects in the pipeline, average prioritization time, and number of projects classified as critical risk. These dashboards will have a dual function — serving as an operational monitoring tool and as a basis for management reports — ensuring transparency for all stakeholders.

Each of these steps is directly aligned with the project's central proposal: to structure and standardize the refinement of demands in the bank, reducing dependencies and increasing the predictability of the process. The creation of models ensures the uniformity and traceability of information. The adaptation of the form and collaboration between different functions ensure that the project's opening phase is more assertive and realistic. And the formation of an organized and prioritized backlog, in turn, allows for more agile execution, supporting the vision of a continuous and transparent flow between business and technology.

These benefits were also recognized by the POs involved in implementing the first stages of the framework. It should be noted that, prior to these changes, even simple tasks required a great deal of effort due to the lack of dedicated communication channels for receiving requests, a unified framework for prioritizing requests, and visibility of potential risks, conditions that could even lead to operational failures or fraud. The pilot adoption of the framework began to address these gaps.

For Step 5 – Demand Investigation, the first documentation activities were initiated using the standardized template created in the bank's Wiki. This step marked the beginning of a more detailed refinement process, in which information was consolidated to give structure and clarity to the selected projects.

The two projects that were initiated were Automatic Order Settlement and Full Risk of the Assignor – Drawee, which is why they are undergoing the refinement process. The completion of the template involved several interactions.

The project overview and basic data were taken from the form completed by the stakeholder and refined in meetings, ensuring that the operational context, risks, and expected results of the initiative were clearly understood. At this stage, the focus remains essentially operational, as the information collected is limited to descriptive fields such as demand details, classification, and risk categories defined in the FELIX framework. Although the possibility of linking these demands directly to BTG’s strategic objectives represents a highly relevant perspective for the future, it is not yet part of the framework’s current scope. Rather, this can be seen as an evolutionary direction: FELIX initially emphasizes standardization and transparency in the discovery of demands at an operational level, while paving the way for a future stage in which project outcomes may be systematically associated with the strategic goals of the bank and its business units.

Process flows were designed in collaboration with the technical lead and subsequently refined with dependent areas to identify system integrations and interconnections that could affect implementation.

Contextual information and risks were discussed with stakeholders, allowing for the identification of dependencies, approval requirements, and points that could represent greater operational or regulatory sensitivity.

The template in the Wiki for the Automatic Order Settlement project was filled out based on different interactions and sources of information, as shown in image 19:

- Initial table (Problem Description, Solution Description, and Systems Used): this was mainly populated with data already submitted in the demand opening form on Monday.com.
- Overview: prepared based on meetings with stakeholders and analysis of the existing exchange process, allowing for contextualization of the challenge of bureaucracy and costs, and making clear the value proposition of automation.
- Highlights: defined in refinement calls with the product and business areas, emphasizing gains such as reduced bureaucracy, transparency, and agility for the customer.
- Points of Attention: listed based on interactions with regulatory and operational teams, raising compliance risks, exchange rate variation, and specific requirements for integration with internal systems.
- Risks and Problems: discussed jointly with stakeholders and technology areas, highlighting bottlenecks that could impact implementation.
- Business Rules: detailed with the requesting team, which defined criteria such as eligible currencies, execution conditions, and applicable charges, all documented in the template.
- Flowchart: developed with support from the Tech Lead, representing the flow of automatic foreign exchange settlement. The flowchart was validated in joint meetings with business and technology to ensure alignment among all involved.

Figure 20 – Monday.com board for Stage 5 – Demand Investigation - Automatic Order Settlement

Liquidação Automática de Ordens

Edit

[Refinamento] Liquidação automática de ordens

Campo	Descrição
Descrição do Problema	Profissionais que recebem salário do exterior pelos provedores [redacted] enfrentam burocracia, demora e custos elevados para converter e receber seus pagamentos em Reais no Brasil.
Descrição da Solução	O projeto oferece uma liquidação automática de câmbio, permitindo que o cliente receba o salário do exterior com conversão rápida, diretamente creditado em sua conta no BTG Pactual, simplificando o processo e reduzindo riscos e custos.
Sistemas Utilizados	[redacted]

🌐 Overview

O projeto "Liquidação automática de ordens" surge para atender a uma demanda crescente de profissionais que recebem pagamentos do exterior e enfrentam dificuldades na conversão e recebimento desses valores em Reais no Brasil. Atualmente, o processo pode ser burocrático, demorado e custoso, impactando a experiência do cliente.

O objetivo é criar um sistema que permita o crédito rápido e transparente dos valores recebidos, diretamente na conta do cliente no BTG Pactual. [redacted] e buscam uma solução simples e segura para converter e acessar seus recursos financeiros. O novo sistema deve facilitar a operação, reduzir a necessidade de intervenções manuais e oferecer transparência nas taxas aplicadas.

🎯 Destaques

- A liquidação automática reduz a burocracia e o tempo de espera, melhorando a experiência do cliente.
- Transparência nas taxas e no processo de câmbio aumenta a confiança do cliente no serviço.
- [redacted]

⚠️ Pontos de atenção

- [redacted]

⁉️ Riscos e Problemas

- [redacted]

📌 Regras de Negócios

Requisitos para Liquidação Automática (todos cumulativos)

- [redacted]

Moedas Elegíveis

- [redacted]

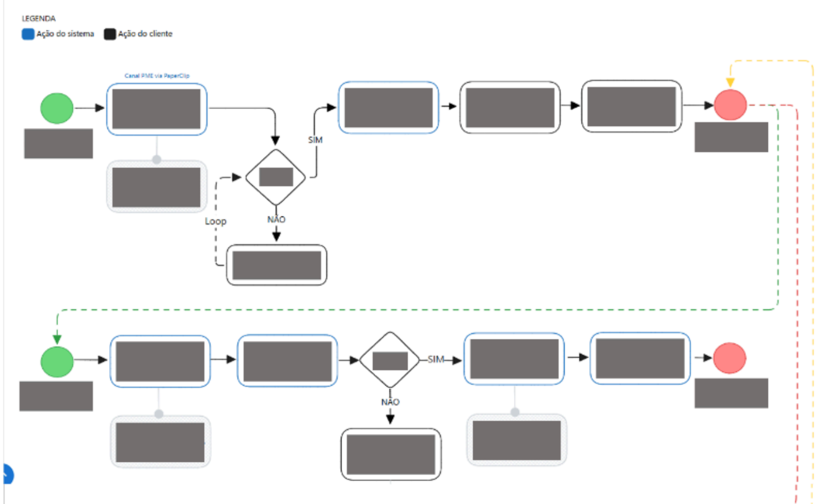
Condições de Execução

- [redacted]

Taxas e Encargos

- [redacted]

🧩 Fluxograma



The template in the Wiki for the Complete Assignor – Drawee Risk project was also filled out according to the same standardization, based on different sources of information, as shown in the image 21:

- Initial table (Problem Description, Solution Description, and Systems Used): once again, the basis provided by the demand opening form was reused, which described the problem faced by customers and the initial solution proposal.
- Overview: prepared in conjunction with the responsible stakeholder, also incorporating a survey of what already existed in the current flow, to contextualize the current scenario and justify the need for improvement.
- Highlights: defined during refinement meetings, when the main value delivery of the project was identified, the implementation of biometrics in the flow without opening a corporate account.
- Points of Attention: raised in calls with the areas involved, pointing out technical and regulatory risks that would need to be observed before implementation.
- Risks and Problems: discussed directly with stakeholders and dependent areas, allowing the identification of potential bottlenecks and critical dependencies.
- Business Rules: mapped with the business team that requested the demand, in order to ensure clarity about the conditions and parameters necessary for the operation in the flow.
- Flowchart: built in partnership with the Tech Lead, illustrating the updated process journey with biometrics. As in the previous project, it was validated in meetings with stakeholders and dependent areas to confirm technical and business consistency.

Figure 21 – Monday.com board for Stage 5 – Demand Investigation - Complete Assignor – Drawee Risk

Cedente Completo - Risco Sacado

Edit

[Refinamento] Cedente Completo [redacted] Risco Sacado

Campo	Descrição
Descrição do Problema	Atualmente, o fluxo de contratação de risco sacado sem abertura de conta PJ já está em operação, porém, sem a utilização de biometria. [redacted]
Descrição da Solução	O projeto permite a implementação de biometria no fluxo já existente da contratação de risco sacado sem a necessidade de abertura de conta PJ.
Sistemas Utilizados	[redacted]

🌐 Overview

O projeto visa aprimorar o fluxo de contratação de risco sacado sem abertura de conta PJ, que já está em operação, mas atualmente não utiliza biometria. [redacted]

🎯 Destaques

- Implementação da biometria no fluxo existente de contratação de risco sacado sem abertura de conta PJ.
- [redacted]

⚠️ Pontos de atenção

- [redacted]

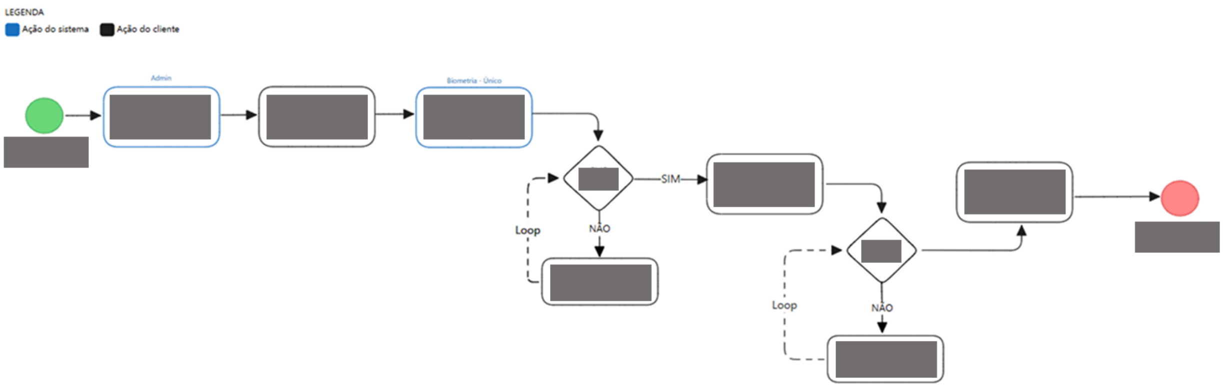
⁉️ Riscos e Problemas

- [redacted]

📌 Regras de Negócios

- [redacted]
- [redacted]
- [redacted]

🧩 Fluxograma



The initial stages of the FELIX framework—Stages 1 through 4—focus on establishing clarity around the business challenge, aligning key stakeholders, and reaching a shared understanding that supports prioritization.

Automatic Order Settlement

The Automatic Order Settlement project provides a practical example of how FELIX is applied during the demand investigation stage. The first interaction took place on August 6, in a meeting between the Product Owner and the business team, focused on the initial understanding of the product. This step corresponds to the beginning of the discovery process, where the context of the request is detailed from the requestor's perspective.

On August 7, a dedicated session was held for the definition of the business flow, mapping the main activities and identifying critical interaction points between systems and areas. This activity aligns with Stage 5 of FELIX, refining process flows and identifying key dependencies.

On August 8, the understanding of business rules was deepened, and the first version of the wireframe was discussed. This visual representation complemented the evolving documentation in the Wiki and enabled clearer communication with design and tech stakeholders.

Complete Assignor – Drawee Risk

The Complete Assignor – Drawee Risk project also followed the FELIX refinement cycle, demonstrating the framework's adaptability across contexts. The first meeting occurred on August 7, between the PO and the business team, to gain a high-level understanding of the project.

On August 13, a deeper investigation into business rules took place, outlining core parameters and constraints, all progressively documented in the Wiki.

Limit Reallocation

The Limit Reallocation project began with an initial alignment on August 6, where the PO and business team discussed the product's context and goals. On August 7, a session was held with representatives from business, credit, and the Tech Lead to create the first version of the credit policy, which served as the base for the initiative's business rules.

These examples illustrate how the early stages of FELIX enable a structured and consistent approach to demand intake and prioritization. By ensuring that each initiative starts with a clear understanding of context, objectives, and initial business rules, the framework sets the stage for efficient refinement and delivery planning. The quality and depth of work done in these first steps not only support informed prioritization decisions but also reduce misalignment and rework in later phases. As demands advance beyond Stage 4, they do so with greater clarity, stakeholder alignment, and readiness for deeper investigation and solution design.

2.9.6 Execution - Part Two (Steps 5, 6 e 8)

In this section, we continue with the implementation of the FELIX Framework, addressing Steps 5, 6, and 8, which correspond, respectively, to the refinement of demands, the validation of refinements made, and the deprioritization of projects, when necessary.

In this sprint, for the Complete Assignor – Drawee Risk and Automatic Order Settlement projects, we advanced with the refinement of their wireframes, conducting validation sessions and finalizing the refinement documentation using the standardized template. This material was consolidated and formally shared via email, ensuring alignment among all stakeholders. As a result, the refinement stage for both projects was successfully completed, and they are now ready to move forward into the development phase.

The process of refining the Credit Taker – Solar project was initiated. The information was consolidated during previously scheduled meetings, with the participation of both the Tech Lead and the business team. This allowed the necessary data to be filled in the standardized template, ensuring clarity regarding the context, business rules, and planned integrations.

Although the refinement was completed, the project was subsequently deprioritized. This was due to an urgency in the Drawn Risk project, which required the same team involved in the development of Credit Taker. Given this overlap, it was decided to reallocate efforts.

Finally, the Limit Reallocation project was completely deprioritized, to the point where the refinement documentation was not even initiated. The decision was driven by a technical impediment, which led to the replacement of this initiative by another project considered as a Version 1 of Limit Reallocation. However, this new project has not yet been prioritized.

This decision, although representing an interruption in the expected flow, is relevant to the Framework evaluation process itself, since deprioritization will also be considered as a metric for monitoring and analyzing the model's effectiveness.

In the next sprint the focus will be on preparing the executive summary, which will consolidate the status of these projects. This summary will serve as the basis for monitoring progress and ensuring alignment with the business desk through dedicated follow-up meetings.

Once demands are prioritized through the FELIX framework, they move into Stage 5 – Demand Investigation and Stage 6 – Project Alignment. These stages mark the beginning of a more structured and collaborative refinement cycle, where the focus shifts to deepening the understanding of business rules, producing formal documentation, assessing technical feasibility, and aligning with key stakeholders such as the business team and Tech Leads. Together, these steps ensure that demands are not only well-documented, but also validated across all dimensions.

Follow-up Meetings

It is important to note how each project evolved within the framework. The overview below describes the main milestones in the refinement and validation of each initiative.

Automatic Order Settlement

The process continued on August 15, with an initial validation of the documentation involving the Tech Lead. This confirmed the shared understanding and alignment across business and technology.

On August 19, the PO collaborated with the UX team, officially submitting the initiative for design elaboration. Then, on August 20, the PO and Tech Lead worked on the definition of features and timeline, marking the beginning of the transition to Stage 6 – Project Alignment.

The final documentation was formalized and sent by the PO on September 1, consolidating the learnings and refinements from previous steps.

Complete Assignor – Drawee Risk

On August 15, the PO worked independently on the drafting of the refinement documentation, which was validated in a session with the Tech Lead and business stakeholders on August 18.

A key moment occurred on August 19, when the PO, Tech Lead, and development team defined the initial set of development features, bridging business needs with technical feasibility.

On August 21, a cross-functional session with the UX team enabled the refinement of wireframes, further clarifying visual and user experience requirements. This was followed by a validation session on August 28, involving UX and devs, ensuring that design deliverables were aligned with the defined rules.

The PO formally submitted the refinement documentation on September 1, closing the cycle of Stage 5 and enabling a structured transition to Stage 6 – Project Alignment.

Limit Reallocation

However, during a follow-up meeting on August 12, involving business and credit stakeholders, it was concluded that the demand would be deprioritized in favor of another initiative. As such, the project was temporarily paused, and the focus shifted to an alternative project that would act as a Version 1 of the same solution.

Despite not progressing beyond Stage 5 – Demand Investigation, the documentation and learning captured during this process remain valuable for future iterations.

Solar Credit Taker

The Solar Credit Taker project followed a more concise refinement cycle within FELIX, beginning with an initial understanding session on August 4 between the PO and the business team. This laid the foundation for further exploration.

On August 15, a second meeting focused on the business rules, identifying key process conditions and decision points.

The PO then worked on the refinement documentation, completed on August 20, and brought this draft to a validation session on August 22 with the Tech Lead and business team.

Finally, the PO submitted the formal documentation on August 26, concluding the refinement phase and preparing the project for the alignment and planning stages.

The cases above illustrate how FELIX Stages 5 and 6 enable demands to evolve from abstract ideas into well-structured, validated, and technically aligned initiatives. By the time a project exits Stage 6, it is not only documented, but also endorsed by key areas and equipped with the inputs needed for development planning. This approach reduces ambiguity, minimizes rework, and increases confidence in delivery—setting the stage for successful execution in the next phases.

Metrics

The adoption of the FELIX framework was not only intended to better structure the process of discovery and refinement of demands, but also to create conditions for the organization to measure the effectiveness of this process. Previously, there was no visibility on key indicators, which made it difficult to analyze efficiency and identify areas for improvement. With the implementation of the framework, it became possible to track objective metrics that reflect the transparency, agility, and assertiveness of decisions made throughout the cycle.

The two main metrics analyzed so far are:

- Deprioritization of demands – which measures the quality and maturity of the prioritizations made.
- Average refinement time – which indicates the efficiency of the process in transforming an initial demand into documentation ready for development.

Deprioritization of demands

Before adopting FELIX, there was no clear control over how many demands were deprioritized throughout the discovery process, which made it difficult to analyze the efficiency of the pipeline and the quality of the prioritizations made by the areas involved. With the structure proposed by the framework, it is now possible to explicitly record and measure deprioritization decisions, including when they occur and the reasons behind them.

Of the four requests to which we have applied FELIX so far, one was deprioritized during the initial stages (the Limit Reallocation project, deprioritized by the Board on August 12). This represents a 25% deprioritization rate, a metric that was not previously visible and can now be tracked in future cycles to assess the maturity of incoming requests and the assertiveness of prioritization decisions.

Average refinement time

Another important metric that the framework now enables is the total time between the start of refinement and the delivery of the final documentation, marking the transition from Steps 5 and 6. With this visibility, it is possible to compare the efficiency of the process before and after FELIX and understand the impact of standardization.

Observation: This comparison was made based on an analysis of past meetings and documented historical data.

Projects without historical basis

Project	**Start**	**End**	**Duration**
Project 1	06/26/2024	08/07/2024	42 days
Project 2	02/11/2025	04/15/2025	64 days
Average	–	–	**53 days**

Legend

The acronyms below represent the names of the projects:

- **SCT** – *Solar Credit Taker*
- **CADR** – *Complete Assignor – Drawee Risk*
- **AOS** – *Automatic Order Settlement*

Projects with FELIX

Project	Start	End	Duration
CCRS	08/07/2025	09/01/2025	25 days
LAO	08/06/2025	09/01/2025	26 days
TCS	08/04/2025	08/26/2025	22 days
Average	–	–	**24.3 days**

One clear area for improvement based on the data from the projects analyzed is the average time required to complete the refinement of a demand. Based on historical data, it was observed that, before the framework was applied, the process took an average of 53 days. After adopting FELIX, the average fell to 24.3 days, a reduction of more than 50% in the total refinement time. This decrease in time does not only mean greater agility; it represents a real gain in efficiency and predictability in the demand pipeline.

2.10. Analysis of Results

After the implementation of the FELIX Framework and the completion of the refinement process for the pilot projects, an evaluative stage was conducted to measure its effectiveness and the perception of those directly involved in its use. A post-implementation survey was applied to Product Owners (POs), Tech Leads, developers, and business representatives, aiming to identify whether the framework had produced tangible improvements in daily operations, communication, and demand management.

FELIX Framework Evaluation Survey

We have recently implemented the FELIX framework to structure and improve the refinement process of demands. The purpose of this survey is to evaluate its impact, relevance, and effectiveness in the daily work of those directly involved (POs, Tech Leads, and Developers). Your feedback will help us understand what worked well, what practical improvements FELIX brought, and where we can make adjustments.

This survey is short and should take no more than 3 minutes to complete. Your honest answers are essential for us to keep improving the framework.

1.Your main role in this project:

( ) Product Owner (PO)

( ) Tech Lead

( ) Developer

2.Project being evaluated (choose one):

( ) Automatic Order Settlement

( ) Solar Credit Taker

( ) Complete Assignor – Drawee Risk

( ) Limit Reallocation

3.Which FELIX stages did you personally use? (check all that apply)

[ ] 1 Project Opening

[ ] 2 Risk Classification

[ ] 3 Backlog

[ ] 4 Prioritization

[ ] 5 Demand Investigation

[ ] 6 Project Alignment

[ ] 7 Executive Summary

[ ] 8 Deprioritization

4.From the stages you selected, which were the most useful? (pick up to 3)

5.How much of the Demand Investigation Template (Stage 5) did you actually use?

( ) 0%

( ) ~25%

( ) ~50%

( ) ~75%

( ) 100%

6.Felix brought clarity to what needs to be done.

( ) Strongly disagree

( ) Disagree

( ) Neutral

( ) Agree

( ) Strongly agree

7.FELIX has reduced rework/ambiguity in my daily work.

( ) Strongly disagree

( ) Disagree

( ) Neutral

( ) Agree

( ) Strongly agree

8.FELIX has improved alignment between areas (business, PO, TL, development).

( ) Strongly disagree

( ) Disagree

( ) Neutral

( ) Agree

( ) Strongly agree

9.Perception of refinement time compared to the previous process:

( ) Much faster

( ) Slightly faster

( ) Practically the same

( ) Slightly slower

( ) Much slower

10.The predictability/visibility of the demand status (backlog, next steps) has improved.

( ) Much faster

( ) Slightly faster

( ) Practically the same

( ) Slightly slower

( ) Much slower

11.FELIX increased my autonomy in the process. (for POs: define/refine; for TL/Devs: receive better prepared inputs).

( ) Strongly disagree

( ) Disagree

( ) Neutral

( ) Agree

( ) Strongly agree

12.Would you recommend using FELIX in other projects?

( ) Definitely yes

( ) Probably yes

( ) Not sure

( ) Probably not

( ) Definitely not

13.What helped you the most in FELIX? (open answer)

14.What would you improve in FELIX? (open answer)

Seven participants completed the survey, ensuring that at least one representative from each project under testing was included. This diversity comprising two Product Owners, three Tech Leads, one Developer, and one Business/Desk representative, provided a balanced perspective that covered the technical, operational, and managerial dimensions of the framework's application. The evaluated projects included Automatic Order Settlement, Solar Credit Taker, Complete Assignor – Drawee Risk, and Limit Reallocation, ensuring that FELIX was tested in varied contexts and levels of complexity. The detailed responses and analyses are presented in **Appendix D – Framework Evaluation**.

Main Findings

The analysis of responses revealed a consistent pattern of positive perception across participants. The stages Demand Investigation (Stage 5) and Project Alignment (Stage 6) were the most used and considered the most useful. This convergence demonstrates that FELIX's greatest impact lies in its early stages, where requirements are clarified, business rules are defined, and alignment between areas takes place. These steps were repeatedly described as crucial for improving communication, ensuring predictability, and minimizing misunderstandings during discovery.

All respondents highlighted that the standardized documentation generated in the Wiki, especially during Stage 5, was 100% useful for project execution. The documentation ceased to be a bureaucratic record and became a practical tool, actively used for decision-making, validation, and follow-up. This shift reinforces one of FELIX's main objectives: to turn documentation into a functional artifact that supports collaboration and traceability.

Questions related to clarity, reduction of rework, and cross-area alignment received unanimous agreement. Participants stated that FELIX reduced ambiguities, facilitated understanding between business and technology, and streamlined discussions. This result validates the framework’s central premise, to act as a bridge between strategic and technical spheres through a structured and transparent flow.

Perceptions about refinement time were generally positive: four participants reported that the process had become faster (two “much faster” and four “slightly faster”), while one said it remained the same and one was neutral. Although not unanimous, this indicates that speed gains are a secondary effect, while the primary benefit lies in clarity, structure, and quality of refinement.

Every participant agreed that the visibility and predictability of the demand status had improved significantly. The organized backlog structure, with defined stages and clear transitions, allowed for better monitoring of priorities and progress, increasing stakeholder confidence and transparency across teams.

The dimension of autonomy also showed strong results. Product Owners reported greater independence to define and refine demands, while Tech Leads and Developers stated that they now received clearer and more consistent inputs, reducing the need for rework. This redistribution of responsibilities within the framework contributed to more fluid collaboration and reduced bottlenecks.

Finally, the acceptance and recommendation rate was high: five respondents said they would “definitely” recommend using FELIX in other projects, and one said “probably yes.” This unanimity reinforces the practical relevance and perceived value of the framework in daily operations.

Qualitative Insights

Open-ended responses highlighted three main strengths:

- Centralization and standardization of information, ensuring that all data are easily accessible and structured in one place (Wiki + Monday).
- Clarity of roles and responsibilities, which clarified “who does what, when, and how.”
- Practical, objective deliverables, serving both as evidence of work performed and as execution guides for the next stages.

As for improvement opportunities, four main suggestions emerged:

- Increase communication focus to avoid dispersion across multiple channels.
- Provide a macro-level view for large and strategic projects. Enhance visibility of activity transitions through automatic notifications.
- Broaden access to the outputs of each stage for all involved parties.

These suggestions point to the next evolutionary cycle of FELIX: expanding from an operational framework for demand refinement to a strategic governance tool, capable of integrating macro-level visibility and communication alignment.

Quantitative Summary (Consolidated)

Indicator	Key Result
Usefulness of Demand Investigation documentation	100% rated as useful
Clarity provided by FELIX	100% agreed or strongly agreed
Reduction of rework	100% agreed or strongly agreed
Improved alignment between areas	100% agreed or strongly agreed
Refinement speed improvement	66.7% reported faster process
Visibility and predictability	100% reported improvement
Increased autonomy	100% agreed or strongly agreed
Recommendation for use in other projects	100% positive responses Interpretation of Results

The results confirm that FELIX fulfills its main objective: to bring structure, clarity, and autonomy to the discovery process of demands. The framework effectively established a common language among business, technology, and product areas, making interactions more predictable and decisions more informed.

The evidence suggests that FELIX's value lies in transforming an informal, reactive discovery process into a formalized, proactive, and traceable one. By creating a standardized method that is adaptable to different project contexts, it strengthens governance while maintaining agility, a balance that is difficult to achieve in complex organizations.

The consistency of responses across different roles reinforces that FELIX is not perceived as an additional layer of bureaucracy, but rather as a facilitator of understanding, alignment, and flow. Its simplicity, combined with adaptability within Monday and the Wiki, enables practical use without the need for heavy technological investment.

2.11. FELIX: Results and Learnings

The application of the FELIX Framework demonstrated throughout its implementation and testing that the model consistently fulfills its role as a structure for discovering, refining, and organizing demands. The results confirm what has already been experienced in daily practice: for products and projects to evolve efficiently, it is essential to have clear, structured, and reliable operational processes, and FELIX delivered precisely that.

From the outset, the framework proved effective in organizing and standardizing the demand refinement process, reducing communication noise, and strengthening the autonomy of Product Owners (POs). Analysis of survey responses from POs, Tech Leads, developers, and business representatives reinforces that FELIX generated tangible benefits, consistently perceived across roles.

The main conclusions of the study indicate that:

- The Demand Investigation and Project Alignment stages were highlighted as the most relevant and useful, confirming that FELIX's greatest impact occurs in the early stages of the discovery process, where problem clarity and cross-area alignment are defined.
- Standardized documentation, particularly when centralized in the Wiki, was considered 100% useful by participants, transforming bureaucratic records into practical tools for execution, communication, and decision-making.
- There was unanimous perception of improved visibility and predictability in demand management, demonstrating that the framework successfully establishes a transparent and traceable structure for project oversight.
- All participants reported increased autonomy: POs gained confidence and ownership in conducting refinements, while Tech Leads and developers received clearer, more consistent inputs, reducing rework and misunderstandings.
- While some participants noted time savings in the refinement process, FELIX's primary contribution was in clarity and quality, with speed emerging as a secondary benefit.

These results demonstrate that the framework fully meets its operational role, making teams' daily work more organized, efficient, and predictable. At the same time, FELIX offers additional potential: it enables a strategic, integrated, and intelligent perspective on demands, allowing decisions and priorities to connect with broader organizational objectives. Limiting FELIX to purely operational use would be a waste of its potential, yet for operational focus, it is fully self-sufficient and highly effective.

The choice of Monday.com as the core platform proved decisive for successful adoption, ensuring low resistance and enabling automation of forms, columns, and flows that structure the framework's stages. Meanwhile, the Wiki served as a centralized repository, consolidating governance and facilitating access to information for all stakeholders.

In summary, FELIX has proven to be a functional, adaptable, and replicable framework, capable of:

- Improving the quality of demand discovery and refinement;
- Strengthening Product Owner autonomy;
- Promoting smoother integration between business, technology, and governance teams;
- Increasing transparency and predictability in prioritization and execution;
- Reducing excessive reliance on individual technical figures in structural decisions.

Looking ahead, the framework shows potential for evolution along three main dimensions:

1. Evolutionary – by incorporating strategic indicators to measure business value and alignment;
2. Technological – through automation and integrated BI dashboards to monitor efficiency and impact metrics;
3. Cultural – by reinforcing the PO's strategic role as a key player in shaping products and solutions, not merely managing tasks.

In conclusion, FELIX proves and demonstrates its operational value in practice, making daily work more efficient and structured, while also enabling a more strategic, integrated, and intelligent approach to demand management. It presents itself not only as a robust operational solution but also as a framework with potential to evolve, supporting strategic decision-making and maximizing organizational impact.

2.12. Data and Strategic Insights

Following the conception, implementation, and analysis of the FELIX Framework, the next step identified was the evolution toward **strategic reporting and data intelligence**. Once the framework successfully organized the demand flow, standardized documentation, and consolidated key information, such as business objectives, risk levels, and demand typology, it became evident that these datasets could be leveraged to generate **automated insights** for different audiences within the organization.

Currently, the team manually produces at least three key reports in PowerPoint, requiring all POs to pause their ongoing work for data collection and presentation updates. These reports include:

- **Monthly Deliverables Report:** lists all deliveries planned for the month.
- **Quarterly Projects Overview:** highlights key projects, their completion percentage, and delivery forecast.
- **Quarterly Outlook:** presents the main new products and features expected in the upcoming quarter.

This manual process, although valuable for leadership visibility, has become increasingly time-consuming and dependent on individual efforts. With FELIX already consolidating operational data, the opportunity arises to transform these manual presentations into automated, dynamic reports that can be generated directly from the framework's database and exposed through an integrated dashboard environment.

This chapter, therefore, presents a proof of concept (PoC) focused on exploring how to:

- Analyze and structure the available data within FELIX for reporting purposes;
- Identify possible data combinations and metrics that best represent strategic performance and delivery progress;
- Segregate and tailor reports for different audiences such as the operational team, area leadership, and partners;
- Automate report generation to reduce manual effort and eliminate redundant tasks;

The PoC aims to demonstrate that, by leveraging the structured data already captured by FELIX, it is possible to establish a data-driven reporting layer capable of delivering consistent, real-time, and stakeholder-oriented insights, reinforcing the framework's evolution.

The first stage of this initiative focuses on identifying the most suitable tool to integrate with the existing FELIX process and data sources. The initial analysis compares Power BI and Monday.com as potential platforms for report generation, considering criteria such as integration capability, automation potential, usability, and visualization flexibility. This evaluation determined which environment better supports the framework's evolution toward automated reporting.

2.12.1 Platform Evaluation: Power BI and Monday

With the consolidation of the FELIX Framework as an operational structure for project discovery and demand refinement, a new challenge emerges: transforming the data generated in Monday.com into strategic information and executive reports capable of supporting decision-making at different organizational levels.

2.12.1.1 Power BI

In this context, Power BI appears as one of the most promising technological alternatives that could address this need, integrating FELIX's operational control with a layer of business intelligence and data analysis.

Currently, BTG Pactual provides the basic Power BI license (Free) to all employees, ensuring broad access to the tool's essential functionalities, such as creating dashboards, building dynamic reports, and integrating with multiple data sources. This licensing policy could facilitate institutional adoption and reduce barriers for expanding BI use within the FELIX context.

For teams that may require more complex analyses or advanced automation, it would be possible to request Power BI Pro (around US$10 per user/month, approximately R$55) or Power BI Premium (about US$20 per user/month, or R$110). These versions would significantly expand storage capacity, refresh frequency, and report performance, while also enabling the use of AI features, machine learning, and predictive modeling.

From a functional perspective, Power BI could provide a set of capabilities directly aligned with the evolving needs of the FELIX Framework:

- Integration with Monday.com and other data sources: the tool connects via API, native connectors, or intermediary files (such as Excel and SharePoint), allowing automatic synchronization of demand, risk, classification and status data that is currently registered inside FELIX.
- Data modeling and transformation: through Power Query and DAX, it would be possible to clean, combine, and standardize data exported from Monday, ensuring consistency for analysis and the creation of strategic indicators.
- Executive visualizations and reporting: Power BI could enable the development of interactive dashboards that clearly and visually represent portfolio status, demand prioritization, and delivery progress. This functionality would directly support the future goal of generating automated executive reports from FELIX's operational data.
- Automation and sharing: reports could be automatically refreshed and securely shared, integrating with the Wiki or internal portals, thus centralizing information governance and promoting organizational transparency.

From a strategic standpoint, adopting Power BI as an extension of FELIX would represent an important step forward. While FELIX structures processes and centralizes operational data in Monday.com, Power BI would act as the analytical and visualization layer, capable of transforming these records into business insights and management dashboards. This integration could enable the creation of automated executive reports, consolidating operational and strategic information, and allowing different organizational levels, tactical and executive, to monitor progress and value generation.

These reports would not only increase operational efficiency, by reducing the manual workload of Product Owners and Tech Leaders in data consolidation, but also strengthen FELIX's role as a source of corporate intelligence, allowing strategic decisions to be based on consistent, traceable, and real-time information.

However, it is important to acknowledge some potential limitations associated with adopting Power BI:

- The Pro license imposes storage and update rate restrictions, which may limit its use in large-scale projects or those requiring continuous monitoring; sice we have to manage 38 boards and expectations are growing, perhaps the free license will not be sufficient to handle the storage needs.

- Integration with Monday.com, while feasible, would require configuration via API or third-party connectors, demanding technical oversight and attention to data governance;

But, from a cost-benefit perspective, Power BI stands out as a high-value, low-cost solution, given that it is already corporately licensed by BTG and could integrate seamlessly into the institution's existing ecosystem. Its learning curve is relatively low, and its widespread internal use would facilitate the dissemination of the dashboards produced.

In summary, Power BI emerges as the most promising tool to support the analytical and strategic evolution of the FELIX Framework. The combination of the two platforms could create a complete and integrated data journey, in which:

- FELIX captures and structures operational information from demands within Monday.com;
- Power BI processes, analyzes, and transforms that data into managerial and executive reports, offering visibility and metrics at multiple levels.

Thus, the integration between FELIX and Power BI would represent the next natural step in the framework's maturity—a progression that could transform operational control into strategic intelligence, promoting greater transparency, efficiency, and alignment between tactical and executive levels within the organization.

2.12.1.2 Monday

Monday.com, although currently the operational basis of the FELIX Framework, could also evolve to perform part of the desired analytical and strategic role, reducing dependence on external tools. This would allow FELIX to operate end-to-end within a single platform, from recording demands to viewing indicators, keeping all teams within the same environment and eliminating additional integrations.

Monday already offers native visualization, automation, and integration features that can be leveraged to generate reports, charts, tactical dashboards, and status tracking. Its simpler, more visual, and intuitive experience could accelerate adoption by squads, especially by non-technical users, who would have more autonomy to consume and manipulate information directly on FELIX boards and dashboards.

From a licensing perspective, Monday would not generate any immediate additional costs, as it is already contracted and part of the official framework structure. The use of native dashboards, internal automations, and visual widgets would be included in the tool's existing capabilities, which represents an advantage in terms of cost and centralization when compared to Power BI's extra license model.

Just as Power BI meets strategic needs, Monday could offer specific benefits to the FELIX context:

- **Native and immediate integration with operational boards:** Monday dashboards consume data directly from boards, without APIs, intermediate files, or connectors, ensuring instant updates and reducing the risk of discrepancies between operations and indicators.
- **Tactical visualizations and day-to-day monitoring:** graph, timeline, workload, and status widgets are useful for operational and short-term analysis, helping the PO and Tech Lead track flow, bottlenecks, batches at risk, and delivery forecasts.
- **Integrated automation:** Monday can automate notifications, move items between boards, trigger responsible parties, and create automatic workflow logic, something that Power BI would not do because it is only analytical.

- **Simpler governance:** dashboards and information are in the same environment where demands are created, reducing friction, accelerating onboarding, and avoiding tool dispersion.

On the other hand, it is necessary to recognize some relevant limitations of Monday in this approach:

- The dashboards are more limited in analytical depth and do not easily support predictive analytics, advanced BI, or complex data modeling;
- The data displayed in Monday depends on the structure of the boards, that is, the level of analysis only goes as far as the modeling of columns and statuses allows;
- The visual features serve tactical purposes well, but may be insufficient for executive reports, portfolio consolidations, or multi-layered analyses.

From a cost-benefit and ease-of-adoption perspective, Monday stands out as the leanest and most straightforward alternative. It reduces layers, eliminates additional integrations, and keeps tracking entirely within the FELIX environment.

In summary, while Power BI emerges as the best path for strategic analysis, executive reporting, and advanced business intelligence, Monday positions itself as the ideal platform for a complete tactical-operational cycle, in which:

- FELIX records, controls, and moves demands;
- Monday itself displays indicators, statuses, and monitoring dashboards, supporting daily decision-making and workflow governance.

2.12.2 Implementation Test: Power BI and Monday

Currently, we have three main reports: 1) Monthly Deliverables Report; 2) Quarterly Projects Overview; and 3) Quarterly Outlook, which use only a fraction of the information we collect to prioritize demands.

These reports have different objectives and, therefore, require different levels of detail:

Monthly Deliverables Report

- Short-term focus. Presents planned and completed deliveries for the month, highlighting status, timeline, and impacted products.
- Required information: demand name, product/team, category (Feature or New Products), timeline, and week of execution.

Quarterly Projects Overview

- Focus on execution and progress for the quarter. Shows the progress of key projects, percentage of completion, launch forecast, and upcoming milestones.
- Required information: status (%), demand name, product/team, launch forecast, forecast for communication with customers and partners.

Quarterly Outlook

- Focus on planning and future vision. Consolidates expected projects and initiatives per quarter, categorizing by type (new product or new features) and product/team, in Gantt chart format.
- Required information: demand name, product, estimated period (timeline), and category.

Although the three reports provide visibility into deliveries, each uses only part of the information gathered in the initial phase. In addition, some reports contain data that is not necessarily related to the refinement of demands or that is not yet collected today. This reinforces the opportunity to standardize and improve the collection and integration of information.

Next, we will demonstrate how these reports perform in Power BI and Monday views to assess which format best meets current reporting demands. This is a proof of concept, using fictitious data for illustrative purposes only. In practice, it will be necessary to:

- Use real data to track demands;
- Make the appropriate connections via API between Power BI and Monday, if Power BI is adopted;
- Or, if the option is to keep only Monday, ensure the complete integration of the necessary data within the platform itself.

2.12.2.1 Power BI

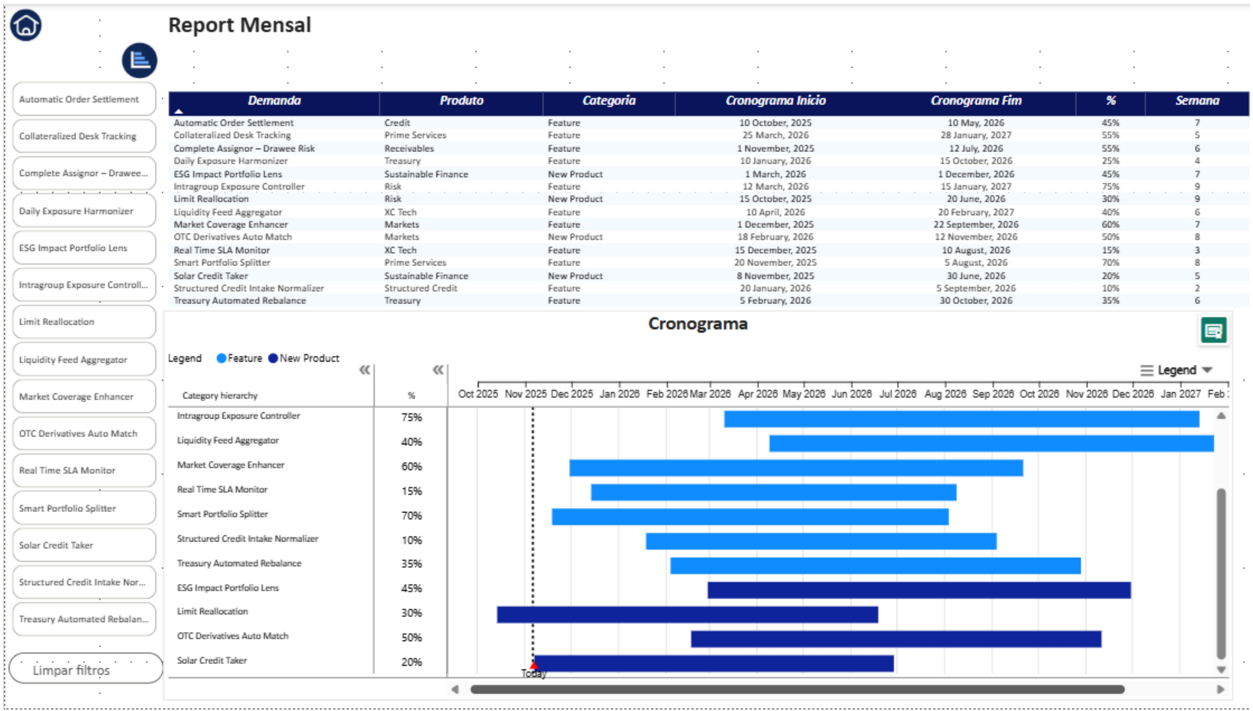
Power BI was evaluated as an alternative for the creation of executive reporting and strategic views derived from the information structured by the FELIX Framework. While Monday serves as the operational platform — controlling the discovery, classification, and prioritization cycle — Power BI acts as a complementary analytical layer capable of consolidating, correlating, comparing, and visualizing these data to support tactical and executive decision-making.

For this stage, a Proof of Concept (PoC) was built using mocked data (not extracted from Monday). The simulated dataset was modeled using the same column structure adopted in FELIX, ensuring representation of realistic scenarios without exposing confidential internal information. This approach reinforces information security while allowing the demonstration of analytical potential in a realistic, risk-free manner.

Based on this simulated dataset, it was possible to construct interactive executive dashboards, in which selecting a specific demand automatically displays its forecasted quarter, associated product, critical notes, completion percentage, and future communication dates. This experience resembles an “Executive Cockpit”, the type of visualization typically used by Heads, Directors, PMOs and governance areas for strategic decision-making.

Figure 22 presents the Monthly Report. This view focuses on short-term horizon and upcoming deliveries within the current execution cycle.

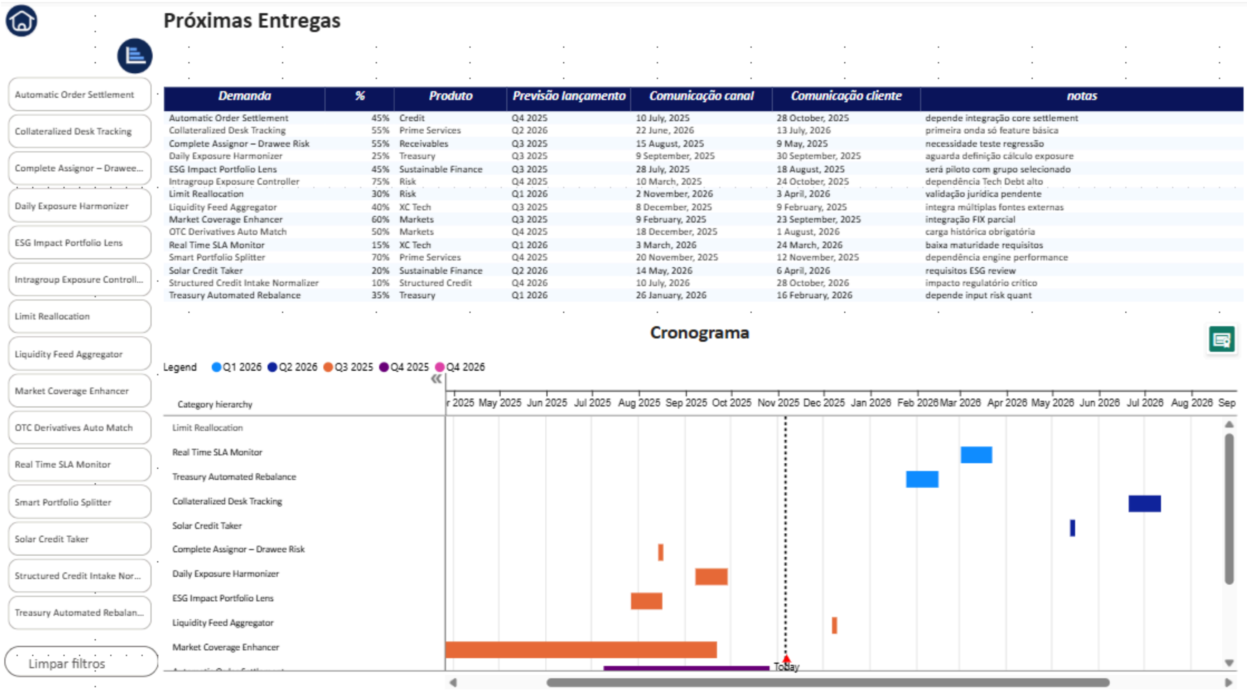
FIGURE 22 – Monthly Report (Power BI)



This panel enables Product Owners and Heads to monitor installed capacity, speed of execution, and delivery tendency without relying on static presentations — promoting continuous and dynamic follow-up.

Figure 23 presents the Next Deliveries panel, oriented to forward-looking strategic view. Here, the central dimension is not the current month, but future context — quarter, initiative category, and pipeline maturity.

FIGURE 23 – Next Deliveries (Power BI)



When a demand is selected, the dashboard automatically displays its forecasted Quarter, notes, projected communication dates and completion percentage. This type of visualization is especially relevant for prioritization, capacity negotiation, scenario evaluation and preparation for external milestones (market communication, partner announcement, regulatory interaction, etc.).

Therefore, even without the real data integration at this moment, Power BI demonstrates strong potential to transform FELIX into a corporate intelligence source. In the future, these dashboards could be automatically fed via Monday API, eliminating the need for manual PPT and spreadsheet consolidation. In this architecture, Monday remains the operational layer of FELIX, while Power BI becomes the analytical and executive layer — enabling FELIX to evolve from a purely operational framework into a strategic, data-driven decision support model.

2.12.2.2 Monday

To create and maintain executive reports in Monday, we use Monday WorkDocs, a collaborative feature that allows us to centralize information, insert charts, pivot tables, and embed visualizations directly from the platform's boards.

WorkDocs acts as a living space for documentation and monitoring, enabling teams to update data in real time without the need for manual exports or static presentations.

Creating a WorkDocs:

To structure the executive reporting environment, a document was created from Monday's Template Center. The step-by-step process was as follows:

1. Access Monday.com with corporate credentials.
2. In the side menu, click on Template Center.
3. Select the Documents category.
4. Choose the “Introduction to Monday.com WorkDocs” template.

From this template, begin customizing the document by adding sections and integrating the necessary widgets (such as tables and visualizations of each Product Owner's boards).

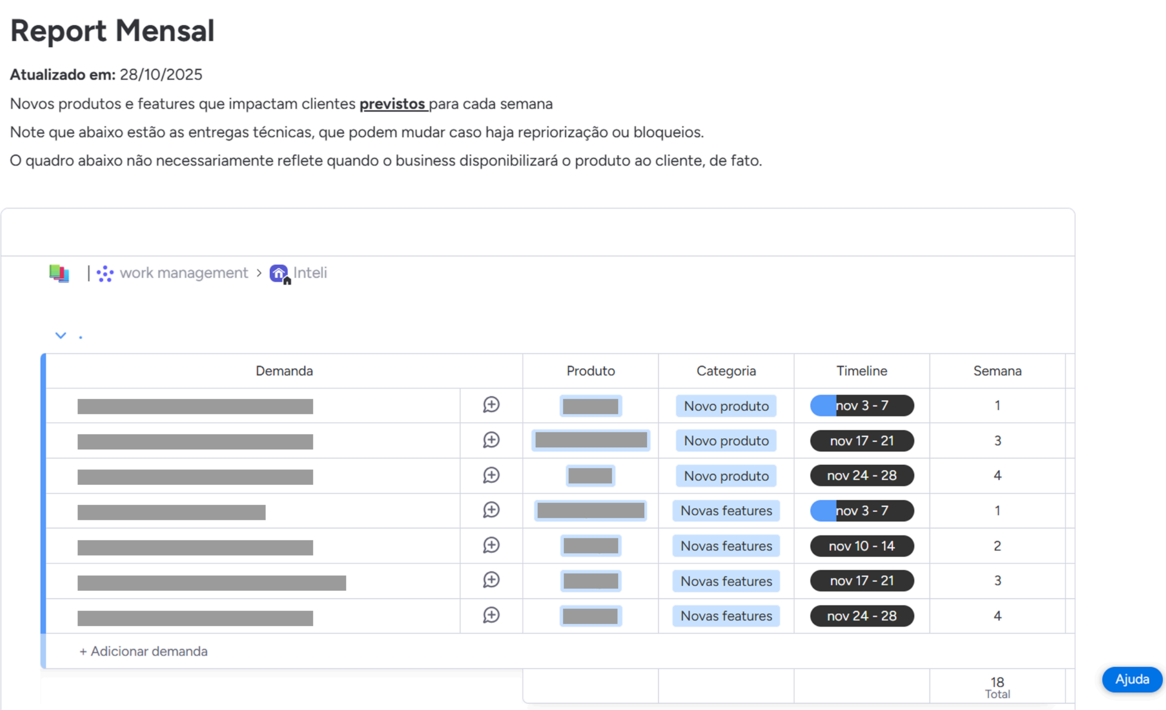
Monthly Deliverables Report

The Monthly Deliverables Report is a short-term report focusing on monthly deliveries. It presents new features and products that impact customers, organized by expected delivery week, as illustrated in Figure 24.

This overview reflects the technical planning of the squads and allows for agile monitoring of execution, considering possible reprioritizations or technical blockages.

The report is automatically populated from the original boards, ensuring that the Head of Product and POs can monitor progress in real time.

Figure 24 – Monthly Deliverables Report

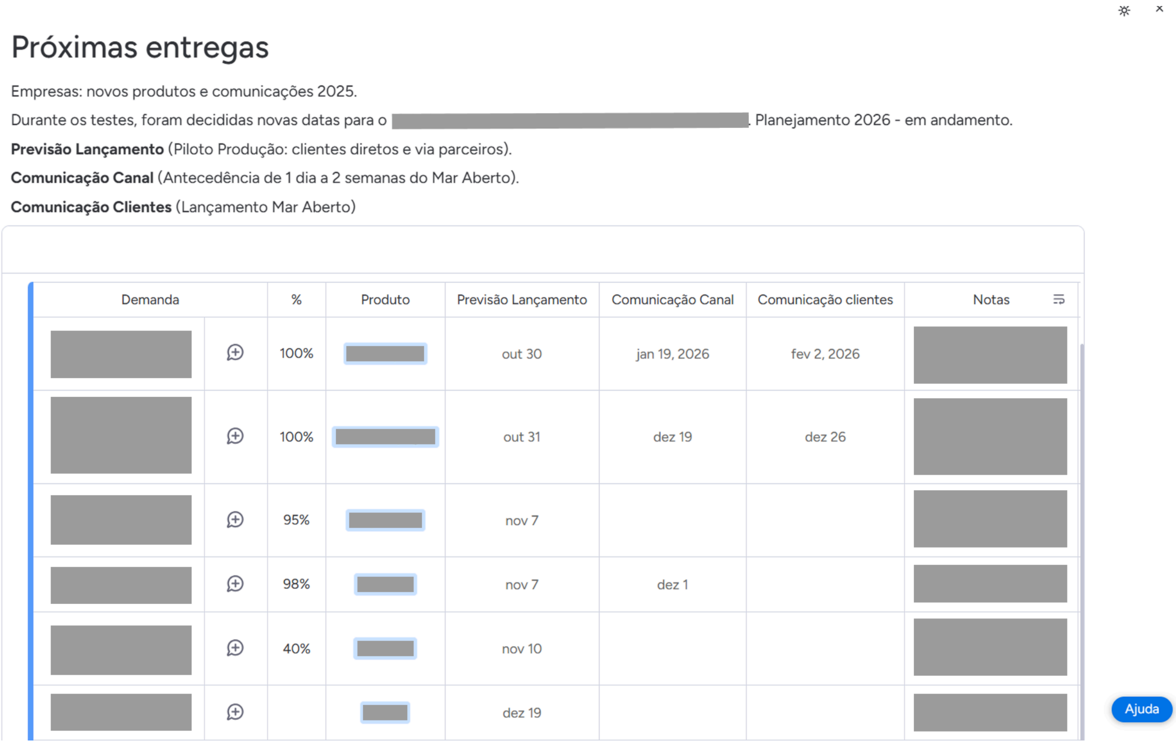


Quarterly Projects Overview

The Quarterly Projects Overview is a quarterly execution and progress report that shows the percentage of completion of demands, release forecasts, and planned communications. The information is grouped by product and category, as shown in Figure 25.

The dates are entered manually, and the report is used in meetings conducted by the Head of Product with the business team. This eliminates the need to manually update presentations, as the content is automatically synchronized with the POs' boards, including the percentage of progress for each demand.

Figure 25 – Quarterly Projects Overview



Próximas entregas

Empresas: novos produtos e comunicações 2025.

Durante os testes, foram decididas novas datas para o [redacted]. Planejamento 2026 - em andamento.

Previsão Lançamento (Piloto Produção: clientes diretos e via parceiros).

Comunicação Canal (Antecedência de 1 dia a 2 semanas do Mar Aberto).

Comunicação Clientes (Lançamento Mar Aberto)

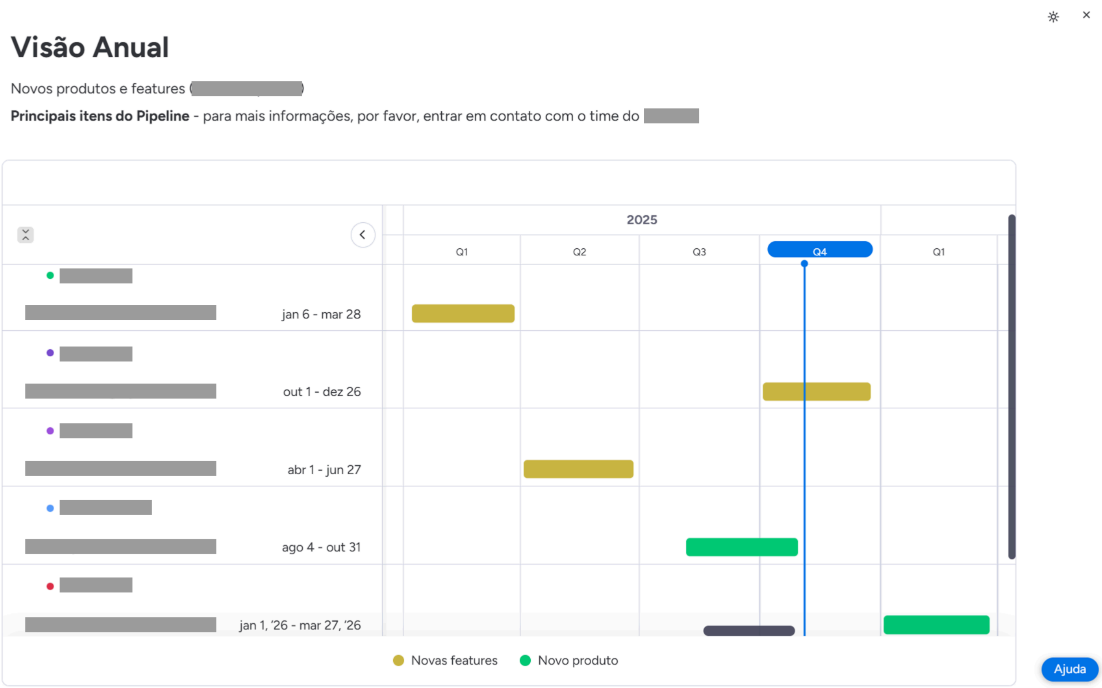
Demanda	%	Produto	Previsão Lançamento	Comunicação Canal	Comunicação clientes	Notas
[redacted]	100%	[redacted]	out 30	jan 19, 2026	fev 2, 2026	[redacted]
[redacted]	100%	[redacted]	out 31	dez 19	dez 26	[redacted]
[redacted]	95%	[redacted]	nov 7			[redacted]
[redacted]	98%	[redacted]	nov 7	dez 1		[redacted]
[redacted]	40%	[redacted]	nov 10			[redacted]
[redacted]		[redacted]	dez 19			[redacted]

Quarterly Outlook

The Quarterly Outlook focuses on future vision and strategic planning. It presents the main items in the pipeline of new products and features, organized by quarter (Q1–Q4) and categorized according to the type of initiative (new product or new feature).

This visualization, shown in Figure 26, provides a consolidated view of the pipeline and facilitates discussion about prioritization and resource allocation.

Figure 26 – Quarterly Outlook



Even though this environment was developed as a proof of concept, it was already connected to real boards from Monday, allowing the retrieval of live data and validating the model's applicability in practice.

This means that the proposed structure could be implemented immediately if needed, since the integration with the boards and dashboards is already configured and functioning. Moreover, the data model is flexible and easily adaptable, making it possible to adjust metrics, views, and aggregation levels according to future reporting needs.

With these three integrated views, Monthly Deliverables Report, Quarterly Projects Overview, and Quarterly Outlook, the use of Monday WorkDocs has consolidated a continuous and collaborative executive reporting model, in which data is updated in real time from operational boards and eliminates the need for static materials.

This ensures greater reliability of information and provides a unique and evolving view of products in an automated way.

Another highlight is its ease of use: by connecting the boards directly to the report views, information is updated automatically and without rework.

In addition, WorkDoc's visual and intuitive format makes report presentation simple and fluid, just scroll through the document to view all sections, allowing for continuous and organized reading of deliveries, progress, and future plans.

In this way, Monday WorkDocs stands out not only as a documentation tool, but as a central hub for monitoring and decision-making, bringing together strategic data, execution status, and medium- and long-term planning in a single environment.

2.12.2.3 Potential use of collected information

In addition to the data currently used in the three main reports, there is a wide range of additional information collected that can be leveraged to strengthen visibility, planning, and decision-making. Fields such as Area Name, Channel Name, Requester, Request Date, Business Driver, Risk, Priority, Discovery Status, among others, make it possible to connect operational deliveries to business strategy.

Strategically, this data can be organized and presented in complementary perspectives:

Operational and performance view:

- Displays metrics related to demand progress, prioritization, risk level, and adherence to deadlines (Status, Risk Rating, Quarter, Desired Quarter, PROD Date).
- Allows you to track delivery predictability, discovery flow, and the balance between planned and executed.

Strategic and business alignment view:

- Uses contextual fields such as Area Name, Channel Name, Business Driver, and Demand Type to identify the origin of demands, their business drivers, and the degree of alignment with organizational objectives.
- Supports portfolio-level analysis, enabling the evaluation of effort concentration and strategic consistency.

Product and experience view:

- Fields such as App, Ux, and Wiki Link allow you to map product evolution, UX involvement, and documentation traceability.
- Strengthens visibility into product maturity, design participation, and knowledge reuse.

These perspectives can be presented through dashboards in Power BI or Monday, depending on the platform adopted. Both allow flexible views by quarter, product, or area, facilitating comparisons between initiatives. This approach transforms the information already collected into a strategic portfolio management tool, offering a broader understanding of how demands are generated, prioritized, and delivered.

In addition, it supports future improvements in data governance, integration, and automation, ensuring that the knowledge produced throughout the demand lifecycle contributes directly to decision-making.

2.12.3 Consolidated Analysis: Academic Evaluation of Executive Report Automation — Monday vs. Power BI

The automation of executive reporting represents a fundamental step in the evolution of the FELIX Framework, expanding its capacity to deliver not only operational organization but also strategic intelligence grounded in data. To enable this transition, it is essential to assess which platform offers the greatest adherence to the objective of transforming the information structured in Monday into consolidated, continuously updated reports that support executive decision-making.

In this context, two platforms were analyzed: Monday.com, currently used as the operational backbone of FELIX, and Power BI, the corporate business intelligence tool adopted across BTG Pactual.

This section presents a detailed, comparative, and academically justified evaluation of which platform is most appropriate for the framework's current maturity, considering process evolution, institutional availability, cost, governance, and adoption barriers.

2.12.3.1 Alignment With Current Use and Process Maturity

One of the decisive elements in choosing the platform is the current operational maturity of the FELIX cycle. Today, all squads, Product Owners, Tech Leads, and business analysts use Monday daily as the single source of truth for registering, tracking, and refining demands. This extensive institutional adoption provides three strategic advantages:

2.12.3.2 Low Adoption Barrier

Because Monday is already part of the daily workflow of all teams, building executive reports directly within the platform:

- reduces the learning curve;
- eliminates the need for additional training;
- minimizes resistance to change
- maintains integration with the existing operational flow.

2.12.3.3 Native Automatic Updates

Since all data in FELIX is created and maintained within Monday, any report built inside the platform:

- updates automatically,
- reflects new information in real time,
- eliminates the need for manual exports,
- reduces human error.

This point is critical: true automation depends on removing manual intervention entirely.

2.12.3.4 Simplified Governance

Monday already centralizes:

- user permissions;
- access controls;
- change history and traceability;
- data versioning.

Moving reporting to an external tool adds risks such as version mismatches, reprocessing requirements, and temporal inconsistencies in indicators. Thus, for the current maturity stage of FELIX, Monday offers superior operational adherence.

2.12.4 Analytical Capacity and Evolution Potential

Although Monday is operationally convenient, Power BI stands out for its analytical robustness. Its architecture allows:

- advanced metric creation using DAX;
- scalable data modeling;
- integration with multiple internal databases;
- historical, predictive, and comparative analyses;
- high-complexity executive dashboards.

These advantages position Power BI as a powerful tool for future expansion. However, its adoption at this stage faces structural limitations:

2.12.4.1 High Technical Dependency

Advanced use of Power BI requires:

- structured datasets;
- update pipelines;
- API/ETL connectors;
- specialized BI/Analytics professionals;
- ongoing maintenance of data models and metrics.

Unlike Monday, Power BI is not plug-and-play and involves significant operational overhead.

2.12.4.2 Licensing Requirements

Although BTG provides Power BI Free to all employees, essential features for executive reporting require:

- Power BI Pro, or Power BI Premium.

These have per-user licensing costs and require formal approval, increasing dependency on corporate governance.

2.12.4.3 Adoption as a Third Layer

Introducing Power BI adds a third layer between:

1. raw operational data (Monday),
2. data transformation,
3. executive visualization.

This increases technical complexity and system dependency. Even though some teams may have the necessary licenses and BI expertise, FELIX — as a newly implemented project discovery framework — has not yet reached the data governance maturity required for such an additional analytical layer.

2.12.5 Consolidated Justification

After examining technical, operational, strategic, and institutional factors, the conclusion is clear: Monday is the most suitable platform for immediate automation of FELIX executive reports. Its advantages include:

- already adopted across all teams,
- seamless organizational fit,
- native real-time updates,
- integrated governance,
- reduced data risk,
- zero rework for updates,
- full support for the three existing reports:
    - Monthly Deliverables Report,
    - Quarterly Projects Overview,
    - Quarterly Outlook.

Power BI remains strongly recommended for a future stage, but not as the initial automation layer. This is because it would require:

- specialized licensing and permissions;
- BI/Analytics governance;
- technical setup and data modeling;
- more complex maintenance;
- changes to the existing operational flow;
- a level of organizational readiness not yet consolidated in FELIX.

The hybrid, phased model maximizes value, reduces risk, and evolves alongside FELIX's maturity. Thus, Monday should serve as the immediate automation layer, while Power BI becomes the natural long-term analytical evolution, strengthening governance, strategic visibility, and decision-making quality across the organization.

3. Conclusion

This study aimed to propose, apply, and evaluate the FELIX Framework as a solution for structuring and governing the demand discovery and refinement process in a complex organizational environment, from an Information Systems perspective. The research was conducted within the context of BTG Pactual, where limitations related to process informality, information fragmentation, and low decision traceability were identified throughout the demand lifecycle.

The FELIX Framework was designed as a socio-technical artifact, integrating defined processes, standardized information artifacts, clearly established organizational roles, and technological support. Its implementation structured the demand flow through formal stages, centralized documentation, and the use of institutionally established platforms, specifically Monday.com and the corporate Wiki. In this way, an information system was established to support both operational execution and managerial monitoring of initiatives.

Framework validation was carried out through pilot projects and the execution of a structured ten-week test plan, encompassing objective criteria, quantitative metrics, and qualitative assessments. The results obtained indicate relevant gains. The average demand refinement cycle time was reduced from 53 days to 24.3 days, representing a reduction of over 50% in the total refinement cycle. In addition, data collected through questionnaires applied to participants showed unanimous agreement regarding improvements in demand clarity, reduction of rework, increased alignment between business and technology areas, enhanced visibility of demand status, and strengthened autonomy of Product Owners.

The results also indicate that the most significant impact of FELIX occurs in the early stages of the process, particularly during the Demand Investigation and Project Alignment phases. In these stages, information formalization contributed to clearer definition of requirements, business rules, and dependencies, reducing ambiguities in subsequent phases. Standardized documentation, centralized in the Wiki, evolved into an active informational resource, supporting execution, cross-area communication, and decision-making processes.

From an Information Systems governance perspective, the framework promoted greater transparency and predictability without introducing excessive complexity. Adoption based on tools already in institutional use reduced entry barriers, minimized resistance to change, and facilitated practical application. Consistency in positive perceptions across different roles involved indicates that FELIX contributed to balancing structural rigor with operational flexibility.

The study also examined the evolution of the framework toward executive reporting automation and strategic use of generated data. A comparative evaluation between Monday.com and Power BI indicated that, considering the current stage of data governance maturity, Monday.com presents greater suitability for immediate automation due to its native integration, real-time updates, and simplified governance. Power BI was identified as an appropriate alternative for a future phase, contingent upon increased organizational maturity, data structuring, and governance consolidation.

This gradual approach aligns with core principles of Information Systems architecture oriented toward organizational capability.

From an academic standpoint, this work contributes by presenting a structured, replicable, and adaptable framework that integrates concepts of demand management, process modeling, governance, and information architecture, applied within a real organizational context. From an organizational perspective, the results demonstrate that the adoption of information systems aligned with well-defined processes and governance can generate concrete gains in efficiency, predictability, and decision quality.

It is therefore concluded that the FELIX Framework met the proposed objectives and produced consistent results within the analyzed context. As future work, the expansion of the framework's application to other organizational scenarios is recommended, as well as the incorporation of additional strategic indicators, historical analyses, and predictive approaches. Nevertheless, within the scope of this study, FELIX proved to be a relevant Information Systems artifact, with potential to evolve from an operational solution into a decision support system, contributing to the strengthening of governance and organizational maturity.

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Appendix

Appendix A - POs survey answers

Survey 1 - 17/02 to 28/02

How do requests reach you?

(Describe where the requests come from, who participates in this process and how the demands are assigned)

Person 1 : Ideally, they would arrive via a list of priorities classified in numerical order, but this didn't happen in practice, so armed with this material, every quarter I would pass on what I understood from internal conversations about what the priorities were, based on the company's strategy, and from this we would do a pre-refinement, classifying the size of the effort and its possible dependencies on other teams. Before the PI, we formalized it via email, understood whether our enablers would also be able to tackle these priorities and, if necessary, reallocated the priorities. By the time the PI took place, everything had already been defined, everyone involved was aligned and all we had to do was formalize the demands that would be worked on during the quarter and what the expected results were, answering what each feature would solve and what the expected result would be.

Person 2: Mostly from the Business/Commercial team.

Person 3: Ideally, the main requests are defined before the start of the year, according to the strategic planning of each product for the year, and broken down in a macro way in the quarters. This way, we already have a forecast of the items and at each sprint review we review the prioritization of demands in a more micro way. However, business stakeholders often have “off-plan” requests, mainly from commercial contacts in which customers report problems or opportunities for improvement. Also, as the competition moves, the business team ends up needing to launch new features/products to keep up with the market, which often leads to replanning. The demands usually come from the business team, but there are also technical demands that the IT team itself prioritizes. Also, partner areas such as customer service, marketing, operations and UX may have unplanned requests that need to be prioritized. The assignment of demands takes place within each squad. Each developer/designer has a few products of greater expertise, and the demands are divided between the focal points of each product, according to their delivery capacity and the complexity/effort of each demand.

Person 4: The demands are defined by the business team

Person 5: As a rule, the demands I work on come through oral requests or messages/emails sent by my manager.

When a demand comes in, is it completely defined or do you need to refine it?
(Explain whether demands come ready-made or whether there is room for improvement of the problem)

Person 1: I don't remember fully defined demands. In refinement, you explore scenarios that were not clear without this step. This is the time to explore success and failure scenarios and, above all, to understand whether the solution is just for the team to build or whether it has dependencies.

Person 2: Mostly refinement is needed.

Person 3: Most of the time, there is plenty of room for refinement. The more complex the demand, the more refinement time is needed. Only minor adjustments/improvements come with complete or almost complete understanding. It usually takes one or several calls of understanding before we arrive at the desired delivery properly refined.

Person 4: Needs refining, from understanding the business requirements, what the business team and the client expect, to refining technical requirements with the development team or other areas involved.

Person 5: Most of the time, the demands arrive ready-made, but in some specific cases, they need to be improved and a greater understanding of the objective and the needs to be met is required.

How do you refine demands on a daily basis?

(Describe your refinement process, what steps you follow, who participates and what tools or methods you use).

Person 1: For the planning of the quarter, we had to wait for confirmation of this list of priorities, which happened about 15 days before PI Planning, but throughout the quarter, we had a recurring schedule of weekly refinements with the teams. At this meeting, we looked at the health of the development that was underway, understanding whether the schedule that had been drawn up when the topic was prioritized was adhering to expectations and also, when possible, taking the opportunity to pull out the increments of our deliveries, for example, in one quarter we were working on an MVP, what the next step was and what to expect from this increment. We also looked at the demands in the prioritization funnel that hadn't been prioritized previously due to lack of capacity. The risk in this second example is that, over the course of the quarter, this backlog loses priority due to any other internal or external factor, such as a regulatory demand, for example.

Person 2: I start by calling in the teams that will provide the input prior to IT, such as Legal for legal validation of what we intend to develop, Fraud to check that the product we are thinking of won't generate security gaps, Finance, Tax, Commissioning (if the product is sold by partners), Billing if the bank chooses to charge the customer fees for using it, always accompanied by the business team. At each meeting, I document what has been agreed in a WIKI and formalize it via email. From then on, I start drawing in BPMN to show what the customer journey will look like and which internal systems will be involved. teams, etc. Usually in the midst of refinement, new needs arise and new teams are activated. Once we have reached a consensus and everyone knows which activities they will be responsible for, I use monday to manage progress and control dates.

Person 3: Refinement always involves the requesting business stakeholder, to explain the need and expected outcome, as well as defining the business rules. Also, the technical lead responsible for the product, to understand the technical needs, solution possibilities and effort involved. Ideally, we also involve someone from UX as soon as possible, so that they are already aware of the needs. The process involves documenting the understanding/rules, as well as designing the flows involved and mapping the people who need to be involved/areas that will be impacted. Example: will there be an impact on security flows? if so, involve the security squad; does this feature/product involve partner remuneration? if so, activate the fee/remuneration team; and so on.

Person 4: Business alignment meeting, technical refinement, meeting with areas involved and planned delivery

Person 5: Refinement is usually done by contacting stakeholders. When this request is being made to someone from another team, I usually contact the person to better understand what their needs are. When the person in question is my manager, I talk to him in parallel to better understand what he wants done.

Do you feel you have enough autonomy to refine the demands in the best way? Why? (Explain whether you have the freedom to adjust demands as necessary or whether there are restrictions that hinder this process).

Person 1: Sometimes you won't have one and your role is to try to involve those people who do, for example, the business people. It's very important to make them feel that they are an integral part of that development and what their role and expected responsibility is. I've always had the practice of defining the first answers to some key questions in order to continue with the refinements with the teams: What are we doing? Why are we doing it now? What do we expect from this delivery? The answers to these questions make the refinement process easier, the team understands what the problem is to be solved and what to expect, with this we gain engagement and they must come from the business based on aligning a customer pain with the company's strategy. Another very important point is that these questions facilitate the reprioritization process, when a demand in the middle of the quartile comes with a request for prioritization and you have to reprioritize it together with the team. This is the role of the business and you as PO or PM can help by making the impacts clear, for example, will this new demand impact the backlog that expects to deliver these results, does this new demand deliver more value than those prioritized?

Person 2: Refinement is joint, but it's up to the PO to coordinate the whole process and keep everyone aware of what we're agreeing. So the process is not dictated by the PO, but flows as the meetings go by and we find new needs. One factor that can limit the work is the regulatory deadline, in which case there is no room for negotiating deadlines or phasing.

Person 3: It depends on the product/stakeholder. There are some that are already more defined (some stakeholders even make a prototype in the PPT), others that are super raw and the PO ends up having more autonomy and protagonism. During the process, if it is necessary to suggest/adjust the demand, I believe that with a good basis, the PO does have the autonomy to adjust the route. It's very important here to provide the reasons for changing the route. For example: suggesting breaking the delivery down into phases, so as to deliver value over time and iterate/validate hypotheses before arriving at the final delivery (which is often the business team's idea of the ideal, but not necessarily what the client needs).

Person 4: Yes and no, it depends on the type of demand, if it's something very specific to the client there's not much room for change, but if it's not, I'm free to give my opinion, change and even change the delivery priorities.

Person 5: In cases where it's necessary to refine the demand, I feel I have the necessary freedom, but I'm not confident enough to go along with my opinion. Therefore, I always contact my manager to align expectations and ensure that what I have imagined/understood is exactly what he wants done.

What are the main challenges you face in refining demands?

(List the main obstacles that hinder more efficient refinement, such as lack of information, time constraints, lack of alignment with stakeholders, etc.)

Person 1: When there are external dependencies on other teams. Everyone has their own priorities and sometimes they don't fit together. Demands without a defined scope of what is expected, so the team becomes demotivated and lost throughout the process. Lack of time aligned with the company's expectations. When it's the business that sets the deadline and not the team, and because of this lack of time, it's not possible to explore all possible scenarios. Team schedules: many times I've had to cancel a refinement due to a service in Production. In my old scenario, the delivery team was the same as the production team. A lot of changes in direction, which makes it difficult to focus. Several times I had to interrupt a refinement to meet a new demand or problem in Production and when I managed to reprioritize this discovery, the team had already lost the timeline and reasoning and we had to take a step back to understand the demand again, thus increasing the discovery time. Requests to start technical refinement without a prototype. This is a lot of rework, because putting together a service architecture without understanding what is expected to be delivered to the end user in most cases means putting effort into something that will probably need to be revisited.

Person 2: The biggest obstacle is the deadline expected by the business and the pressure generated to get everything out very quickly, which hinders the refinement that often continues to take place while development is underway.

Person 3: I believe that the root cause is always a lack of clarity about the OBJECTIVE of the delivery. What are we doing this for? What is the expected result at the end? I often feel that we are doing and doing, and we don't have a clear objective. Once the objective is clear, we can suggest other ways of getting there. I feel like we're working on the WHAT and not the WHY? Having the WHY, we can redefine the HOW and have more autonomy. In addition, there is often a lack of information about the product, about the need, and time for refinement is tight because of the deadline set for delivery.

Person 4: Failure to align with everyone involved, asking for priority/urgency on a demand that the person doesn't even know what they want done.

Person 5: I believe that, as a rule, the main problem is fully understanding all the stages of the task before starting it. Most of the time, I only discover the complexity of the task once I've started it. Before that, it's very difficult to have that dimension and be able to work in the best way.

Survey 2 - 17/03 a 28/03

How do requests reach you?

(Describe where the requests come from, who participates in this process and how the demands are assigned)

Person 1: It currently comes from the bank's internal areas and also from the leaders in my area

Person 2: They come via the client, usually at a meeting to discuss the project. But sometimes it comes from the leaders of the team that worked on it, like a top down.

Person 3: Through management tools with agile dashboards (Scrum and Kanban)

Person 4: The service areas are determined by the strategic leadership of the team and requests come directly from the customer, either by opening tickets, requesting a message on Teams or email or by actively prospecting for demands.

Person 5: Requests originate from emails, Whatsapp messages, work orders, calls or virtual meetings and generally come from customers. Sometimes, the request arises from a need identified by the team as a result of other demands. In the vast majority of cases, the demand is assigned by the team's techlead, which in our case can be three: backend, frontend or QA.

Person 6: I work as a PO in a software factory, where we deliver solutions to the client through commercial partners. Therefore, the demands come through the client's project management tool, Azure, in collaboration with the business partner's requirements team.

When a demand comes in, is it completely defined or do you need to refine it?
(Explain whether demands come ready-made or whether there is room for improvement of the problem)

Person 1: It's necessary to refine, usually we only know the central theme when it arrives, and then if we see that it's a good project, we book in to understand it in more detail.

Person 2: It depends, but for the most part it's necessary to refine, to go into the granular, to understand the business rules, the systems used and the context of the project. Few times does the client come with rounded information; when they did, it was using the AGNS specification and the project was small.

Person 3: Most of the time it needs to be refined

Person 4: It is necessary to refine the problem, design the sub-process affected by the requested demand and, occasionally, contact other teams related to the requesting client for technical and business alignment.

Person 5: They don't always arrive with a satisfactory level of detail. Sometimes we are free to improve the description of the requirement (when it's a simpler feature) and sometimes we meet with the client to get a better understanding. In the case of very large demands, the client is required to go into as much detail as possible, which often doesn't rule out the need for alignment meetings.

Person 6: The vast majority of requests come with a macro definition of the business rules and therefore require an in-depth study of the business rules and possible impacts on the system before the request is taken to the development team.

How do you refine demands on a daily basis?
(Describe your refinement process, what steps you follow, who participates and what tools or methods you use).

Person 1: We usually arrange a meeting with the owner of the routine to understand how it is currently done, so that we can understand in detail where we can act.

Person 2: Through meetings with clients, recorded, and groups/chats for questions during the specification. After these understandings, they are documented in the confluence.

Person 3: The refinement process usually takes place after the Daily meetings, in which the development tasks are discussed and the situation they are in. These meetings take place with the client.

Person 4: The specification process involves active or passive discovery (depending on the origin of the demand) of the entire macro-process directly with the client. Once all the materials have been collected, be they documents, videos, spreadsheets, presentations, etc., the entire process is documented, a flowchart drawn and a technical solution proposed. The correct cases and possible ramifications are mapped out to avoid errors.

Person 5: It depends on the complexity of the demand. If it's something simple, I myself access the application in which the improvement will be applied, understand the feasibility and register the demand in Jira, for later allocation of a dev to work on it. If the demand was more complex, it was necessary to meet with other business analysts (optional) and also with the techlead of the areas involved (essential). The construction of flowcharts is also a relevant activity in some cases. Some of our applications require prototyping in Figma. Others only require written guidelines or simple changes to the system's screenshot. These are attached to the demand card to guide the dev. There are cases in which, after detailing the card, the dev comes to us to reinforce the understanding of the activity for a pre-validation and we also have the SCRUM ceremonies, the sprint planning meeting is also an occasion to refine the understanding.

Person 6: We hold a business refinement meeting, where I make a point of taking part in all the requirements gathering agendas. This agenda is attended not only by me, but also by the commercial partner and the client. We currently use Scrum in the project, so the flow is as follows: we raise the priority demands with the client, we measure the effort and cost of development, we pass this information on to the client and the client tells us which demands we should develop first.

Do you feel you have enough autonomy to refine the demands in the best way? Why?
(Explain whether you have the freedom to adjust demands as necessary or whether there are restrictions that hinder this process).

Person 1: No, currently to carry out more in-depth technical refinement, I turn to my technical partner to think about the architecture of the solution

Person 2: No, but because of the technical dependence on the protype team. I still need them to have conversations and understandings with the IT teams and also to think about solutions for the project.

Person 3: Yes. I always have a communication channel open with the client at any time to refine the demands.

Person 4: Yes, but the time available to consider a demand as refined often hinders the process of strategic planning and questioning that could lead to functionalities that are more adherent to a good product or sustainable solutions in the long term.

Person 5: We have freedom up to a point, when the demands don't involve a lot of work at backend level, integration, etc. What makes the process difficult is that we don't have the technical vision of the impact that certain requests will have on the applications and the functioning of the systems. That's why it's necessary to meet with the techleads to go over what has been demanded by the client and reach a common denominator and measure the effort of the deliveries.

Person 6: Yes. In order to make any adjustments to the demands, we depend on the client's acceptance and, in some cases, the commercial partner's approval.

What are the main challenges you face in refining demands?
(List the main obstacles that hinder more efficient refinement, such as lack of information, time constraints, lack of alignment with stakeholders, etc.)

Person 1: technical refinement and full understanding of the routine by the requesting area

Person 2: Getting the technical information from the IT team, getting to the detail of the information with the from/to filled in.

Person 3: The main problem is the lack of code documentation aligned with the business documentation to minimize verification time for a requirement or demand.

Person 4: Time constraints, lack of alignment with leadership on strategy and incomplete information, forcing direct refinement with the client to be prolonged.

Person 5: Lack of clear information from the client and even indecision about what they really need; time constraints, both because of possible client urgencies and because of the techleads' scarce time for alignment meetings; lack of knowledge of contractual details, because sometimes what the client wants will cost X hours worked and the client may not have this expected guarantee of payment; Little technical knowledge of the effort needed to implement a feature or suggest alternatives to it.

Person 6: Time constraints, sometimes we want to start developing the demand as soon as possible and we end up compromising an effective survey and refinement of the specifics of the demands.

Appendix B - Tech leads and solution architects survey answers

Survey 1 - 17/03 a 28/03

How do technical demands reach you? *Describe the process for receiving technical demands and how these demands are communicated to you (meetings, management platforms, direct communication with stakeholders, POs, etc.).*

Person 1: Demands come to me in 3 ways: from the PO responsible for the areas I serve, directly from clients in the areas of the bank I serve or have served and from my boss (usually internal demands from the area).

Person 2: Currently, requests come through a platform called Confluence, where the PO writes the specification of the problem we want to address. In addition, before the technical demands are ready, there are specification meetings with the client, in which some POs involve the tech lead responsible for the area or even the DEV who is interested or who will take on the demand.

Person 3: Alignment meetings with stakeholders

Person 4: Today there are stakeholders who are “owners of the business” and ask for something, as well as other peer leads who also ask, and there are also technical debts and today, as a business owner, we are also able to generate demands for ourselves, from large projects to small fixes

Person 5: Sometimes they come from the business team, when they see an opportunity that is worth exploring, other times from analysis and monitoring of metrics and system logs, which provide indications of the problems that users face on a day-to-day basis and, consequently, points for improvement in the system.

Person 6: In the last quarter, the technical demands came directly from contacting the client and specifying the routine. In the course of mapping the As Is, the requester comments that they need to consume data from X systems, so the next step is to seek out the person responsible for the systems and make an appointment with them to understand how to consume the information.

How do you define and prioritize the technical approach for each demand you receive? *What criteria do you use to decide on the best technical solution to meet the demand? How do you decide between different approaches or technologies?*

Person 1: I try to understand the problem well, look at the scalability required and possible changes or problems. Then I look at the solutions we've already developed and see how this new demand fits best with the technologies we've already used and that we know work well. If we haven't done anything like that yet, I try to search the internet for the best technology and solution to the problem.

Person 2: As our Automation area is a tactical area, we should prioritize an architecture for which we already have templates designed, with the aim of delivering quickly. Our solutions consist of lambdas and cronjobs, and we understand which one is the most suitable for what we are going to do. In addition, we evaluate the complexity of each demand to decide whether it is best to use a relational database or DynamoDB.

Person 3: It is analyzed whether the technical solution can scale beyond the basic scenarios envisaged, whether there is technological know-how within the resources and how complex maintenance is and what the cost is to keep the solution running.

Person 4: I think that in today's context we first evaluate a factor of complexityXtimeXvalue of delivery. Then we always design what we think is ideal, after understanding all the business needs and translating this into technology, what is the volume? criticality of the flow? can it be asynchronous? And according to the factor above, we understand what we can attack, try an mvp of the solution, or something like that

Person 5: Ease of maintenance of the planned solution, future scalability and assertiveness of the input data, to avoid miscalculations and possible losses for the client and the company.

Person 6: First, I understand the client's needs, considering the performance required and the method of consuming external data from other systems. I also take into account the need for exception flow mapping, understanding how the system needs to behave if any stage fails.

Do you have the autonomy to define the architecture and technical solutions to the demands or are there limitations? *Are there external factors (e.g. budget constraints, deadlines, stakeholder requirements) that affect your ability to define the ideal solution?*

Person 1: Most of the time I have the autonomy to decide, but I always end up choosing the quickest and most efficient solution, since I work on tactical projects that should be short term.

Person 2: I have the autonomy to design the technical solution we're going to tackle in the best way possible, but as we have delivery targets throughout the year, we have to prioritize agility and efficiency. That's why we rely heavily on solutions that have already been developed and that make our day-to-day work easier. With regard to external factors, we always have to be careful that the solutions we adopt are not excessively expensive if they are used inappropriately, such as the use of Glue Job, Athena and other tools. However, we have this initial control when defining the solution.

Person 3: Limited by budget

Person 4: There are certain limitations, budget is not so much a problem, but there are non-approved technologies and technologies that some higher tech stakeholders don't like, and this creates a taboo and an impossibility of being able to think autonomously about which technologies to use

Person 5: We have autonomy as long as the solution doesn't deviate from the design standards defined by the team, in order to guarantee easy learning by other colleagues if they need to work on our systems.

Person 6: I have a lot of autonomy to define the architecture and create proofs of concept of my ideas, but always with care for costs.

What tools or methodologies do you use to coordinate the execution of technical demands with your team? *Tell us about the tools such as Jira, Azure DevOps, Trello and others that you use to manage the technical execution of demands.*

Person 1: To keep track of my individual tasks I use OneNote, to write down everything I've talked about with the client and all the rationale for what I have to do on the project. To keep track of all the projects, the overall status and dates I use Azure DevOps.

Person 2: With regard to technical demands, we use Confluence for both the specification and the final design of the solution. In other words, once the DEV has finished, he has to document the code in Confluence. With regard to deadlines and tasks created, we use Azure DevOps for control. In addition, for my personal control over the developers, I use OneNote for general notes.

Person 3: Azure, Excel

Person 4: Monday and Azure Devops

Person 5: Monday to define the macro scope of activities, Azure DevOps to define the micro scope, including features and user histories, MS Teams/Outlook to control agendas and meetings. We follow some rites of the SCRUM methodology, such as daily meetings, planings and code reviews.

Person 6: I use Azure, creating tasks that should ideally last up to 2 days of development.

What is the biggest difficulty you encounter when communicating technical needs and priorities to the development team? *Are there any challenges in communicating clearly or setting expectations, both internally within the team and with other departments (POs, stakeholders, etc.)?*

Person 1: The biggest difficulty I experience is foreseeing all the technical problems that may arise or possible unforeseen integrations

Person 2: The biggest difficulty I encounter when communicating technical needs and priorities to the development team is mainly in integrations and the way data consumption is carried out. For example, if the consumption is via API, the estimated time is X; however, if it is via messaging, the time increases by 10 due to the complexity of saving the data and consuming it later. This difference in complexity is not always clear to everyone involved, especially stakeholders and POs, which can lead to challenges in communication and aligning expectations. When these integrations and complexities are already well defined in the initial project specification, I can better estimate deadlines and development effort.

Person 3: There can be noises in the understanding of demands in cases of alignment where the necessary people have not been involved.

Person 4: Today, as I sometimes end up playing the role of a stakeholder, not just a tech stakeholder, I find it a little difficult to change hats quickly, and this is also a little more burdensome than normal, and ends up making it difficult to write the tasks better, even when refining previous definitions

Person 5: The greatest difficulty lies in the technical understanding of the proposed solutions, which needs to be very well explained and aligned with all the teams involved in order to identify possible gaps in the systems and planned integrations. It is necessary for those involved to have a high technical level and to know their products and/or systems well so that the scopes are well defined, thus avoiding teams developing solutions that are already offered by other existing systems and known problems going unsolved.

Person 6: Stakeholders who don't understand the technical side of development often don't agree with certain deadlines. This alignment with clients is always a challenge.

Have you ever faced challenges related to changing requirements or priorities during the development cycle? *How do you deal with unexpected changes and how do they affect the proposed technical solution? How do you adjust the team's planning to adapt?*

Person 1: I've had to deal with scope changes on several projects, and they have a significant impact on the time it takes to deliver the solution. This often leads to frustration on the part of the client because they don't really understand that it wasn't mapped out, since it's very obvious to them certain steps in their daily lives. In these cases, I understand the new demand well, price this adjustment and pass it on to the team. I understand together with the team the development to be done, the stages and how long it would take

Person 2: Changes during development are quite common, often due to information that the client didn't include in the initial specification. This can cause anxiety among developers, but as a tech lead, we have to take a lot of the responsibility and come up with new approaches to deal with these situations. Generally, we increase the delivery time and assess what is feasible in the time available, leaving the rest for a version 2 of the project.

Person 3: Constant, the solutions are designed to be easily changed and evolved so that depending on the type of change, it shouldn't have such an impact on development time.

Person 4: It even happens frequently, we usually stop everything and replan as soon as possible, the fact that the development cycle is not so respected “helps” in this regard

Person 5: Yes, frequently, and this is part of the agile methodologies used by the team. When this happens, we bring the team together to discuss possible solutions and adjustments to the scope and find out who is responsible for the solution in the other teams involved in the project, so that we can then contact the right people, ensuring assertive and effective communications. Finally, we study possible extensions to pre-defined deadlines and align delivery date expectations to ensure that everyone is on the same page.

Person 6: Yes. The guideline is that if the proposed change prevents the use of the mapped solution in production, this point is formalized with the leaders of the requesting area and included in the delivery. Otherwise, the change is barred from the current delivery and allocated to the next version of the project.

What is your collaboration process with POs and stakeholders throughout the development cycle? *When do you interact with POs or stakeholders during the process? How do these interactions impact the definition and execution of technical solutions?*

Person 1: I interact with the PO from the moment he has understood the problem with the client and put together a specification, even if it's just a sketch. I help in the process with the technical parts, designing the solution and checking the feasibility of the solution.

Person 2: The collaboration process is usually quite effective. Both the PO and the stakeholders usually explain any business rules in detail or clarify any doubts with the developer responsible for the requirement, involving the IT teams when necessary. These interactions, depending on the case, can alter the initial scope of the project due to requested changes or improve initial understanding, which in turn speeds up delivery.

Person 3: Surveying the effort and tasks of a given demand, and also understanding the demand in full with the stakeholders.

Person 4: We interact a lot in refinements, it helps a lot, it ends up having an impact on how the task is going to be defined, even before the refinement, and it's essential, whenever this triad of tech, product and business happens, it's ideal

Person 5: Interaction with stakeholders happens all the time through project progress reports, release notes with the latest features and scheduled and spontaneous meetings to resolve doubts. This constant interaction is essential so that expectations are always aligned, those responsible are always well identified and any deviations are not taken by surprise, avoiding friction between the areas involved.

Person 6: Mainly in mapping the As Is and defining the screen prototype

Appendix C - Developer survey answers

Survey 1 - 17/02 to 28/02

1. How do work demands come to you on a daily basis?

(Describe the process, including where you receive it and how it is assigned).

Person 1: I have routine demands such as sending daily reports, and other operations that arrive by e-mail, Bloomberg terminal, automation project demands I usually ask if there is anything to do, and momentary demands usually an analyst comes to my desk and asks for it

Person 2: These demands come in two ways. Either from meetings I have with clients in my business area (the clients are internal), in which they request new features or projects, or from conversations with my boss, in which he does the same thing as above, only privately, without me. In short, it all comes down to talking to people in the business area, either myself or my boss, and passing it on to me. In the end, I usually record this myself on an Azure DevOps board, so that I have a clear flow of tasks, as well as development time and delivery deadlines.

Person 3: My manager currently has a conversation with the project's stakeholders and they define the scope to be worked on and the demands of that scope. After this conversation between them, my manager sets up a meeting with me and gives me the context of the demand, how I have to act on it and what the delivery deadlines are.

Person 4: Tech Lead

Person 5: Mainly at the request of the business team, but also by analyzing system logs and taking a critical look at the portal, which allows us to understand the customer journey and which points need improvement. Tasks are assigned by analyzing the workload of the team members (Azure's backlog helps a lot here) and each person's previous experience with the topic to be developed.

Person 6: Demand from stakeholders

Person 7: It can either come from the commercial or operational team, in some cases there may be projects that come from a partner, but a good part of it is also proactive, always taking into account the success of the products and the benefit for those impacted.

Person 8: By the squad leaders, usually the teams. It's usually passed on by them in the same way.

Person 9: The demands come mainly through conversations, without a centralized system to keep track of the tasks. When a new project starts, we hold a meeting to understand the requirements, and from there, I carry out the work until the final delivery, with no structured monitoring of progress.

Person 10: It usually happens on two fronts. The product team can segment some demands, either regulatory terms or business needs, or these demands can come from QA (some defect or fix needed), or from some Product Manager in the tech area. We usually receive these demands by e-mail, and the assignment is made by creating a JIRA by our VP (Lead) or a Senior and redirecting it to the DEVs.

Person 11: Arrives via PO and Tech Lead

2. Do you think that the way demands come to you currently works well or could it be improved?

- It works well, I don't see any need for change.
- It works, but it could be better.
- It doesn't work well, it needs improvement.
- Others...

Person 1: It works, but it could be better.

Person 2: It works, but it could be better.

Person 3: It doesn't work well, it needs improvement.

Person 4: It works well, I don't see any need for change.

Person 5: Other: Works well for the size of the team and scope of the product. Excessive control can be detrimental and lead to micromanagement and a consequent decrease in morale.

Person 6: Works well, no need for change.

Person 7: It works, but it could be better.

Person 8: It doesn't work well, it needs improvement.

Person 9: It works, but it could be better.

Person 10: Works, but could be better.

Person 11: Works, but could be better.

3. If you think it could be improved, how could this process be made more efficient?

Person 1: A list of tasks and pending items, with priorities

Person 2: It would be ideal to have a system capable of listening to the meetings, automatically organizing the ideas discussed and mapping out new flows requested. At the end of the meeting, the developers or tech leads could review, edit and approve the suggestions. Once approved, the system would integrate the flows directly into management tools such as Azure DevOps, Monday, Jira, Trello, among others.

Person 3: I think that if I had direct contact with the stakeholders I could have a better contextualization of the problem and how the solution applies to it, as well as being able to clear up any doubts and propose improvements to what has been worked on, giving me more prominence within the work rather than just being a conveyor belt for producing demands.

Person 4: No answer

Person 5: No answer

Person 6: No answer

Person 7: I actively participate in business decisions to evolve the products and, to a certain extent, the software I work with, which I believe evolves well without deviating from the ideal strategy, so demands are 'micro-managed' with care, but I believe that urgent demands with a general impact on a larger scope should be passed through a committee of technical managers to assess the general risks and effectively plan the implementation of these changes. Today the teams act individually and decentrally.

Person 8: I believe that better organizational methods could be used. Using agile methods and tools to organize these activities, prioritizing the really urgent ones.

Person 9: The process would be more efficient with a tool to record and monitor demands. This would help to better organize the work, give more visibility to the progress of tasks and improve communication.

Person 10: The scope of the change could be a little more detailed and include the possible impacts of the change. In addition, the demand deadline is usually not feasible with the development “effort”, which is something that is discrepant when it comes to deliveries

Person 11: It would be interesting if the PO had more choice in the activities to be prioritized per sprint, with the Tech Lead being in charge of defining the best developer. Perhaps this is the case and I haven't noticed.

Survey 2 - 17/03 to 28/03

1. How do work demands come to you on a daily basis?

(Describe the process, including where you receive it and how this assignment happens).

Person 1: They come to me through my leader who has already evaluated the specification and arrived at the point where it is feasible.

Person 2: I receive demands from my Tech Lead in calls or dailys, which before reaching my tech lead are raised by the clients and specified by the PO. But there have been cases where I've listened to the pain directly from the client and managed to map out the solution and develop it.

Person 3: Through my line manager or the robot that sends the issues

Person 4: The demands come through the PO/Tech lead. This happens more than once a week. The PO specifies the projects and aligns them with the Tech Lead who distributes them among the available devs.

Person 5: They arrive via specification reports, these reports are included in our Confluence and then shared with the developers assigned to the project, meetings are usually held to go over the specification with the devs.

Person 6: Through Confluence, I receive how the flow is and how I should set up the solution. Then I can assemble the cards according to what I want to develop.

Person 7: The demands arrive via the PO and Tech Lead and the Developers create the tasks.

Person 8: Demands usually arrive after a conversation with clients or collaborators, and are then separated by priority for each team member.

Person 9: Through management tools with agile dashboards (Scrum and Kanbam)

Person 10: Via team chat or allocation from the superior via new project specifications

Person 11: Through tech leads and requirements analysts

Person 12: Card assignment in the Jira system. In the Daily Meetings we have, we talk about completed demands, future demands, cards are created, assigned to the managers of each team (techleads), and 'child' cards are created to assign to whoever picks up the task.

Person 13: Receipt occurs through Jira, most of the time I identify the tasks that need to be done and assign myself to them by notifying the tech lead. There are also tasks that are defined with the team, tasks aimed at improving workflow, improving the company's projects or activities aimed at improving the delivery of demands.

Person 14: I usually look in Jira and check with the Tech Lead about the possibility of starting the task

Person 15: I ask my Tech Lead if there's a task I can take on, or I look at 5 Jira boards, one for each client, and ask them if I can take on the task that's available.

2. Do you think that the way demands come to you currently works well or could it be improved?
- It works well, I don't see any need for change.
- It works, but it could be better.
- It doesn't work well, it needs improvement.
- Others...
-

Person 1: It works well, I don't see any need for change.

Person 2: It works, but it could be better.

Person 3: It works, but it could be better.

Person 4: It works, but it could be better.

Person 5: It works well, I don't see any need for changes.

Person 6: It works, but it could be better.

Person 7: It works, but it could be better.

Person 8: It works, but it could be better.

Person 9: It works, but it could be better.

Person 10: It works, but it could be better.

Person 11: It works, but it could be better.

Person 12: It works well, I don't see any need for change.

Person 13: It works well, I don't see any need for change.

Person 14: It works, but it could be better.

Person 15: It works, but it could be better.

3. If you think it could be improved, how could this process be made more efficient?

Person 1: No answer

Person 2: I believe that the specification should be read by the tech lead with a more critical eye. Sometimes my fellow developers and I have had to figure out where to look for certain information, such as databases and so on, which ends up hindering the delivery of the project.

Person 3: Tasks could arrive by robo, currently they arrive by e-mail

Person 4: I think the tasks could be divided according not only to the availability of the devs, but also to their level of experience.
It would be nice to have greater visibility of quick wins, so that any time gap in the sprint could be converted into a task, without idle time between one project and another or large projects being blocked.

Person 5: No answer

Person 6: the cards could be ready-made, with a complete step-by-step description of what needs to be done. For example, test the integration with table x in the lake, match the data in the table with file y. This way, I could just worry about putting what was requested into the logic of the code. That way, I could just worry about putting what was requested into the logic of the code. It would also be interesting to have homologation cards to make testing easier and to deliver the card in develop done closer to what it should be.

Person 7: I think it would be better for POs to create tasks for each User Story. That way, the project would be better monitored.

Person 8: Understand all the points before starting to pass on demands!

Person 9: Documentation showing what is being done and how it is being done in the code in line with the business specification documentation. This would optimize the time that is wasted on any checks that need to be made.

Person 10: Specifications with greater depth of flow and the participation of a developer in the discovery process. Improve project scope limitation and better filter out projects that make sense.

Person 11: No answer

Person 12: At the moment I think this treadmill is quite functional.

Person 13: No answer

Person 14: sprint start and duration notifications could be shown in a clearer and more accessible way through a spreadsheet, table or any other tool that would help to clearly inform the time of action, especially in scenarios where several sprints occur in parallel. The process of notifying demands is still very “manual”, there could be a platform with all the tasks containing an option for the dev to “signal” that they want to start that activity and then the person responsible (in my case the Tech Lead) would be notified with the option of approving or rejecting that request.

Person 15: If someone inserts a new card, inform them that there is a new task, instead of me looking board by board to see if there is a new card, there are days when there is no new card all day.

4. How often do you receive work demands?
- Daily
- Weekly
- Monthly
- Other...

Person 1: Other: I receive demands like a treadmill, where a new project arrives only when I've finished the current one.

Person 2: Other: Every two weeks

Person 3: Daily

Person 4: Daily

Person 5: Weekly

Person 6: Monthly

Person 7: Daily

Person 8: Daily

Person 9: Daily

Person 10: Daily

Person 11: Weekly

Person 12: Weekly

Person 13: Other: It depends with the number of tasks and sprint, the sprint is over and requirements are still being raised and cards are being created I am free from Jira board tasks, but there are still tasks defined with the team that need to be touched.

Person 14: Other: almost daily. I start an activity after finishing another, which takes an average of 1 to 4 days depending on its complexity.

Person 15: Daily

5. Do you feel you have the autonomy to organize your own demands or are you totally dependent on others?

- I have complete autonomy
- I have some autonomy, but I follow a fixed direction
- I am totally dependent on others

Person 1: I have total autonomy

Person 2: I have some autonomy, but I follow a fixed direction

Person 3: I have some autonomy, but follow a fixed direction

Person 4: I have some autonomy, but follow a fixed direction

Person 5: I have complete autonomy

Person 6: I have complete autonomy

Person 7: I have total autonomy

Person 8: I have total autonomy

Person 9: I have some autonomy, but I follow a fixed direction

Person 10: I have total autonomy

Person 11: I have some autonomy, but I follow a fixed direction

Person 12: I have some autonomy, but I follow a fixed direction

Person 13: I have some autonomy, but I follow a fixed direction

Person 14: I have some autonomy, but I follow a fixed direction

Person 15: I am totally dependent on third parties

Appendix D - Framework Evaluation:

Question1 : What is your main role in this project?

Person 1: Developer

Person 2: Business | Desk

Person 3: Product Owner (PO)

Person 4: Product Owner (PO)

Person 5: Tech Lead

Person 6: Tech Lead

Question 2: Project to be evaluated (choose one):

Person 1: Solar Credit Taker

Person 2: Automatic Order Settlement

Person 3: Automatic Order Settlement

Person 4: Complete Assignor – Drawee Risk

Person 5: Automatic Order Settlement

Person 6: Limit Reallocation

Question 3: Which FELIX stages did you participate in? (Select all applicable options)

Person 1: Demand Investigation, Project Alignment

Person 2: Project Kick-off, Backlog, Prioritization, Demand Investigation, Project Alignment

Person 3: Backlog, Prioritization, Demand Investigation, Project Alignment, Executive Summary

Person 4: Prioritization, Demand Investigation, Project Alignment, Executive Summary

Person 5: Project Kick-off, Risk Classification, Backlog, Prioritization, Demand Investigation, Project Alignment

Person 6: Backlog, Prioritization, Demand Investigation, Project Alignment, Deprioritization

Question 4: Of the stages you selected, which ones were the most useful? (choose up to 3)

Person 1: Demand Investigation and Project Alignment

Person 2: Project Kick-off, Prioritization, Demand Investigation

Person 3: Demand Investigation and Project Alignment

Person 4: Demand Investigation and Project Alignment

Person 5: Project Alignment, Demand Investigation, Prioritization

Person 6: Demand Investigation and Project Alignment

Question 5: How much of the documentation generated by "Demand Investigation (Stage 5)" did you actually find useful for the project?

Person 1: 100%

Person 2: 100%

Person 3: 100%

Person 4: 100%

Person 5: 100%

Person 6: 100%

Question 6: FELIX brought clarity on what needs to be done.

Person 1: Strongly agree

Person 2: Strongly agree

Person 3: Strongly agree

Person 4: Strongly agree

Person 5: Agree

Person 6: Agree

Question 7: FELIX reduced rework/ambiguity in my daily work.

Person 1: Strongly agree

Person 2: Agree

Person 3: Strongly agree

Person 4: Agree

Person 5: Agree

Person 6: Strongly agree

Question 8: FELIX improved alignment between areas (business, PO, TL, development).

Person 1: Strongly agree

Person 2: Strongly agree

Person 3: Agree

Person 4: Strongly agree

Person 5: Strongly agree

Person 6: Strongly agree

Question 9: Perception of refinement time compared to the previous process:

Person 1: Much faster

Person 2: Much faster

Person 3: A little faster

Person 4: A little faster

Person 5: A little faster

Person 6: Almost the same

Question 10: Predictability/visibility of demand status (backlog, next steps) improved.

Person 1: Much faster

Person 2: A little faster

Person 3: Much faster

Person 4: Much faster

Person 5: A little faster

Person 6: Much faster

Question 11: FELIX increased my autonomy in the process. (for POs: define/refine; for TL/Devs: receive better-prepared inputs).

Person 1: Strongly agree

Person 2: Agree

Person 3: Agree

Person 4: Strongly agree

Person 5: Agree

Person 6: Strongly agree

Question 12: Would you recommend using FELIX in other projects?

Person 1: Definitely yes

Person 2: Definitely yes

Person 3: Definitely yes

Person 4: Definitely yes

Person 5: Probably yes

Person 6: Definitely yes

Question 13: What helped you the most in FELIX? (open-ended)

Person 1: Having documentation not only as a presentation but also as proof of work, for evaluations, business questioning, etc. This enriches the delivered work, making the effort clear.

Person 2: Documenting the project steps.

Person 3: Concentrating all project information in one place (business rules, flows, screens) and making it easier to explain to new people from other areas.

Person 4: Having everything well-structured in one place, always updated, linked to meetings, with pre-configured fields in the Wiki that made documentation easier and standardized validation with business.

Person 5: Activities and responsibilities were more clearly defined, and expectations were better aligned across all teams.

Person 6: The final materials from each stage, which make it clear what needs to be developed, how, and what does not need to be. Discovery processes ensure total alignment of expectations and delivery.

Question 14: What would you improve in FELIX? (open-ended)

Person 1: —

Person 2: Concentrate communication in just one group, although I understand the difficulty due to multiple meeting groups.

Person 3: A higher-level vision for macro projects, like reporting to executives/senior stakeholders.

Person 4: None.

Person 5: State transitions of activities from one team are not easily perceived by others. A notification system when an activity moves to Done would be useful.

Person 6: Since it’s still in validation, outputs from each stage could be more visible to everyone.